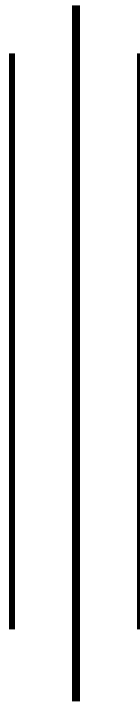


INSTITUTE OF ENGINEERING

PULCHOWK CAMPUS

DEPARTMENT OF MECHANICAL ENGINEERING



**FUNDAMENTALS OF
THERMODYNAMICS AND HEAT
TRANSFER**

(BE FIRST YEAR)

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CHAPTER 1

INTRODUCTION

1.1 Definition and Scope of Engineering Thermodynamics

Thermodynamics is defined as the study of energy, its forms and transformations, and the interactions of energy with matter. Hence, thermodynamics is concerned with

- the concept of energy,
- the laws that govern the conversion of one form of energy into another,
- the properties of the working substance or the media used to obtain the energy conversion.

In engineering or applied thermodynamics the scope is restricted to the study of heat and work and the conversion of one into the other. Thermodynamic laws are applied to work producing and work absorbing devices in order to understand their functioning and improve their performance.

Thermodynamics has extremely wide range of applications. For example it is used by the mechanical engineer in the design of energy converting devices such as steam and gas turbines, internal combustion engines, fuel cells, thermoelectric generators as well as refrigerators and air-conditioning equipment.

1.2 Value of Energy to Society

The availability of energy and people's ability to harness that energy in useful ways have transformed our society and the energy requirements of almost all countries are ascending up inexorably. It is very hard to imagine the present life without electricity and other forms of energy. The energy available and consumed data exhibit the perspective picture of the economic condition and scope and the level of advancement of living people's civilization.

A few hundred years ago, the greatest fraction of the population struggled to subsist by producing food for local consumption. Now, in many countries a small fraction of the total work force produces abundant food for the entire population, and much of the population is freed for other pursuits. We are able to travel great distances in short times by using a choice of conveyances; we can communicate instantaneously with persons anywhere on earth; and we control large amounts of energy at our personal whim in the form of automobiles, electric tools and appliances, and comfort conditioning in our dwellings. All these changes are as results of the development of the science and application of thermodynamics, our ability to obtain energy, transform it, and apply it to society's needs.

1.3 Macroscopic versus Microscopic Viewpoint

There are two points of view from which the behavior of matter can be studied: the macroscopic and the microscopic. The description of a system or matter using a few of its measurable bulk properties constitutes a point of view called macroscopic. For example, consider a diesel engine cylinder with air in it. The air occupies a certain volume for each piston the position. The volume for any position is easily measured. The temperature and

pressure are also easily measured. The state or condition of the system (air in the cylinder) is completely described by means of the above large scale characteristics or properties of the system. Such properties are called macroscopic point of view. Different system requires different properties for complete description of the state. Some common attributes of such description are: (a) These characteristics do not involve any assumptions about the structure of the material, (b) these are readily measured, and (c) Only few such properties are required to completely describe the system.

In the above example the air in the cylinder is assumed to consist of a very large number of particles (all having the same mass) and each moving with an independent velocity when a microscopic point of view is adopted. The description of the position of each molecule requires three coordinates and the three velocity components. At least six variables are required to describe the state of each molecule. To describe the state of all molecules is almost impossible. The pressure due to the air is assumed to be caused by the moving molecules striking the wall. To determine the pressure, the change in momentum of all the molecules striking the wall per unit area should be summed up. Similarly to find the internal energy of the system, the kinetic energy of all the molecules should be summed up. Though microscopic study of a system may provide data for fixing the state, it is very complex for ordinary purposes. The macroscopic point of view is sufficient as far as applied thermodynamics is concerned.

All the results of classical or macroscopic thermodynamics can, however, be derived from the microscopic and statistical study of matter.

1.4 Concepts and Definitions

Before the basic structure of thermodynamics can be outlined, it is necessary to define some of the fundamental concepts thoroughly.

1.4.1 System and Surroundings

Whenever a change is to be analyzed it is necessary to specify the region under study in order to avoid any confusion. In thermodynamics defining the boundary around the region of consideration does this. This boundary may be real or imaginary, moreover, it may be fixed or it may change in shape as well as volume. Everything within the boundary is called the system, while the region external to the boundary is called the surroundings. All transfers of mass and energy between the system and surroundings are evaluated at the boundary.

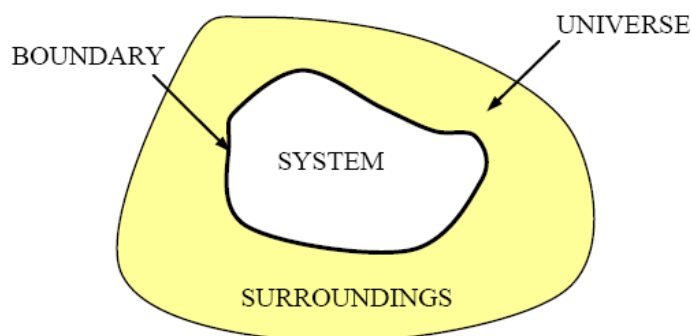


Figure 1.1: System, Surroundings, Boundary and Universe

Hence, the system is a specified region wherein changes due to transfers of mass and energy or both are to be studied. It is not necessary that the volume or shape of the system should remain fixed.

A system and its surroundings together comprise a universe which can be defined as totality of matter that exists.

The thermodynamic system may be classified into the following three groups: (a) Closed system; (b) Open system; and (c) Isolated system.

If the mass within the boundary of the system remains constant, it is called a closed system as shown in **Figure 1.2**. Thus a closed system does not permit any mass transfer across its boundary but it permits transfer of energy. Closed system is sometimes also referred to as control mass (CM).

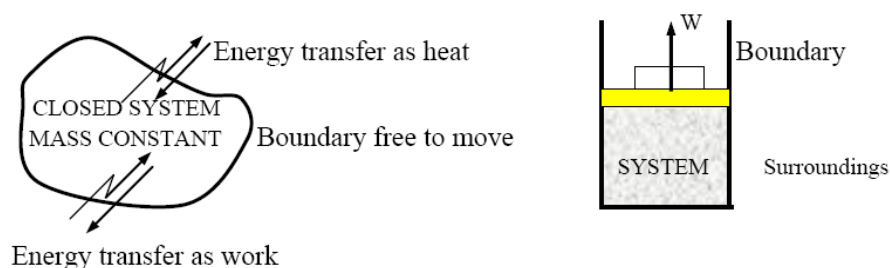


Figure 1.2: Closed System

If both mass and energy cross the boundary of a system it is called an open system as shown in **Figure 1.3**. An open system permits both mass and energy across the boundaries, and the mass within the system may not remain constant. For thermodynamic analysis of an open system, such as an air compressor, attention is focused on a certain volume in space surrounding the compressor, known as the control volume, bounded by a surface called the control surface.

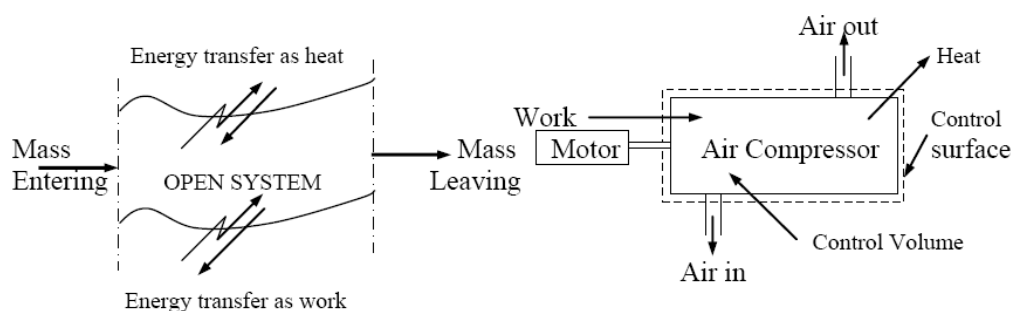


Figure 1.3: Open System and Control Volume

If neither mass nor energy is allowed to cross the boundary of a system it is called an isolated system. Thus in an isolated system does not have transfer of either mass or energy with the surroundings. Any closed rigid insulated box can be taken as an example of an isolated system.

1.4.2 Property

A thermodynamic system would have many characteristics, directly or indirectly measurable, which would serve to describe or identify the system. For example, mass,

physical composition, temperature, pressure, volume, surface area, velocity, elevation and electrical potential are some such characteristics. In addition a large number of other characteristics of the system can be defined. These characteristics or quantities, the values of which describe the system are called thermodynamic properties or thermodynamic coordinates of the system.

The thermodynamic properties of a system may be divided into the following two general classes: (a) Intensive properties, and (b) Extensive properties.

If the value of a property is independent of the mass of the system, it is called an intensive property. The pressure, temperature, density, velocity, height, viscosity, etc. are examples of intensive properties.

If the value of a property is proportional to the mass of the system, it is called an extensive property. The volume, surface area, energies of all kinds, electric charge, magnetization, etc. are examples of extensive properties.

The ratio of an extensive property to the mass is called the specific value of that property. Thus, if the total energy of a system is denoted by E , and the mass by m , then the specific total energy or total energy per unit mass is given by

$$e = \frac{E}{m} \quad \dots\dots\dots (1.1)$$

In general, capital letters will denote extensive properties, and small letters will be used to represent the specific values of these properties.

1.4.3 Thermodynamic Equilibrium

While talking about the properties of a system we had assume implicitly made the assumption that the value of a property is the same at all points in the system. Evidently it is meaningless to speak of the temperature or the pressure of a system unless the temperature or the pressure has the same value throughout the system. In other words, we implied that the system is in a state of thermodynamic equilibrium.

Hence, a system is said to exist in a state of thermodynamic equilibrium when no change in any macroscopic property is registered, if the system is isolated from its surroundings. Therefore, there can be no spontaneous change in any macroscopic property if the system exists in an equilibrium state. Thermodynamics studies mainly the properties of physical systems that are found in equilibrium states.

A system will be in a state of equilibrium, if the conditions for the following three types of equilibrium are satisfied: (a) mechanical equilibrium; (b) chemical equilibrium; and (c) thermal equilibrium.

In the absence of any unbalanced force within the system itself and also between the system and surroundings, the system is said to be in a state of mechanical equilibrium. If an unbalanced force exists, either the system alone or both the system and the surroundings will undergo a change of state till the mechanical equilibrium is attained.

If there is no chemical reaction or transfer of matter from one part of the system to another, such as diffusion or solution, the system is said to be in a state of chemical equilibrium.

A system is said to be in thermal equilibrium, when there is no temperature difference between the parts of the system or between the system and surroundings.

1.4.4 Thermodynamic State

Each unique condition of system is called a state. Thus state of a system (when the system is in thermodynamic equilibrium) is the condition of a system at an instant of time which can be as described by the statement of its properties, such as pressure, volume, temperature etc. The number of properties, which are required to describe a system, depends upon the nature of the system.

It follows from the definition of state that each property has a single value at each state. Stated differently, all properties are state or point functions. Therefore all properties are identical for identical states.

On the basis of the above discussion we can determine if a given variable is a property or not by applying the following tests:

- (a) A variable is a property if, and only if, it has a single value at each equilibrium state.
- (b) A variable is a property if, and only if, the change in its value between any two prescribed equilibrium states is single valued.

1.4.5 Thermodynamic Process

When a system changes from one equilibrium state to another equilibrium state, then the path of successive states through which the system has passed is known as thermodynamic process. In **Figure 1.4**, 1 – 2 represents a thermodynamic process.

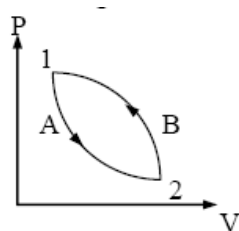


Figure 1.4: Thermodynamic process or Cyclic process

When a process or processes are performed on a system in such a way that the final state is identical with the initial state, it is known as a thermodynamic cycle or cyclic process. In **Figure 1.4**, 1 – A – 2 and 2 – B – 1 are processes whereas 1 – A – 2 – B – 1 is a thermodynamic cycle.

Let us consider a system of gas contained in a cylinder as shown in **Figure 1.5(a)**. The system initially is in equilibrium state, represented by the properties P_1 , V_1 , T_1 . The weight on the piston just balances the upward force exerted by the gas. If the weight is removed, there will be an unbalanced force between the system and the surroundings, and under gas pressure, the piston will move up till it hits the stops. The system again comes to an equilibrium state, being described by the properties P_2 , V_2 , T_2 . But the intermediate states passed through by the system are nonequilibrium states which cannot be described

by thermodynamic coordinates. **Figure 1.5(b)** shows point 1 and 2 as the initial and final equilibrium states joined by a dotted line, which has got no meaning otherwise.

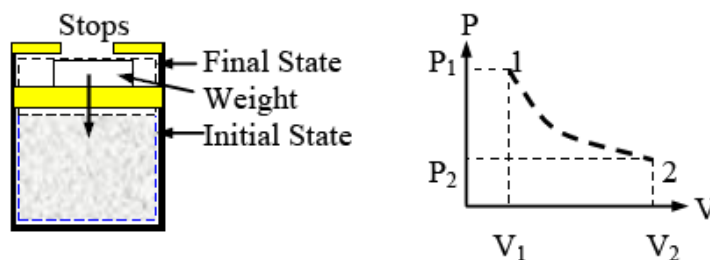


Figure 1.5 (a): Transition between two equilibrium states by an unbalanced force
(b): Plot representing the transition between two equilibrium states

Hence, if a system passes through a sequence of non-equilibrium states during a process, these states can not be located on any property diagram because each property does not have a unique value in the entire system. Such a process is called an irreversible process. It can always be assumed, however, that the system is in a state of equilibrium before and after the process, so that irreversible processes may be represented on a property diagram by a broken line joining the initial and final equilibrium states. The broken line indicates the intermediate states visited by the system during the process are indeterminate.

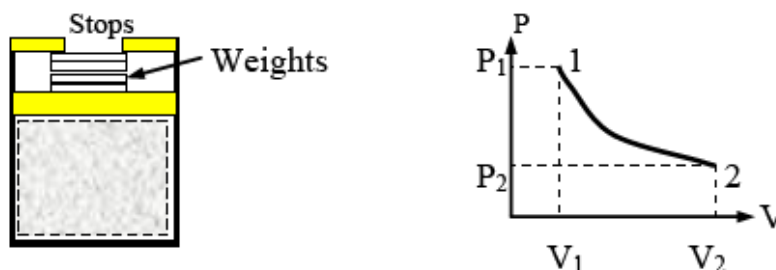


Figure 1.6 (a): Infinitely slow transition of a system by infinitesimal force
(b): A quasi-equilibrium process

Now if the single weight on the piston is made up of many very small pieces of weights as shown in **Figure 1.6(a)**, and these weights are removed one by one very slowly from the top of the piston, at any instant of the upward travel of the piston, if the gas system is isolated, the departure of the state of the system from the thermodynamic equilibrium state will be infinitesimally small. So every state passed through by the system will be an equilibrium state. Such a process, which is locus of all the equilibrium points passed through by the system, is known as quasi-static or quasi-equilibrium process. Infinite slowness is the characteristic feature of a quasi-static process. A quasi-static process is thus a succession of equilibrium states and represented by a continuous line as shown in **Figure 1.6(b)**. A quasi-static process is also called a reversible process.

1.4.6 Some Common Properties

(a) Pressure

Because many systems studied in thermodynamics involve gases or vapors, pressure is a useful property for describing the state of the system. Pressure is defined as the normal force exerted per unit area on a real or a fictitious surface.

With the restriction that area over which the force is applied must not smaller than some minimum value a , the mathematical definition of the local pressure is

$$P = \lim_{\Delta A \rightarrow a} \frac{\Delta F}{\Delta A} \quad \dots\dots\dots (1.2)$$

For engineering work, pressures are often measured with respect to atmospheric pressure rather than with respect to perfect vacuum. The pressure relative to the atmosphere is called gauge pressure. The pressure relative to a perfect vacuum is called absolute pressure. Hence, the gauge pressure is related to the atmospheric pressure by

$$P_{abs} = P_{atm} + P_{gauge} \quad \dots\dots\dots (1.3)$$

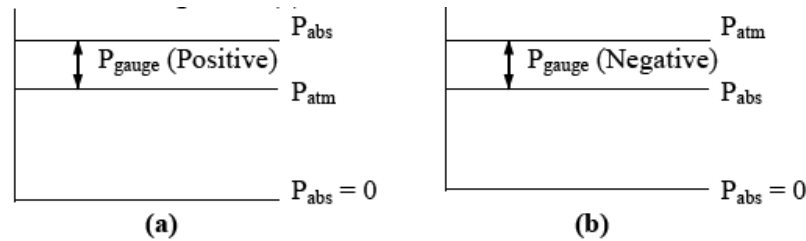


Figure 1.7: Relation between absolute, atmospheric and gauge pressure

This relation is used for pressures above atmospheric pressure, as shown in **Figure 1.7(a)**. For pressures below atmospheric pressure, the gauge pressure will be negative. This negative gauge pressure is known as vacuum pressure. Therefore

$$P_{abs} = P_{atm} - P_{gauge} \quad \dots\dots\dots (1.4)$$

This relation is shown in **Figure 1.7(b)**. The standard value of atmospheric pressure is taken as 101.3 kPa (or 760 mm of Hg) at sea level.

(b) Specific Volume

Volume V is the space occupied by a substance and is measured in m^3 . This is an extensive property. The specific volume (v) of a substance is defined as the volume per unit mass and is measured in m^3/kg . Thus mathematical definition of specific volume is

$$v = \lim_{\Delta m \rightarrow \mu} \frac{\Delta V}{\Delta m} \quad \dots\dots\dots (1.5)$$

where μ is a minimum amount of mass that is large compared with the mass of the individual molecules of which it is composed. Hence, specific volume is the reciprocal of the density, i.e.,

$$v = \frac{1}{\rho} \quad \dots\dots\dots (1.6)$$

(c) Temperature

It is an intensive thermodynamic property, which determines the degree of hotness or the level of heat intensity of a body. A body is said to be at a high temperature or hot, if it shows high level of heat intensity in it. Similarly, a body is said to be at a low temperature or cold, if it shows a low level of heat intensity.

The temperature of a body is measured with the help of an instrument known as thermometer. For most temperature measurements, it is convenient to define a scale that is a linear function of some measurable quantity, at least over some range of temperature between two fixed points. Hence all temperature scale are based on two fixed points known as freezing point of water under atmospheric pressure or ice point and the boiling point of water or steam point.

The two scales most useful in thermodynamics are so called absolute scales. The absolute scale for SI is the Kelvin scale. This scale is a one point scale based on the second law of thermodynamics. The single point is the triple point of water, where ice, liquid water, and water vapour coexist in a closed system in the absence of air. The Kelvin scale replaced the original scale which was based on a linear function between two selected points. The Kelvin unit does not use the degree symbol, but simply the symbol K. The other absolute scale is called the Rankine scale. It is related to Kelvin scale by

$$1.8^{\circ}R = 1K \quad \text{..... (1.7)}$$

where a degree Rankine is denoted by $^{\circ}R$.

Two other commonly used scales are the Fahrenheit scale and the Celsius scale. The Fahrenheit scale is also linear, and it was originally based on two define point: 32° as the temperature at which a system of air-saturated water and ice coexist and 212° as the temperature of a system containing water and steam at a pressure of 1 atm. The symbol for the Fahrenheit degree is $^{\circ}F$. This original definition has been replaced by the following relations to the Kelvin scale:

$$T^{\circ}C = T K - 273.15 \quad \text{..... (1.8)}$$

$$T^{\circ}F = (1.8T)^{\circ}C + 32 \quad \text{..... (1.9)}$$

where a degree on the Celsius scale is denoted by $^{\circ}C$.

The Celsius scale was originally defined by a single point and a defined size for the degree. The fixed point was the triple point of water, and it was defined to have a temperature of $0.01^{\circ}C$. The size of the degree chosen for Celsius scale comes from the Kelvin scale and yields the temperature of the steam point to be $100.00^{\circ}C$ at 1 atm. This original definition was replaced by the relation with Kelvin scale given in **equation (1.8)**.

Figure 1.8 compares the various scales. Note that the size of 1 K is the same as 1 degree on the Celsius scale; similarly, the size of the degrees on Rankine and Fahrenheit scales are equal.

1.5 Equality of Temperature and Zeroth Law of Thermodynamics

Equality of Temperature

Let two bodies, one hot and one cold, be placed in contact with each other, while isolated from all other bodies. In course of time, the hot body becomes colder, and the cold body becomes hotter. In other words, both bodies experience a change in one or more of their properties. Finally, all changes in the properties of the two bodies ceases and the bodies are said to be in thermal equilibrium with each other or are at same temperature. Thus the condition of equality of temperature is as follows:

Two systems have equal temperatures if there are no changes in their properties when they are brought in thermal contact with each other.

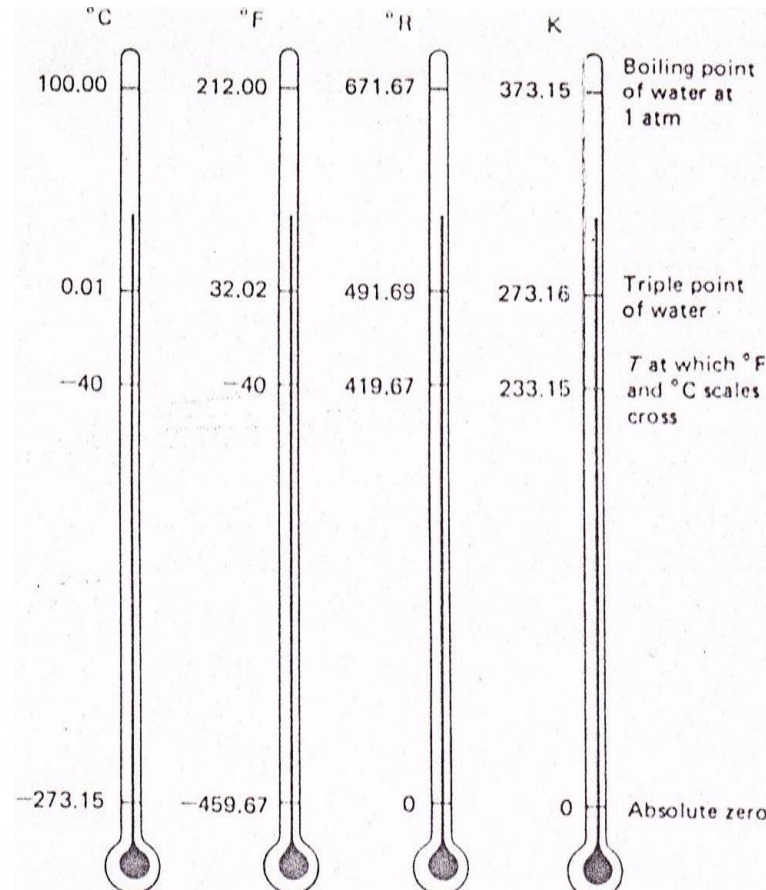


Figure 1.8: Comparison of temperature scales

Zeroth Law of Thermodynamics

It is a matter of experience that when two bodies are in thermal equilibrium with a third body, they are in thermal equilibrium with one another. This statement, which is sometimes called the *Zeroth law of thermodynamics*, is tacitly assumed in every measurement of temperature. Thus, if we want to know if two bodies are at the same temperature, it is not necessary to bring them into contact and see whether their observable properties change with time, as described previously. It is necessary only to see if they are individually in thermal equilibrium with a third body. The third body is usually a *thermometer*.

CHAPTER 2

ENERGY AND ENERGY TRANSFER

2.1 Energy and its Meaning

In classical mechanics, energy is defined as the capacity of doing work. Energy can also be defined as the capacity to exert a force through a distance. Energy input always produces some effect on the matter in the system.

Energy may be classified into two general types as stored energy in materials and as energy in transition from one system to another system or the surroundings. The stored energy is the energy possessed by a system within its boundaries. The potential energy, kinetic energy and internal energy are the examples of stored energy. The energy in transition is the energy possessed by a system which is capable of crossing its boundaries. The heat, work and electrical energy are the examples of transit energy.

It may be noted that only the stored energy is a thermodynamic property whereas the transit energy is not a thermodynamic property as it depends upon the path.

2.1.1 Potential Energy

It is the energy possessed by a body or a system for doing work, by virtue of its position above the ground level. For example, a body raised to some height above the ground level possesses potential energy because it can do some work by falling on the earth's surface. Potential energy of a system is given by

$$PE = Wz = mgz \quad \dots\dots\dots (2.1)$$

where,

- W = Weight of the system,
- m = Mass of the system,
- z = Distance through which the system falls, and
- g = Acceleration due to gravity.

2.1.2 Kinetic Energy

It is the energy possessed by a body or a system, for doing work, by virtue of its mass and velocity of motion. Kinetic energy of a system is given by

$$KE = \frac{1}{2} m \bar{V}^2 \quad \dots\dots\dots (2.2)$$

where,

- m = Mass of the system, and
- $\bar{V}$ = Velocity of the system.

2.1.3 Internal Energy

It is the energy possessed by a body or a system due to its molecular arrangement and motion of the molecules. It is usually represented by U . In the study of thermodynamics,

we are mainly concerned with the change in internal energy (dU) which depends upon the change in temperature of the system.

$$dU = mc_v dT \quad \dots\dots\dots (2.3)$$

where,

- m = Mass of the system,
 c_v = Specific heat at constant volume,
 dT = Change in temperature of the system

2.1.4 Total Energy

The total energy of the system (E) is equal to the sum of the above three types of energies. Mathematically,

$$E = PE + KE + U = mgz + \frac{1}{2}m\bar{V}^2 + U \quad \dots\dots\dots (2.4)$$

For unit mass, the above expression is written as

$$e = gz + \frac{1}{2}\bar{V}^2 + u \quad \dots\dots\dots (2.5)$$

where, u is specific internal energy.

2.2 Energy Transfer

A closed system and its surroundings can interact in two ways: (a) by work transfer, and (b) by heat transfer. These may be called energy interactions and these bring about changes in the properties of the system. Thermodynamics mainly studies these energy interactions and the associated property changes of the system.

2.2.1 Heat transfer

Heat is that form of energy in transit the driving force for which is temperature difference. Heat is transferred across a boundary from a system at higher temperature to a system at lower temperature by the mechanisms of conduction, convection and radiation. A precise definition of heat has been given by Obert:

Heat is energy transferred, without transfer of mass, across the boundary of a system because of a temperature difference between system and surroundings.

With the above definition of heat the following points should be noted

- Heat is transient quantity which can be identified only as it crosses the boundary of a system. It exists or occurs only during a transfer of energy into or out of a system. When the heat transfer is over, no heat is present, only the result of heat transfer-energy.
- Heat is not a property of a system. It is therefore incorrect to talk of heat contained in a system. The correct term will be energy.

The symbol for heat is Q . By convention heat flowing into a system is considered positive while heat flowing out of a system is considered negative.

2.2.2 Work transfer

In mechanics, work is defined as the product of a force and the distance moved in the direction of the force. In mathematical terms, the amount of work δW done by a force $\mathbf{F}$ in moving through a differential distance $d\mathbf{s}$ directed outward from the system boundary is

$$\delta W = \mathbf{F} \cdot d\mathbf{s} \quad \dots\dots\dots (2.6)$$

The work done over a finite path between points s_1 and s_2 is

$$W_{12} = \int_1^2 \mathbf{F} \cdot d\mathbf{s} \quad \dots\dots\dots (2.7)$$

However, the above definition of work is not adequate in the study of thermodynamics. The reason is that thermodynamics is concerned with interactions or energy transfers between a system and its surroundings, and these energy transfers are of two kinds, namely, heat and work. We would therefore like to make a distinction between heat and work. The definition of work as used in mechanics fails to provide this distinction because we can use heat to cause a force to act through a distance. For example suppose we place a hot body in contact with water so that the water begins to boil. The steam produced can then drive a steam engine which, in turn, can lift a weight through a distance. One way to arrive at a general definition of work is to paraphrase the definition of heat. In the words of Obert:

Work is energy transferred, without transfer of mass, across the boundary of a system because of an intensive property difference other than temperature that exists between system and surroundings.

This definition includes all forms of work and makes a clear distinction between work and heat. In general the intensive property difference encountered in engineering problems is pressure difference. The pressure difference (between system and surroundings) at surface of the system gives rise to a force and the action of this force over a distance is termed mechanical work.

In some cases the intensive property difference may be the difference in electrical potential between the system and surroundings. The resulting energy transfer across the system boundary is called electrical work. Another definition of work has been given by Keenan:

Work flows from a system (and to another) during a given operation if the sole effect external to the system could be reduced to the rise of a weight.

The above definition of work has a wider scope than the definition used in mechanics and it also makes a distinction between work and heat. This will be clear from what follows.

The system shown in **Figure 2.1 (a)** illustrates a storage battery. External to the system, the terminals are connected to a resistance by means of a switch. Let us close the switch for a certain interval of time. The current will flow through the battery and the resistance and as a result the resistance will get warmer. Clearly the system (battery) has interacted

with the surroundings, i.e., an energy transfer has taken place between the system and surroundings because the state of each has changed.

Analyzing this system according to the mechanics of work, we conclude that no work is done because no force has moved through a distance. But according to the thermodynamic definition, work has been done because we can replace the resistance by an electric motor which winds up a rope on which a weight is suspended. This is shown in **Figure 2.1 (b)**. If the current flow is the same in both cases, as far as the system (battery) is concerned, there is absolutely no difference in the two situations. However, in **Figure 2.1 (b)**, the sole effect external to the system has been the rise of a weight. Hence, the system has done work.

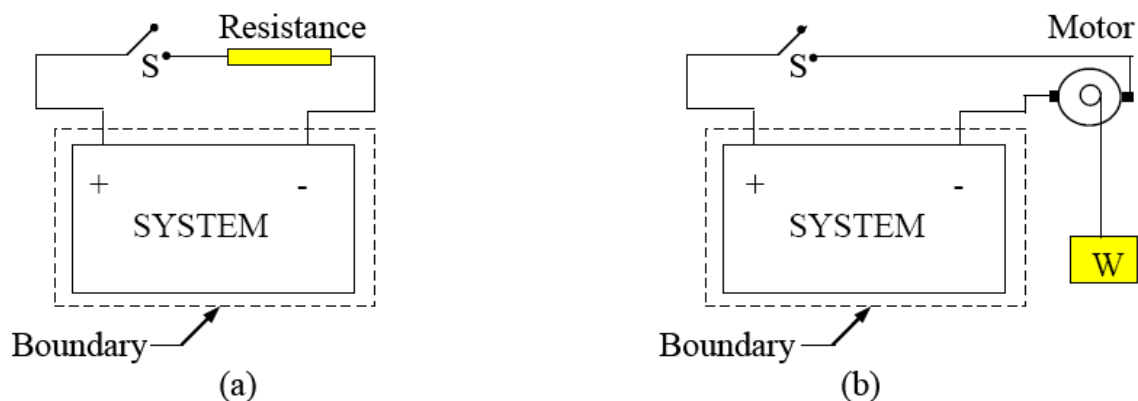


Figure 2.1: Heat and work Transfer

Note that unless the path is specified from the initial to the final state of the system, we are not able to calculate the work done. In other words, the work done in going between the initial and final states can have any value, depending on the path chosen. Work is therefore called a path function in thermodynamics and an inexact differential quantity in mathematics. It is therefore denoted by δW . The value of work depends not only on the initial and final states of the system but also on the path followed. Work is not a property, since it cannot be specified from knowing only the state of the system.

The sign convention generally adopted for thermodynamic work is that work done by the system on the surroundings is positive, while work done on the system by the surroundings is negative. Thus, any energy crossing the system boundary from the system into the surroundings in the form of work has a positive sign.

2.3 Expression for Displacement Work Transfer

Many problems of interest in thermodynamics are carried out in terms of P , V , and T rather than in terms of force and distance. So it is convenient to express the mathematical relations for work in terms of variables P , V , and T .

$$W_{12} = \int_1^2 \mathbf{F} \cdot d\mathbf{s} = \int_1^2 \mathbf{P}A \cdot d\mathbf{s} = \int_1^2 P dV \quad \dots\dots\dots (2.8)$$

With the definition of pressure as the normal force per unit area and with $d\mathbf{s}$ denoted by $d\mathbf{x}$ which is in the direction of force $\mathbf{F}$, the dot product in the integrand becomes PdV .

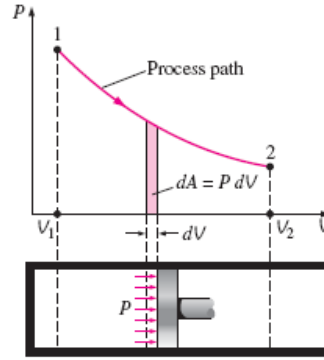


Figure 2.2: The area under the process curve on a $P - V$ diagram represents the boundary work.

If **Equation (2. 8)** is divided by mass, then the work per unit mass w_{12} is

$$w_{12} = \int_1^2 P dv \quad \dots\dots\dots (2.9)$$

Equation (2.8) reveals that the area under the process curve on a $P - V$ diagram is equal, in magnitude, to the work done during a quasi-equilibrium expansion or compression process of a closed system. (On the $P - v$ diagram, it represents the boundary work done per unit mass.)

Magnitudes of work transfer in various quasi – static processes are derived below.

2.3.1 Constant Volume (Isochoric) Process

Heating or cooling of a working substance inside a rigid container can be taken as a constant volume process. Work transfer during a constant volume process is

$$W_{12} = \int_{V_1}^{V_2} P dV = 0 \quad \dots\dots\dots (2.9)$$

2.3.2 Constant Pressure (Isobaric) Process

Expansion or compression of a working substance inside a cylinder with freely moving piston can be taken as a constant pressure process. Work transfer during a constant pressure process is

$$W_{12} = \int_{V_1}^{V_2} P dV = P(V_2 - V_1) \quad \dots\dots\dots (2.10)$$

2.3.3 Constant Temperature (Isothermal) Process

For an ideal gas undergoing constant temperature process, pressure and volume relationship is given by

$$PV = \text{Constant } (C)$$

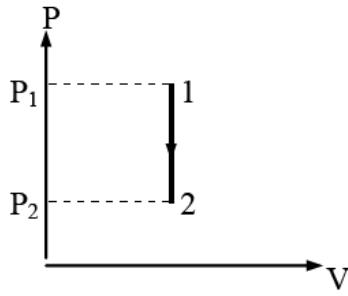


Figure 2.3: Constant volume process

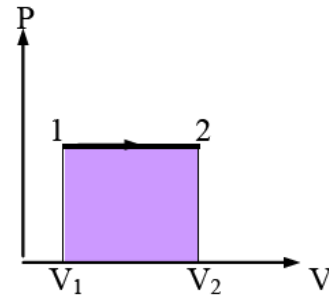


Figure 2.4: Constant pressure process

Pressure at any intermediate state is then

$$P = \frac{P_1 V_1}{V}$$

Substituting this into **Equation (2.9)**, we get

$$W_{12} = \int_{V_1}^{V_2} P dV = \int_{V_1}^{V_2} \frac{P_1 V_1}{V} dV = P_1 V_1 \int_{V_1}^{V_2} \frac{1}{V} dV = P_1 V_1 \ln\left(\frac{V_2}{V_1}\right) \quad \dots\dots\dots (2.11)$$

2.3.4 Polytropic Process

Thermodynamic process which follows the relation $PV^n = \text{constant}$ is called a polytropic process and the index n is called polytropic index.

Pressure at any intermediate state during polytropic process is

$$P = \frac{P_1 V_1^n}{V^n}$$

Substituting this into **Equation (2.9)**, we get

$$\begin{aligned} W_{12} &= \int_{V_1}^{V_2} P dV = \int_{V_1}^{V_2} \frac{P_1 V_1^n}{V^n} dV = P_1 V_1^n \int_{V_1}^{V_2} \frac{1}{V^n} dV \\ &= \frac{P_1 V_1^n}{1-n} (V_2^{1-n} - V_1^{1-n}) = \frac{P_2 V_2 - P_1 V_1}{1-n} \quad \dots\dots\dots (2.12) \end{aligned}$$

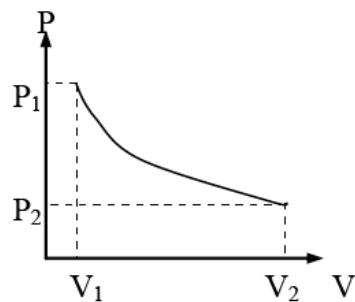


Figure 2.5: Constant temperature process

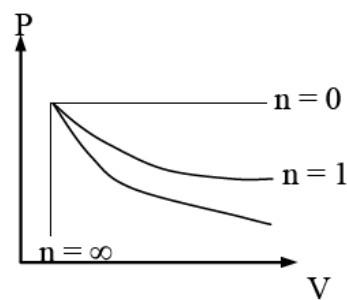


Figure 2.6: Polytropic process

2.4 Comparison of Heat and Work

There are many similarities between heat and work. These are:

- The heat and work are both transient phenomena. The systems do not possess the heat or work. When a system undergoes a change, heat transfer or work done may occur.
- The heat and work are boundary phenomena. They are observed at the boundary of the system.
- The heat and work represent the energy crossing the boundary of the system.
- The heat and work are path functions and hence they are inexact differentials. They are written as δQ and δW .

2.5 Power

To design practical devices, not only is it important to know how much energy is needed to drive a device or how much energy can be obtained from the device, but also we need the rate at which energy can be obtained. The rate at which energy is transferred is called the power. We denote the rate of various energy forms by placing a dot over the symbol; i.e., the rate of doing work is

$$\dot{W} = \lim_{\Delta t \rightarrow 0} \frac{\delta W}{\Delta t} \quad \dots\dots\dots (2.13)$$

and the rate of heat transfer across a boundary is

$$\dot{Q} = \lim_{\Delta t \rightarrow 0} \frac{\delta Q}{\Delta t} \quad \dots\dots\dots (2.14)$$

In SI, the unit of power is the watt (W), which is equal to an energy rate of 1 J/s.

CHAPTER 3

PROPERTIES OF COMMON SUBSTANCE

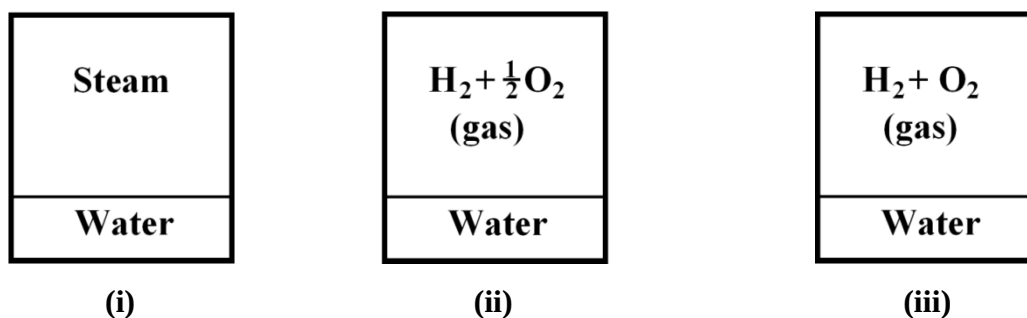
3.1 Pure Substance and State Postulate

Pure implies substances composed of a single chemical species. Hence a pure substance is a system which is

- (a) *homogeneous in composition,*
- (b) *homogeneous in chemical aggregation, and*
- (c) *invariable in chemical composition.*

Homogeneous in composition means that the composition of each part of the system is the same as the compositions of every other part. Composition means relative proportion of the chemical elements into which the sample can be analyzed. It does not matter how these elements are combined.

In **Figure 3.1** for example, system (i), comprising steam and water, is homogeneous in composition, since chemical analysis would reveal that hydrogen and oxygen atoms are present in the ratio 2:1 whether the sample be taken from the steam or from the water. The same is true for system (ii), containing uncombined hydrogen and oxygen gas in the atomic ratio 2:1 in the upper part, and water in the lower part. System (iii) however is not homogeneous in composition; for the hydrogen and oxygen are present in the ratio 1:1 in the upper part, but in the ratio 2:1 (as water) in the lower part.



Satisfies condition (a) Satisfies condition (a) Does not satisfies condition (a)
 Satisfies condition (b) Does not satisfies condition (b)

Figure 3.1: Illustrating the definition of a pure substance

Homogeneous in chemical aggregation means that the chemical elements must be combined chemically in the same way in all parts of the system. Consideration of Figure 3.1 shows that system (i) satisfies this condition also; for steam and water consists of identical molecules. System (ii) on the other hand is not homogeneous in chemical aggregation since, in the upper part of the system, the hydrogen and oxygen are not combined chemically (individual atoms of H and O are not uniquely associated), whereas in the lower part of the system the hydrogen and oxygen are combined in the form of water.

Invariable chemical aggregation means that the state of the chemical combination of the system does not change with *time*. Thus a mixture of hydrogen and oxygen, which

changed into steam during the time that the system was under consideration, would not be a pure substance.

The important characteristic of a pure substance is that it is invariable in chemical composition even though there may be a change of phase. Thus, a system consisting of a mixture of various phases of water viz. water and ice, water and steam is a pure substance. Similarly, a system consisting of oxygen as a vapor; a liquid or a combination of these is also a pure substance. Air, however, though a mixture of several gases is considered as a pure substance only as long as it is all gas or all liquid. A mixture of dry gaseous air and liquid air is not a pure substance, because chemical composition of liquid phase is different from that of vapor phase.

State Postulate (Two Property Rule)

For a pure simple compressible substance, repeated observations and experiments show that two independent properties are necessary and sufficient to establish the stable equilibrium state of a system. This observation is not derivable from more basic theorem using classical approaches to thermodynamics; but no violations have been found.

The observed behavior of a pure simple compressible substance is summarized in the state postulates:

The values of any two independent thermodynamic properties are sufficient to establish the stable thermodynamic state of a control mass composed of a pure simple compressible substance.

Thermodynamic properties are properties such as pressure, temperature, and volume, not geometric properties such as shape or elevation and not velocity. Stable state means an equilibrium state, so that state postulate gives no information concerning nonequilibrium states. The above statement is for a control mass or a specified amount of matter. The state postulate can also be interpreted on a per unit basis or an intensive property basis. Recalling that extensive properties can be expressed in an alternative intensive form; the state postulate can be stated in a very useful form:

The values of any two independent intensive thermodynamic properties are sufficient to establish the stable thermodynamic state of a pure simple compressible substance.

The state postulate indicates the information needed to specify the state of the substance in a particular system. The properties considered thus far are P , v , T , and u . The state postulate says that if two of these properties are independent and known, then all others are uniquely specified. If T and v are known for a pure compressible substance, then P and u have unique known values. Mathematically this is given as

$$P = P(T, v) \quad \text{..... (3.1)}$$

and

$$u = u(T, v) \quad \text{..... (3.2)}$$

The choice of independent variables in the expressions is arbitrary as long as they are independent, so that equivalent alternative statements are

$$P = P(u, T) \text{ or } P = P(u, v) \quad \text{..... (3.3)}$$

and

$$u = u(P, T) \quad \text{or} \quad u = u(P, v) \quad \dots\dots\dots (3.4)$$

3.2 Ideal Gas

A perfect gas or an ideal gas is defined as a gas having no forces of molecular attraction. A gas which follows the gas laws at all ranges of pressures and temperatures can be considered as an *ideal gas* but no such gas exists in nature. However, real gases tend to follow these laws at low pressures or high temperatures or at both. This is because the molecules are far apart at reduced pressures and elevated temperatures and the force of attraction between them tends to be small. At low pressures, the gases behave nearly as ideal gases. The range of this low pressure is different for different gases.

3.2.1 Boyle's Law

This law was enunciated by Robert Boyle in 1661 on the basis of his experimental results. Boyle's law may be stated as follows:

If the temperature remains constant, the volume of a given mass of gas is inversely proportional to the pressure.

$$v \propto \left(\frac{1}{P}\right)_T \quad \dots\dots\dots (3.5)$$

3.2.2 Charles' Law

This law, also known as Gay-Lussac's law, was enunciated in 1787. It may be stated in parts:

(a) *If the pressure is held constant, the volume of a given mass of gas varies directly as the absolute temperature.*

$$v \propto (T)_P \quad \dots\dots\dots (3.6)$$

(b) *If the volume is held constant, the pressure of a given mass of gas varies directly as the absolute temperature.*

$$P \propto (T)_v \quad \dots\dots\dots (3.7)$$

A relation between all the three variables P , v , and T may be obtained by combining Boyle's law and Charles' law. Referring to **Figure 3.2**, consider unit mass of a gas in state 1 and let it undergo an arbitrary change to state 2. From state 1, draw a constant pressure line, and from state 2 draw a constant temperature line. The two lines intersect at point A.

Now applying Charles' law for the process 1-A

$$\begin{aligned} \frac{v_1}{v_A} &= \frac{T_1}{T_A} \\ v_A &= \frac{T_A}{T_1} v_1 = \frac{T_2}{T_1} v_1 \quad \dots\dots\dots (3.8) \end{aligned}$$

Applying Boyle's law for the process A - 2,

$$\frac{v_2}{v_A} = \frac{P_A}{P_2}$$

$$v_A = \frac{P_2}{P_A} v_2 = \frac{P_2}{P_1} v_2 \quad \dots\dots\dots (3.9)$$

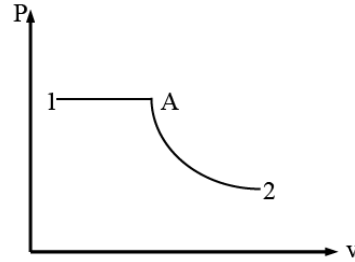


Figure 3.2: Relation between P , v , and T

Equating the two relations (3.8) and (3.9)

$$\frac{P_1 v_1}{T_1} = \frac{P_2 v_2}{T_2} \quad \dots\dots\dots (3.10)$$

Since states 1 and 2 are selected arbitrarily, it follows that

$$\frac{P_1 v_1}{T_1} = \frac{P_2 v_2}{T_2} = \frac{P_3 v_3}{T_3} = \frac{Pv}{T} = R$$

$$Pv = RT \quad \dots\dots\dots (3.11)$$

where R is a constant for a given gas. The constant R is called the *characteristic gas constant* and has the units of J/kg.K or m.kgf/kg.K.

For a system consisting of a mass of m kg of gas, **Equation (3.11)** becomes

$$PV = mRT \quad \dots\dots\dots (3.12)$$

3.3 Two Phase (Liquid and Vapor) System

Working substance in the form of mixture of liquid and vapor are used in many engineering systems including power plants, many refrigeration systems, and thermal distribution systems using water or steam to transport energy.

3.3.1 Saturation Curve for Two Phase Mixture on T-v Diagram

Consider a piston–cylinder device containing liquid water at 20°C and 1 atm pressure (state 1, **Figure 3.3a**). Under these conditions, water exists in the liquid phase, and it is called a *compressed liquid*, or a *subcooled liquid*, meaning that it is *not about to vaporize*. Heat is now transferred to the water until its temperature rises to, say, 40°C. As the temperature rises, the liquid water expands slightly, and so its specific volume increases. To accommodate this expansion, the piston moves up slightly. The pressure in the cylinder remains constant at 1 atm during this process since it depends on the outside barometric pressure and the weight of the piston, both of which are constant. Water is still a compressed liquid at this state since it has not started to vaporize.

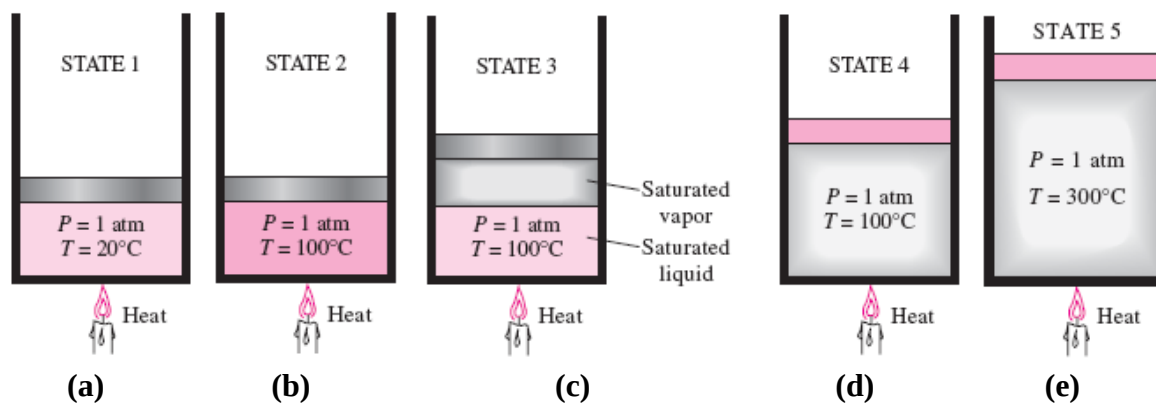


Figure 3.3: Heating of liquid water at constant pressure

As more heat is transferred, the temperature keeps rising until it reaches 100°C (state 2, **Figure 3.3b**). At this point water is still a liquid, but any heat addition will cause some of the liquid to vaporize. That is, a phase-change process from liquid to vapor is about to take place. A liquid that is *about to vaporize* is called a *saturated liquid*. Therefore, state 2 is a saturated liquid state.

Once boiling starts, the temperature stops rising until the liquid is completely vaporized. That is, the temperature will remain constant during the entire phase-change process if the pressure is held constant. This can easily be verified by placing a thermometer into boiling pure water on top of a stove. At sea level ($P = 1 \text{ atm}$), the thermometer will always read 100°C if the pan is uncovered or covered with a light lid. During a boiling process, the only change we will observe is a large increase in the volume and a steady decline in the liquid level as a result of more liquid turning to vapor.

Midway about the vaporization line (state 3, **Figure 3.3c**), the cylinder contains equal amounts of liquid and vapor. As we continue transferring heat, the vaporization process continues until the last drop of liquid is vaporized (state 4, **Figure 3.3d**). At this point, the entire cylinder is filled with vapor that is on the borderline of the liquid phase. Any heat loss from this vapor will cause some of the vapor to condense (phase change from vapor to liquid). A vapor that is *about to condense* is called a **saturated vapor**. Therefore, state 4 is a saturated vapor state. A substance at states between 2 and 4 is referred to as a **saturated liquid–vapor mixture** since the *liquid and vapor phases coexist* in equilibrium at these states.

Once the phase-change process is completed, we are back to a single phase region again (this time vapor), and further transfer of heat results in an increase in both the temperature and the specific volume (**Figure 3.3e**). At state 5, the temperature of the vapor is, let us say, 300°C ; and if we transfer some heat from the vapor, the temperature may drop somewhat but no condensation will take place as long as the temperature remains above 100°C (for $P = 1 \text{ atm}$). A vapor that is *not about to condense* (i.e., not a saturated vapor) is called a **superheated vapor**. Therefore, water at state 5 is a superheated vapor. This constant-pressure phase-change process is illustrated on a T - v diagram in **Figure 3.4**.

If the entire process described here is reversed by cooling the water while maintaining the pressure at the same value, the water will go back to state 1, retracing the same path, and in so doing, the amount of heat released will exactly match the amount of heat added during the heating process.

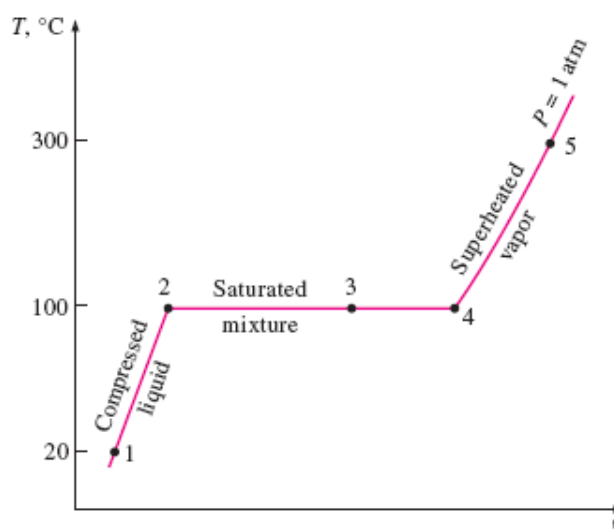


Figure 3.4: T - v diagram for the heating process of water at constant pressure

In our daily life, water implies liquid water and steam implies water vapor. In thermodynamics, however, both water and steam usually mean only one thing: H_2O .

Now we repeat this process at different pressures to develop the T - v diagram. Let us add weights on top of the piston until the pressure inside the cylinder reaches 1 MPa. At this pressure, water has a somewhat smaller specific volume than it does at 1 atm pressure. As heat is transferred to the water at this new pressure, the process follows a path that looks very much like the process path at 1 atm pressure, as shown in **Figure 3.5**, but there are some noticeable differences. First, water starts boiling at a much higher temperature (179.9°C) at this pressure. Second, the specific volume of the saturated liquid is larger and the specific volume of the saturated vapor is smaller than the corresponding values at 1 atm pressure. That is, the horizontal line that connects the saturated liquid and saturated vapor states is much shorter.

As the pressure is increased further, this saturation line continues to shrink, as shown in **Figure 3.5**, and it becomes a point when the pressure reaches 22.06 MPa for the case of water. This point is called the *critical point*, and it is defined as *the point at which the saturated liquid and saturated vapor states are identical*.

The temperature, pressure, and specific volume of a substance at the critical point are called, respectively, the *critical temperature* T_{cr} , *critical pressure* P_{cr} , and *critical specific volume* v_{cr} . The critical-point properties of water are $P_{\text{cr}} = 22.06 \text{ MPa}$, $T_{\text{cr}} = 373.95^\circ\text{C}$, and $v_{\text{cr}} = 0.003106 \text{ m}^3/\text{kg}$.

At pressures above the critical pressure, there is not a distinct phase change process (**Figure 3.6**). Instead, the specific volume of the substance continually increases, and at all times there is only one phase present. Eventually, it resembles a vapor, but we can never tell when the change has occurred. Above the critical state, there is no line that separates the compressed liquid region and the superheated vapor region. However, it is customary to refer to the substance as superheated vapor at temperatures above the

critical temperature and as compressed liquid at temperatures below the critical temperature.

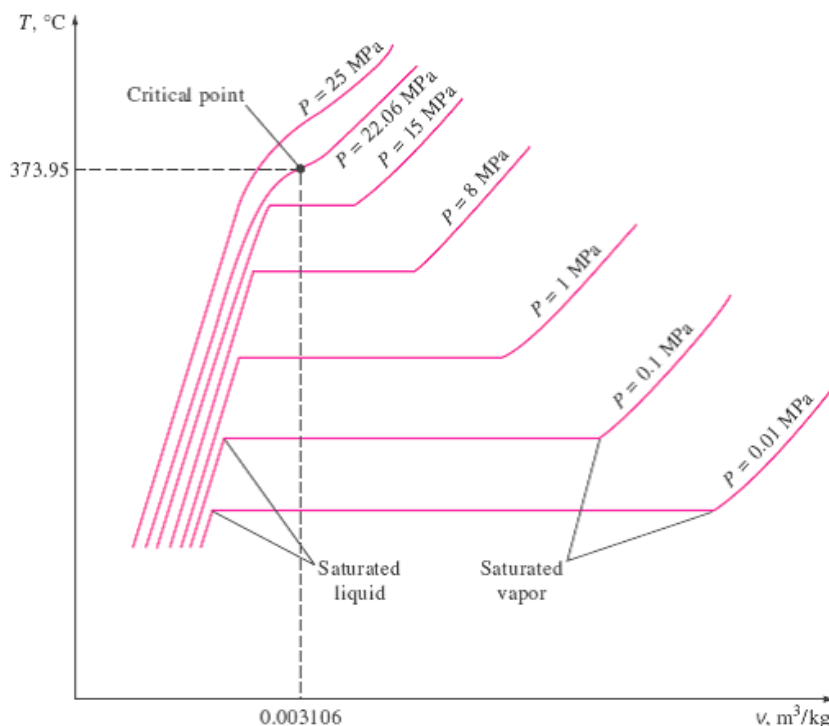


Figure 3.5: T - v diagram of constant-pressure phase-change processes of a pure substance at various pressures (numerical values are for water).

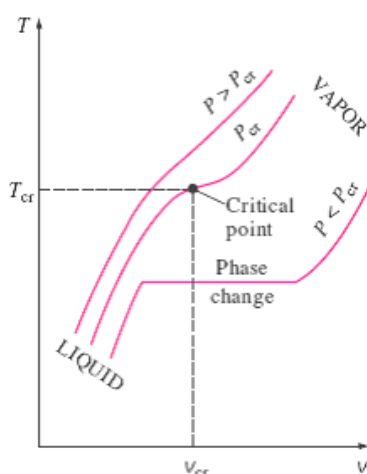


Figure 3.6: At supercritical pressures ($P > P_{cr}$), there is no distinct phase-change (boiling) process.

The saturated liquid states in **Figure 3.5** can be connected by a line called the *saturated liquid line*, and saturated vapor states in the same figure can be connected by another line, called the *saturated vapor line*. These two lines meet at the critical point, forming a dome as shown in **Figure 3.7**. All the compressed liquid states are located in the region to the left of the saturated liquid line, called the *compressed liquid region*. All the superheated vapor states are located to the right of the saturated vapor line, called the *superheated vapor region*. In these two regions, the substance exists in a single phase, a liquid or a

vapor. All the states that involve both phases in equilibrium are located under the dome, called the *saturated liquid–vapor mixture region*, or the *wet region*.

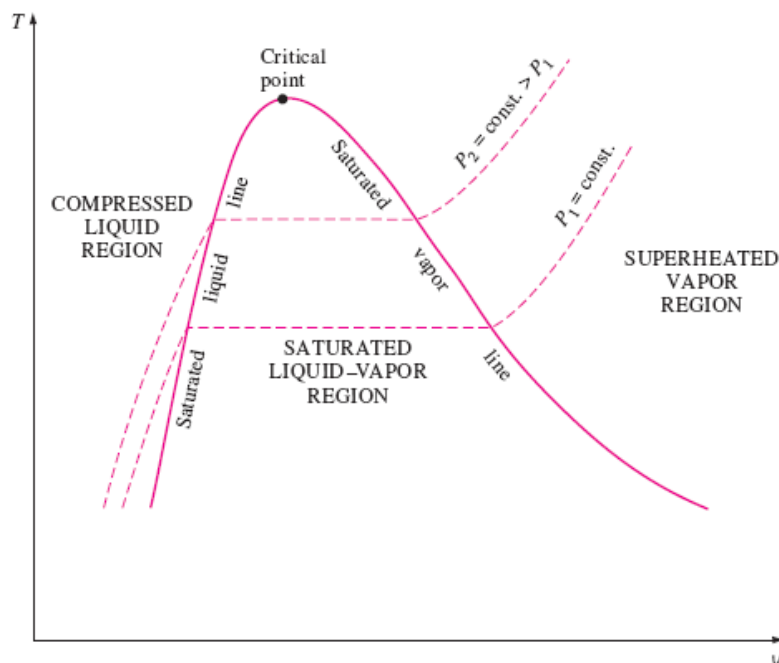


Figure 3.7: Saturation curve on T-v diagram

3.3.2 Saturation Curve for Two Phase Mixture on P-v Diagram

The general shape of the P - v diagram of a pure substance is very much like the T - v diagram, but the $T = \text{const.}$ lines on this diagram have a downward trend, as shown in **Figure 3.8**.

Consider again a piston–cylinder device that contains liquid water at 1 MPa and 150°C. Water at this state exists as a compressed liquid. Now the weights on top of the piston are removed one by one so that the pressure inside the cylinder decreases gradually. The water is allowed to exchange heat with the surroundings so its temperature remains constant. As the pressure decreases, the volume of the water increases slightly. When the pressure reaches the saturation-pressure value at the specified temperature (0.4762 MPa), the water starts to boil. During this vaporization process, both the temperature and the pressure remain constant, but the specific volume increases. Once the last drop of liquid is vaporized, further reduction in pressure results in a further increase in specific volume. Notice that during the phase-change process, we did not remove any weights. Doing so would cause the pressure and therefore the temperature to drop [since $T_{\text{sat}} = f(P_{\text{sat}})$], and the process would no longer be isothermal.

When the process is repeated for other temperatures, similar paths are obtained for the phase-change processes. Connecting the saturated liquid and the saturated vapor states by a curve, we obtain the P - v diagram of a pure substance, as shown in **Figure 3.8**.

3.3.3 Saturation Temperature and Saturation Pressure

It probably came as no surprise to you that water started to boil at 100°C. Strictly speaking, the statement “water boils at 100°C” is incorrect. The correct statement is

“water boils at 100°C at 1 atm pressure.” The only reason water started boiling at 100°C was because we held the pressure constant at 1 atm (101.325 kPa). If the pressure inside the cylinder were raised to 500 kPa by adding weights on top of the piston, water would start boiling at 151.8°C. That is, *the temperature at which water starts boiling depends on the pressure; therefore, if the pressure is fixed, so is the boiling temperature.*

At a given pressure, the temperature at which a pure substance changes phase is called the *saturation temperature* T_{sat} . Likewise, at a given temperature, the pressure at which a pure substance changes phase is called the *saturation pressure* P_{sat} . At a pressure of 101.325 kPa, T_{sat} is 99.97°C. Conversely, at a temperature of 99.97°C, P_{sat} is 101.325 kPa.

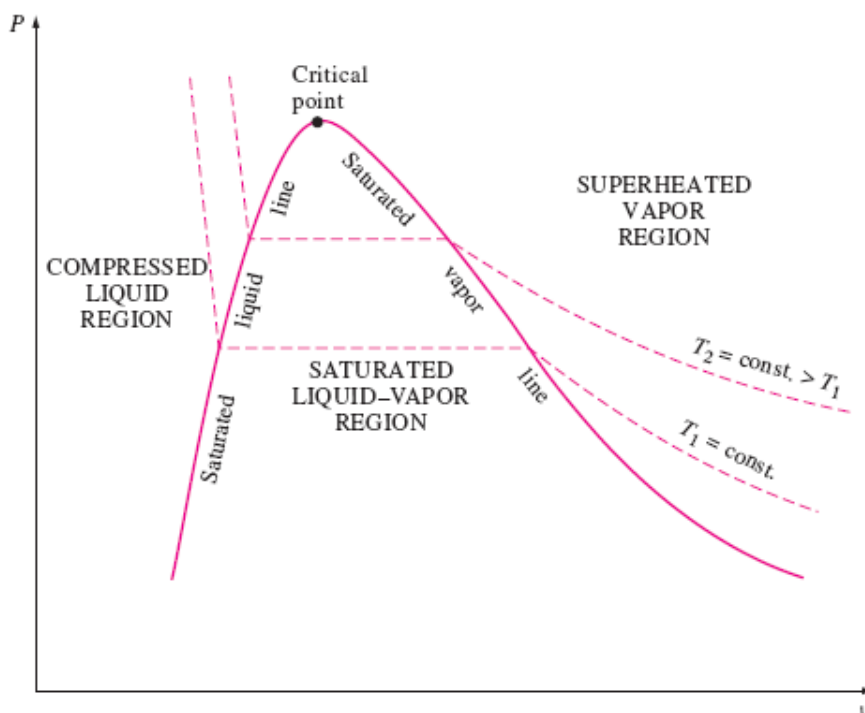


Figure 3.8: Saturation curve on P-v diagram

3.4 Property of a Two Phase Mixture

Each of the two phase is at the boiling point at the existing control mass pressure at every state along line 24 (**Figure 3.4**). Thus, if we measure the specific volume of the liquid phase at any state along line 24, we find the same value: the specific volume of saturated water at the control mass temperature, $v_1(T)$, is equal to v_2 . Similarly, the specific volume of the vapor at any state along line 24 is found to be the specific volume of the saturated vapor, $v_g(T) = v_4$. The subscripts l and g denote the liquid and vapor phases, respectively.

It is relatively simple to tabulate the specific volume of saturated liquid and saturated vapor as a function of temperature. It would thus be useful to define the specific volume v of the total mass defined by the control mass at any point along the two-phase line 24 (**Figure 3.4**) in terms of the specific volumes of the individual phases v_l and v_g . To do this, we introduce a property called *quality* of the two-phase mixture. The quality is defined as the mass of the vapor present in the two-phase mixture divided by the total mass, and it is given the symbol x . Thus

$$x = \frac{m_g}{m} \quad \dots\dots\dots (3.13)$$

For a two-phase control mass, the quality ranges from 0 for a control mass composed of only saturated liquid to 1 for a control mass composed of only saturated vapor. Quality is also often expressed as a percentage. Note that quality is undefined except for a two-phase mixture of vapor and liquid.

The system volume along the two-phase line 24 (**Figure 3.4**) is

$$V = V_l + V_g \quad \text{..... (3.14)}$$

Dividing by the total mass gives

$$\begin{aligned} \frac{V}{m} = v &= \frac{V_l}{m} + \frac{V_g}{m} \\ &= \frac{m_l}{m} \cdot \frac{V_l}{m} + \frac{m_g}{m} \cdot \frac{V_g}{m} \\ &= (1 - x)v_l + xv_g \end{aligned}$$

$$\text{or,} \quad v = v_l + xv_{lg} \quad \text{..... (3.15)}$$

where v_{lg} is defined as the difference between the specific volumes of the saturated vapor and the saturated liquid, or $v_g - v_l$.

Because the specific volume and other specific properties of the control mass are determined once x is specified at a given P or T , it is common practice to use x and one other specific property to specify the state along the two-phase line.

Similarly other properties are evaluated as:

$$u = u_l + xu_{lg} \quad \text{..... (3.16)}$$

$$h = h_l + xh_{lg} \quad \text{..... (3.17)}$$

$$s = s_l + xs_{lg} \quad \text{..... (3.18)}$$

3.5 Other Thermodynamic Properties

3.5.1 Internal Energy and Enthalpy

Temperature, pressure, and specific volume are not only the properties of interest in the study of thermodynamics. For example, internal energy denoted by U is also one of the important thermodynamic property. *Internal energy incorporates the microscopic forms which result from the molecular motions.* The specific internal energy $u = U / m$ can be found for any simple compressible substance as a function of any two independent variables according to the state postulate.

In many problems in thermodynamics, the group of properties $u + Pv$ occurs repeatedly. To evaluate this group of at a given state, values of u and v must be found from some data source. For convenience, we can define a new property, called *enthalpy* and given the symbol h , as

$$h = u + Pv \quad \text{..... (3.19)}$$

Because all the properties on the right hand side of **Equation (3.19)** can be found as a function of two independent variables, it follows that h can be tabulated as a function of the same two variables. Thus, we can evaluate and tabulate h directly at any given state, saving a great deal of effort in retrieving property data for the problems where $u + Pv$ appears.

3.5.2 Specific Heats

Suppose we wish to find the change in internal energy for a substance undergoing a process. By using **Equation (3.2)**, the change in internal energy is expressed as

$$du = \left(\frac{\partial u}{\partial T}\right)_v dT + \left(\frac{\partial u}{\partial v}\right)_T dv \quad \text{..... (3.20)}$$

Complete specification of the state is assumed by the state postulate if T and v are independent. Any two independent thermodynamic properties could have been used in **Equation (3.20)**, yet T and v are two very useful measurable properties. The partial and derivatives in **Equation (3.20)** are needed and will be important to thermodynamic analysis. The first partial derivative is particularly important and is considered here. The change in internal energy per degree of temperature change for a constant-volume process is this first partial derivative and is called the *specific heat at constant volume*. It is given by symbol c_v and, from the above discussion, is defined mathematically as

$$c_v = \left(\frac{\partial u}{\partial T}\right)_v \quad \text{..... (3.21)}$$

If the change in enthalpy is required, then this is considered in the same manner as the internal energy. Take the two independent variables to be P and T (this change will be quite useful), so that

$$h = h(T, P) \quad \text{..... (3.22)}$$

and the change in the enthalpy is

$$dh = \left(\frac{\partial h}{\partial T}\right)_P dT + \left(\frac{\partial h}{\partial P}\right)_T dP \quad \text{..... (3.23)}$$

For a constant-pressure process, the first partial derivative is significant. This property is the *specific heat at constant pressure*, defined as

$$c_P = \left(\frac{\partial h}{\partial T}\right)_P \quad \text{..... (3.21)}$$

3.6 Development of Property Data

Each pure substance has its distinctive P , v , T , u , h , c_v , and c_P data, and usually these data must be determined experimentally. It is possible to generate the values of all properties from some minimum set of experimental measurements. Also the properties of some substances that have simple microscopic structure can be determined for some regions of the phase diagram without the use of direct experiment if some auxiliary spectroscopic data are available. This can be achieved by application of the methods statistical thermodynamics. Nevertheless, for most substances, careful and detailed

experimental measurements are necessary to obtain the values of the minimum set of dependent properties as a function of a chosen set of independent properties.

Presentation of the experimental data relating all the properties in all regions of interest for each substance is obviously a difficult task. We need to compress as many data as possible into a small space. Many methods have been tried, and some of the most useful are described below.

3.6.1 Graphical Data Presentation

It is very useful to be able to visualize thermodynamic processes on coordinates of various pairs of thermodynamic properties. We have done so on the $P - v$ diagram in developing the background and definitions. Graphical representation of processes on the phase diagrams plotted in terms of other property pairs enables visualization and analysis of many practical systems such as engines.

One very useful phase diagram can be plotted on coordinates of pressure versus enthalpy. An example appears in **Figure 3.9**. On the $P - h$ diagram, we can immediately see the change in enthalpy that occurs during any constant-pressure process, since the change in h then appears as the length of a horizontal line. Such insights are very helpful when we begin the study of thermodynamic processes and cycles. Diagrams on other property coordinates are also useful.

3.6.2 Tabular Data

When detailed property data are to be presented for a wide range of states, so that a single equation of state is not sufficient, it is customary to present the data in tabular form. Because the commonly measured properties are temperature and pressure, usually data are presented with T and P used as the independent properties and the other properties treated as dependent. The intervals between tables entries in complete tabulations are usually chosen so that linear interpolation will provide accurate values for states that fall between entries.

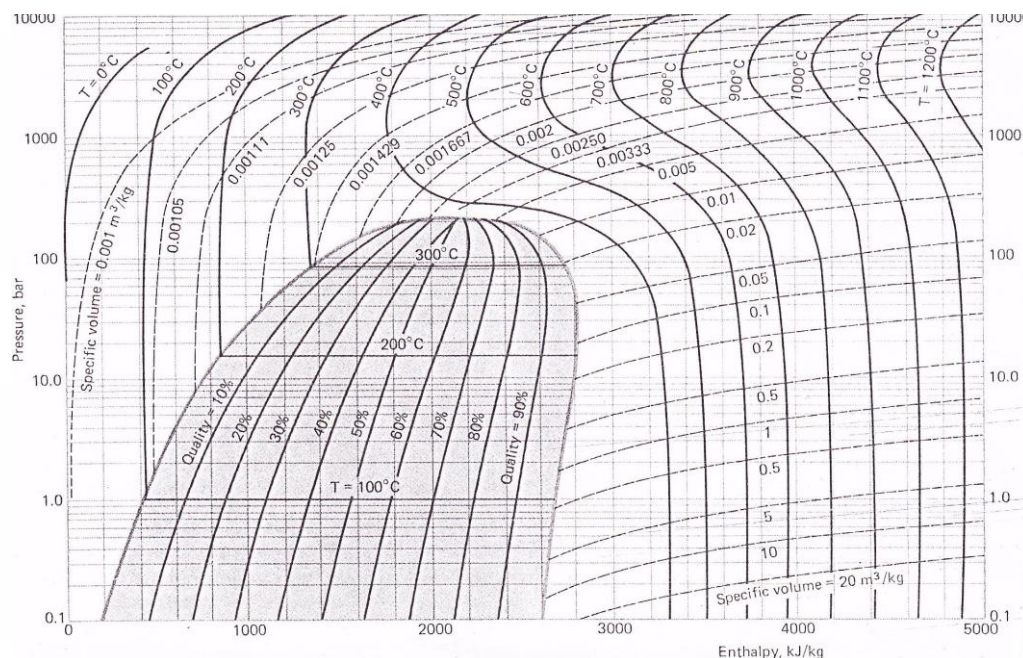


Figure 3.9: P-h diagram

CHAPTER 4

FIRST LAW OF THERMODYNAMICS

4.1 First Law of Thermodynamics for a Control Mass

Conservation relations form the foundation of thermodynamics. The concept of a conserved quality is important in many areas of science and is based on common experiences. We cannot prove it mathematically; however, no experiment has violated it. Conservation of energy and the first law of thermodynamics are different names for the same principle.

To determine the heat and work crossing the boundary of a system, a clear picture of relationship between the states of the system and energy transfer is required. For a simple compressible substance, two independent intensive properties specify the state. The system changes from one state to another by an energy transfer in the form of work and/or heat transfer. This energy transfer must cross a system boundary. The amount of energy transfer depends on the path; i.e., work depends on the specific path followed in performing the work. The state of a substance depends not on how the state is reached, but on its properties only.

4.1.1 Conservation of Mass for a Control Mass

First we will analyze the energy interaction for a control mass. Thus, we consider the mass within a boundary surface which can undergo translation, rotation, and deformation, but no mass crosses the boundary. The substance contained within a cylinder with sliding piston is a typical example.

As we know that in control mass no mass crosses the boundary, this conservation principle can be stated as follows:

The mass of a control mass never changes.

This can be expressed mathematically as

$$dm = 0 \quad \text{..... (4.1)}$$

$$\text{or,} \quad m = \text{constant} \quad \text{..... (4.2)}$$

The change in the mass of the control mass between state 1 and state 2 is

$$m_2 - m_1 = 0 \quad \text{..... (4.3)}$$

The conservation of mass can also be expressed on a rate basis by considering the time interval dt , i.e.,

$$\frac{dm}{dt} = 0 \quad \text{..... (4.4)}$$

4.1.2 Conservation of Energy for a Control Mass

This law is alternatively termed the conservation of energy. This energy balance is stated as follows:

A change of the total energy (kinetic, potential, and internal) of a control mass is equal to the heat transfer to the control mass minus work done by the control mass.

This is expressed mathematically as

$$dE = \delta Q - \delta W \quad \text{..... (4.5)}$$

The sign convention is that all heat transfer into the system is positive, while work done by the system on the surroundings is positive, as shown in **Figure 4.1**.

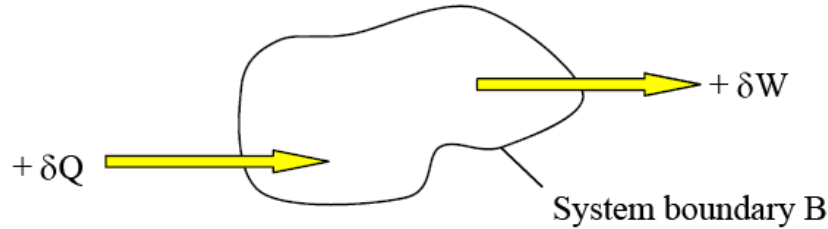


Figure 4.1: Sign Convention

Thus, the change in energy of the control mass from state 1 to state 2 is equal to the heat transfer to the control mass for any path minus the work done by the control mass, again, along any path. Between these two states, **Equation (4.5)** can be integrated to obtain

$$E_2 - E_1 = {}_1Q_2 - {}_1W_2 \quad \text{..... (4.6)}$$

The total energy in all the above equations is the sum of the internal, kinetic, and potential energy. These are included explicitly to yield

$$dU + d(KE) + d(PE) = \delta Q - \delta W \quad \text{..... (4.7)}$$

Integrating between states 1 and 2 results in

$$(U + KE + PE)_2 - (U + KE + PE)_1 = {}_1Q_2 - {}_1W_2 \quad \text{..... (4.8)}$$

Substituting in the expressions for kinetic and potential energy gives the conservation of energy as

$$\left(U + \frac{1}{2}m\bar{V}^2 + mgz\right)_2 - \left(U + \frac{1}{2}m\bar{V}^2 + mgz\right)_1 = {}_1Q_2 - {}_1W_2 \quad \text{..... (4.9)}$$

If the mass within the volume of the control mass is uniform in space, then these equations are alternatively expressed on a specific or unit mass basis as

$$d(me) = \delta(mq) - \delta(mw) \quad \text{..... (4.10)}$$

But the mass is constant in a control mass, so

$$de = \delta q - \delta w \quad \text{..... (4.11)}$$

For a process between states 1 and 2, the conservation of energy relation becomes

$$e_2 - e_1 = {}_1q_2 - {}_1w_2 \quad \text{..... (4.12)}$$

With the internal, kinetic and potential energy expressions **Equation (4.9)** reduces to

$$(u + \frac{1}{2}\bar{V}^2 + gz)_2 - (u + \frac{1}{2}\bar{V}^2 + gz)_1 = {}_1q_2 - {}_1w_2 \quad \dots\dots\dots (4.13)$$

As in **Equation (4.9)**, the left-hand side of **Equation (4.13)** is an integration of an exact differential which represents the change in properties of the control mass. This change in total energy is independent of the path between the states. The right hand sides of these equations are integrals of inexact differentials and generally depend on the particular path.

The conservation of energy expression has been given for a process between states 1 and 2. If the control mass undergoes a complete cycle that takes the control mass from state 1 back to state 1, then **Equation (4.5)** yields

$$\oint dE = \oint \delta Q - \oint \delta W \quad \dots\dots\dots (4.14)$$

But total energy is a property, and according to the mathematical attributes of a property, change in energy for a complete cycle is zero, i.e.,

$$\begin{aligned} \oint dE &= 0, \text{ so} \\ \oint \delta Q - \oint \delta W &= 0 \quad \dots\dots\dots (4.15) \end{aligned}$$

Hence, the first law of thermodynamics can also be stated as follows:

If the system is carried through a cycle then the summation of the work delivered to the surroundings is equal to the summation of heat taken from the surroundings.

The conservation of energy can also be presented on a rate basis. For a time interval of Δt , the expression is

$$\frac{dE}{dt} = \frac{\delta Q}{\Delta t} - \frac{\delta W}{\Delta t} = \dot{Q} - \dot{W} \quad \dots\dots\dots (4.16)$$

4.2 First Law of Thermodynamics for a Control Volume

Although devices such as turbines, pumps, and compressors through which mass flows can be analyzed in principle by studying a particular quantity of matter, as it passes through the device, it is normally simpler to adopt the control volume point of view. A control volume is a region in space studied in a particular analysis. The surface that bounds the control volume may be called a boundary or a control surface. The boundary, which is defined relative to a specified coordinate frame may be fixed or may deform. Mass and energy may cross the boundary. As in the case of a closed system, energy transfer can occur by means of work and heat. In addition, another type of energy transfer must be accounted for: the energy accompanying mass as it enters or exits.

4.2.1 Conservation of Mass for Control Volume

The mass flowing into a specified control volume must equal the mass leaving the control volume; otherwise, the mass within the control volume must change. This is stated as follows:

The change of mass within a control volume is equal to the mass entering minus the mass leaving.

Let the symbol $\dot{m}$ stand for the mass flow rate. All flow streams entering the control volume across the system boundary are added to get the total rate at which mass enters the system:

$$\dot{m}_{in} = \sum_{in} \dot{m} \quad \dots\dots\dots (4.17)$$

Similarly for the outlet streams, the mass flow rate at the outlets is

$$\dot{m}_{out} = \sum_{out} \dot{m} \quad \dots\dots\dots (4.18)$$

Now the mass balance on the control volume (CV) becomes simply

$$\frac{dm_{CV}}{dt} = \dot{m}_{in} - \dot{m}_{out} = \sum_{in} \dot{m} - \sum_{out} \dot{m} \quad \dots\dots\dots (4.19)$$

This states that the rate of change of mass within the control volume is equal to the mass that enters the control volume per unit time minus the mass that leaves the control volume per unit time. This is shown schematically in **Figure 4.2**.

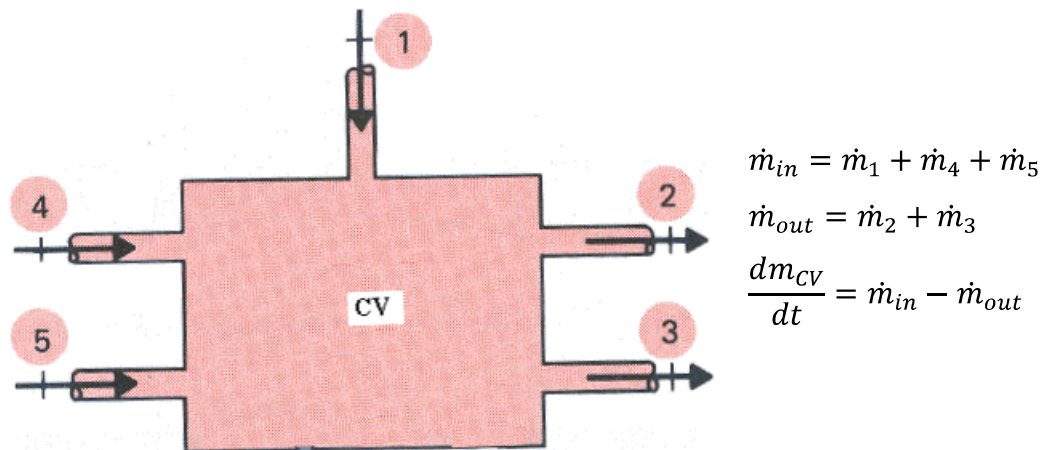


Figure 4.2: Conservation of mass for a CV

The inlet and outlet mass flow rates at the boundaries can be expressed in terms of the density, velocity and area. The volume swept out for a time increment Δt as the mass moves through a port (as shown in **Figure 4.3**) is

$$\dot{V} = A \frac{\Delta L}{\Delta t} \quad \dots\dots\dots (4.20)$$

As Δt tends to zero, $\Delta L/\Delta t$ is the magnitude of velocity, denoted by $\bar{V}$. This is the velocity of the fluid relative to the control volume, so

$$\dot{V} = A\bar{V} \quad \dots\dots\dots (4.21)$$

and the mass flow rate for a port is given by the volume swept out times the density, i.e.;

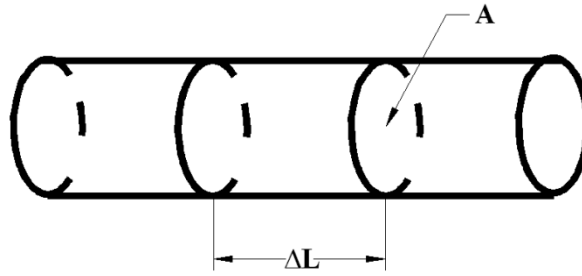


Figure 4.3: Mass flow at a port

$$\dot{m} = \rho A \bar{V} = \rho \dot{V} \quad \dots\dots\dots (4.22)$$

The area of the crossing boundary at the inlet is denoted by A_{in} and at the outlet by A_{out} . The mass flow rates are

$$\dot{m}_{in} = (\rho A \bar{V})_{in} \quad \dots\dots\dots (4.23)$$

and

$$\dot{m}_{out} = (\rho A \bar{V})_{out} \quad \dots\dots\dots (4.24)$$

The conservation of mass statement is then

$$\frac{dm_{cv}}{dt} = \sum_{in} (\rho A \bar{V})_{in} - \sum_{out} (\rho A \bar{V})_{out} \quad \dots\dots\dots (4.25)$$

When there is no mass entering or leaving the control volume, the conservation of mass reduces

$$\frac{dm_{cv}}{dt} = 0 \quad \dots\dots\dots (4.26)$$

or, by integrating with respect to time,

$$dm_{cv} = 0 \quad \dots\dots\dots (4.27)$$

This is exactly the expression for a control mass.

4.2.2 Conservation of Energy for Control Volume

The mass flowing into the control volume carries energy with it, and the same is true for the outlets from the control volume. The net gain in energy from the inlets and outlets must equal the change in energy within the control volume plus the heat transfer minus the work done. The work term requires special attention since the fluid moving into and out of the control volume is doing work. This conservation statement is stated as:

The change in energy within the control volume is equal to the net energy transported to the control volume plus the heat transferred minus the work done by the control volume.

This is shown schematically in **Figure 4.4**. Let E , termed the energy rate, denote the energy transported with the fluid at the inlets and outlets. The energy within the control volume is denoted by E_{cv} , so the conservation of energy is

$$\frac{dE_{CV}}{dt} = \dot{E}_{net} + \dot{Q} - \dot{W} \quad \dots\dots\dots (4.28)$$

or, in terms of inlets and outlets,

$$\frac{dE_{CV}}{dt} = \dot{E}_{in} - \dot{E}_{out} + \dot{Q} - \dot{W} \quad \dots\dots\dots (4.29)$$

The energy change within the control volume is given by dE_{cv}/dt . This represents the change of total energy within the control volume and is often called the energy storage term, since any accumulation of energy within the control volume is included in this expression.

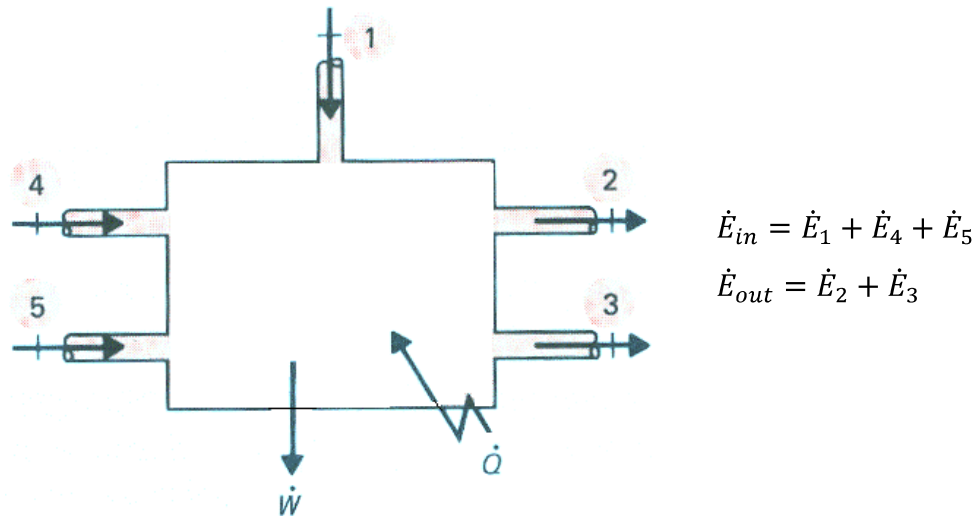


Figure 4.4: Conservation of energy for a CV

The mass flow transports energy into and out of the control volume. These contributions are denoted by $\dot{E}_{in}$ and $\dot{E}_{out}$, respectively. The development of the mass flow rate term to for the conservation of mass is extended to include the energy transport. The fluid carries energy across each element, so the energy transported with the fluid element is given by **Equations (4.23) and (4.24)**, multiplied by the specific total energy, or

$$\dot{E}_{in} = (e\dot{m})_{in} \quad \dots\dots\dots (4.30)$$

and

$$\dot{E}_{out} = (e\dot{m})_{out} \quad \dots\dots\dots (4.31)$$

Therefore, the net energy transported into the control volume by fluid motion is

$$\dot{E}_{net} = (e\dot{m})_{in} - (e\dot{m})_{out} \quad \dots\dots\dots (4.32)$$

Substituting $\dot{E}_{net}$ from **Equation (4.32)** into **Equation (4.28)**, the energy conservation equation becomes

$$\frac{dE_{CV}}{dt} = (e\dot{m})_{in} - (e\dot{m})_{out} + \dot{Q} - \dot{W} \quad \dots\dots\dots (4.33)$$

For multiple inlets and outlets, this equation becomes

$$\frac{dE_{cv}}{dt} = \sum_{in} (\dot{em})_{in} - \sum_{out} (\dot{em})_{out} + \dot{Q} - \dot{W} \quad \dots\dots\dots (4.34)$$

Heat Transfer Contribution

The heat transfer to the control volume is the energy transport across any portion of the boundary because of temperature difference between the control volume and the surroundings. This energy transfer is the same as that considered for the control mass. It generally occurs across the inside boundary rather than crossing the boundary of the control volume. Hence, we can write

$$\dot{Q} = \dot{Q}_{cv} \quad \dots\dots\dots (4.35)$$

Work Transfer Contribution

The work term in the conservation of energy includes various types of works. The contributions of PdV work (expansion or compression of boundary), electrical work, and other work forms which transport energy across the boundaries are incorporated into the energy balance.

If a control volume includes a shaft that crosses the boundary and interacts with the fluid through a shearing motion, this contribution is termed as shaft work. For example, shaft work is the contribution resulting from the rotation of a paddle wheel or turbine rotor within the control volume. The shaft crosses the boundary and transfers energy between the control volume and the surroundings. This shaft work term is truly a result of shear forces acting on the paddle wheel, yet the measurable response from the control volume is in the rotation of the shaft. This work is usually desired result in a work calculation because it is directly usable.

To obtain a relation for flow work, consider a fluid element of volume V as shown in **Figure 4.5**. The fluid immediately upstream forces this fluid element to enter the control volume; thus, it can be regarded as an imaginary piston. The fluid element can be chosen to be sufficiently small so that it has uniform properties throughout.

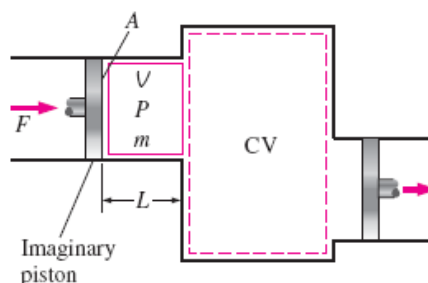


Figure 4.5: Schematic for flow work

If the fluid pressure is P and the cross-sectional area of the fluid element is A (**Figure 4.6**), the force applied on the fluid element by the imaginary piston is

$$F = PA \quad \dots\dots\dots (4.36)$$

To push the entire fluid element into the control volume, this force must act through a distance L . Thus, the work done in pushing the fluid element across the boundary (i.e., the flow work) is

$$W_{flow} = FL = PAL = PV \quad \dots\dots\dots (4.37)$$

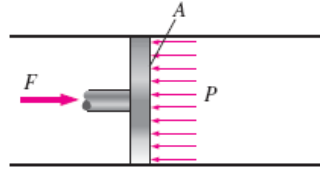


Figure 4.6: In the absence of acceleration, the force applied on a fluid by a piston is equal to the force applied on the piston by the fluid.

The flow work per unit mass is obtained by dividing both sides of this equation by the mass of the fluid element:

$$w_{flow} = Pv \quad \dots\dots\dots (4.38)$$

The flow work relation is the same whether the fluid is pushed into or out of the control volume.

With the mass flow rate expression in **Equations (4.23)** and **(4.24)**, we get

$$\dot{W}_{flow} = -(\dot{m}Pv)_{in} + (\dot{m}Pv)_{out} \quad \dots\dots\dots (4.39)$$

The rate of work done in the control volume formulation is the sum of the shaft and flow work plus the general forms.

$$\begin{aligned} \dot{W} &= \dot{W}_{shaft} + \dot{W}_{gen} + \dot{W}_{flow} \\ &= \dot{W}_{shaft} + \dot{W}_{gen} - (\dot{m}Pv)_{in} + (\dot{m}Pv)_{out} \end{aligned} \quad \dots\dots\dots (4.40)$$

The term $\dot{W}_{gen}$ includes the PdV work, electrical work, etc and it is combined with $\dot{W}_{shaft}$ and expressed in the following equations as $\dot{W}_{CV}$. Thus,

$$\dot{W} = \dot{W}_{CV} - (\dot{m}Pv)_{in} + (\dot{m}Pv)_{out} \quad \dots\dots\dots (4.41)$$

Conservation of Energy Expression

The conservation of energy for a control volume is obtained from **Equation (4.34)** with **(4.41)**. This substitution yields

$$\frac{dE_{CV}}{dt} = [(e + Pv)\dot{m}]_{in} - [(e + Pv)\dot{m}]_{out} + \dot{Q}_{CV} - \dot{W}_{CV} \quad \dots\dots\dots (4.42)$$

This conservation statement says that the change of energy within the control volume is equal to the entering energy contribution minus the leaving energy contributions. The energy balance is also expressed in terms of multiple inlet streams and outlet streams as

$$\frac{dE_{CV}}{dt} = \sum_{in} [(e + Pv)\dot{m}]_{in} - \sum_{out} [(e + Pv)\dot{m}]_{out} + \dot{Q}_{CV} - \dot{W}_{CV} \quad \dots\dots\dots (4.43)$$

Both the **Equations (4.25)** and **(4.43)** describe the change in property within the control volume- the property is mass in **Equation (4.25)** and **Equations (4.43)**. These changes

are equal to incoming contributions minus the outgoing contributions. In the case of the conservation of mass, contributions are mass flow rates. For conservation of energy, the incoming contribution is heat transfer, energy flux, and flow work, and the outgoing contribution is the energy flux, control volume work, and flow work.

A very important term appears in the flux terms. It is the sum of the specific total energy and the product Pv . The term Pv results from the flow work performed as fluid enters and leaves the control volume. The total energy is the sum of the internal energy, kinetic and potential energy, so

$$e + Pv = \frac{E}{m} + Pv = u + \frac{1}{2}\bar{V}^2 + gz + Pv \quad \dots\dots\dots (4.44)$$

Enthalpy is defined as $h = u + Pv$, so

$$e + Pv = h + \frac{1}{2}\bar{V}^2 + gz \quad \dots\dots\dots (4.45)$$

This indicates another important application for enthalpy. **Equation (4.43)** can be written as

$$\begin{aligned} \frac{dE_{CV}}{dt} = & \sum_{in} \left[\left(h + \frac{1}{2}\bar{V}^2 + gz \right) \dot{m} \right]_{in} \\ & - \sum_{out} \left[\left(h + \frac{1}{2}\bar{V}^2 + gz \right) \dot{m} \right]_{out} + \dot{Q}_{CV} - \dot{W}_{CV} \end{aligned} \quad \dots\dots\dots (4.46)$$

In short, what stays within is equal to what goes in minus what goes out.

For the special case of no mass entering or leaving the control volume all $\dot{m}$'s are zero, and **Equation (4.46)** reduces to

$$\frac{dE_{CV}}{dt} = \dot{Q}_{CV} - \dot{W}_{CV} \quad \dots\dots\dots (4.47)$$

which is the control mass form. If there is no energy transfer, the result is

$$\frac{dE_{CV}}{dt} = 0 \quad \dots\dots\dots (4.48)$$

which is the isolated system result.

4.3 Control Volume Analysis

The analysis of a system requires detailed consideration of the inlets and outlets to the control volume as well as the state within the control volume. The spatial considerations are addressed as both inlets and outlets and the volume itself.

Uniform properties are spatially constant but may vary with time. At inlets and outlets this means that the state across the crossing boundary is described by at any time by single values of the properties. Uniform properties within the control volume indicate that all elements within the control volume have the same value for the properties at any time. Nonuniform properties vary spatially. The time variation must also be addressed at the inlets and outlets as well as within the control volume. Steady state flow conditions at the

inlets and outlets mean that there is no variation of the properties with time. The density, velocity, specific internal energy, etc. have constant values for all times. Steady state properties within the control volume indicates that there is no accumulation within the control volume that causes the properties within the control volume to vary with time. Unsteady state conditions imply time variation either at the inlets and outlets or within the control volume. In summary, uniform and nonuniform describe the spatial variation while steady state and unsteady state describes the time variation.

4.3.1 Steady State Analysis

The analysis of the long-term operation of a particular component or complete system eliminates the need for describing the initial start-up period. This permits consideration of the steady state operation where no mass or energy accumulates within the control volume. Thus

$$\frac{dm_{CV}}{dt} = 0 \quad \dots\dots\dots (4.49)$$

$$\frac{dE_{CV}}{dt} = 0 \quad \dots\dots\dots (4.50)$$

The conservation of mass and of energy in **Equations (4.19) and (4.46)** are further reduced by assuming the steady state conditions and uniform properties at each inlet and outlet to yield

$$\dot{m}_{in} = \dot{m}_{out} \quad \dots\dots (4.51)$$

and

$$\sum_{in} \left[\left(h + \frac{1}{2} \bar{v}^2 + gz \right) \dot{m} \right]_{in} + \dot{Q}_{CV} = \sum_{out} \left[\left(h + \frac{1}{2} \bar{v}^2 + gz \right) \dot{m} \right]_{out} + \dot{W}_{CV} \quad \dots\dots (4.52)$$

These expressions are the basis of much of the cycle analysis. In short, both these expressions state, "What goes in equals what comes out."

4.3.2 Unsteady State Analysis

Time variation occurs in filling and discharging problems as well as in start-up and shut down considerations. Unsteady state analysis is needed when there is accumulation or storage within the control volume and/or time variation at the inlets and outlets. Integrating the conservation statements in **Equations (4.19) and (4.46)** from t_1 and t_2 yield

$$\int_{t_1}^{t_2} \frac{dm_{CV}}{dt} dt = \int_{t_1}^{t_2} \dot{m}_{in} dt - \int_{t_1}^{t_2} \dot{m}_{out} dt \quad \dots\dots\dots (4.53)$$

$$\text{or,} \quad m_2 - m_1 = m_{in} - m_{out} \quad \dots\dots\dots (4.54)$$

and

$$\int_{t_1}^{t_2} \frac{dE_{CV}}{dt} dt = \int_{t_1}^{t_2} (e\dot{m})_{in} dt - \int_{t_1}^{t_2} (e\dot{m})_{out} dt + \int_{t_1}^{t_2} \dot{Q}_{CV} dt - \int_{t_1}^{t_2} \dot{W}_{CV} dt \quad \dots\dots (4.55)$$

If the properties are considered to be uniform (at the inlets and outlets and within the control volume), the inlets and outlets are steady, and the kinetic energy of the inlet and outlet streams are negligible in comparison with the enthalpy and potential energy of these streams, then

$$\int_{t_1}^{t_2} \dot{Q}_{CV} dt = {}_1Q_2 \quad \dots\dots\dots (4.56)$$

and

$$\int_{t_1}^{t_2} \dot{W}_{CV} dt = {}_1W_2 \quad \dots\dots\dots (4.57)$$

where the right-hand side represent the heat transferred and work done for $t_2 - t_1$.

This results in the conservation of energy as

$$E_{CV,2} - E_{CV,1} = (em)_{in} - (em)_{out} + {}_1Q_2 - {}_1W_2 \quad \dots\dots\dots (4.58)$$

or,

$$\begin{aligned} \left(u + \frac{1}{2}\bar{v}^2 + gz\right)_2 m_2 - \left(u + \frac{1}{2}\bar{v}^2 + gz\right)_1 m_1 \\ = \left(h + \frac{1}{2}\bar{v}^2 + gz\right)_{in} m_{in} \\ - \left(h + \frac{1}{2}\bar{v}^2 + gz\right)_{out} m_{out} + {}_1Q_2 - {}_1W_2 \end{aligned} \quad \dots\dots\dots (4.59)$$

This equation includes the unsteady storage term within the control volume and all inlet and outlet flow terms. In short, the conservation of energy states that the change of energy within equals energy in minus energy out.

4.4 Control Volume Applications

There are three major subdivisions to thermodynamic analysis of a control volume.

1. *Control Volume Location.* The control volume must be indicated on the system drawing. Carefully distinguish between the crossing boundary and inside boundary. Denote the energy transfers. The boundary must be drawn so that the transfer process of interest (mass flow, work, heat transfer) crosses the boundary if the solution for that quantity is sought. The conservation statements connect the desired result to other quantities. These other quantities must be known if the desired quantity is to be evaluated.
2. *Conservation Statements.* The conservation of mass and that of energy are applied to the control volume. The first point to consider is the time variation of the problem, i.e., to decide whether steady or unsteady analysis is suitable. The second point is the spatial variation of the inlets and outlets as well as within the control volume, i.e., to decide whether properties are uniform at inlets and outlets and within the control volume.
3. *Properties.* Two independent properties for a simple compressible substance are required to determine each state. Tabular information or equations of state are needed to obtain the results.

4.4.1 Steady State Work Application

Turbines, compressors, pumps and fans are typical steady state devices that have a work interaction with the surroundings. These devices generally have a single inlet and a single outlet. Also the change in potential energy from inlet to outlet is usually very small. Thus conservation equations reduce to

Mass:

$$\dot{m}_{in} = \dot{m}_{out} \quad \dots\dots\dots (4.60)$$

Energy:

$$\left(h + \frac{1}{2}\bar{V}^2 + gz\right)_{in} \dot{m}_{in} + \dot{Q}_{CV} = \left(h + \frac{1}{2}\bar{V}^2 + gz\right)_{out} \dot{m}_{out} + \dot{W}_{CV} \quad \dots\dots\dots (4.61)$$

or,

$$\left(h + \frac{1}{2}\bar{V}^2 + gz\right)_{in} + q_{CV} = \left(h + \frac{1}{2}\bar{V}^2 + gz\right)_{out} + w_{CV} \quad \dots\dots\dots (4.62)$$

A turbine, as shown in **Figure 4.7(a)**, is a power-producing component which does work through the rotation of a shaft. The high-pressure fluid expands to a low pressure, doing work against the turbine blades. The kinetic and potential energy changes are usually, but not always, small. The heat transfer to the surroundings is also generally small compared to the work produced.

Compressors, as shown in **Figure 4.7(b)**, require a work input to produce a high pressure at the outlet from the low pressure at the inlet. They are usually used for a gaseous working fluid. Fans, as shown in **Figure 4.7(c)**, are completely employed to move fluid and alter velocities rather than produce large pressure changes. Pumps, as shown in **Figure 4.7(d)**, also require work input and are generally associated with liquids. All three devices require work input and typically have small heat transfers.

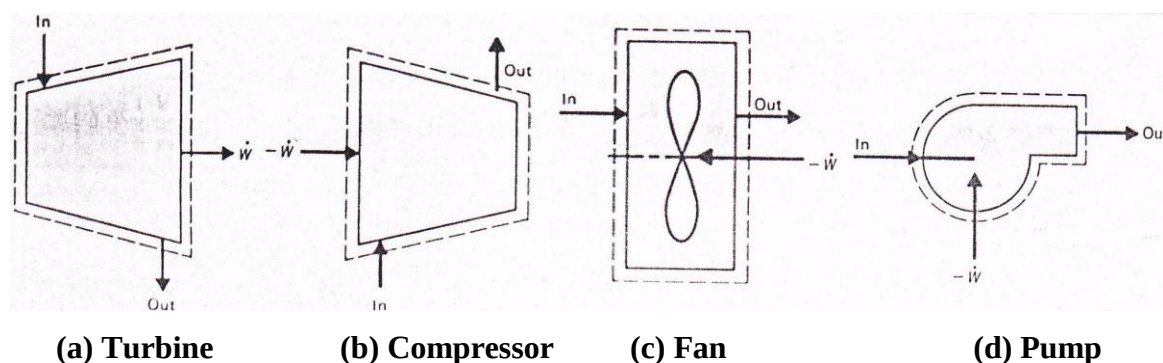


Figure 4.7: Steady state work applications

4.4.2 Steady State Flow Applications

Heat exchangers, condensers, steam generators, diffusers, and nozzles, throttling valves, and pipes are devices that operate at steady state and do not produce or consume work. Most experience very small changes in potential energy, so

Mass:

$$\dot{m}_{in} = \dot{m}_{out} \quad \text{..... (4.63)}$$

Energy:

$$\left(h + \frac{1}{2}\bar{V}^2 + gz\right)_{in} \dot{m}_{in} + \dot{Q}_{CV} = \left(h + \frac{1}{2}\bar{V}^2 + gz\right)_{out} \dot{m}_{out} \quad \text{..... (4.64)}$$

A heat exchanger, as shown in **Figure 4.8(a)**, is used to transfer energy from one fluid to another. There is no work interaction with the surroundings, and the changes in potential and kinetic energies are typically small. The heat transferred is computed from the enthalpy difference, and the analysis is very dependent on the choice of the control volume. **Figure 4.8 (a)** indicates two possible control volumes, and the specific one used depends on the desired and known information. For the control volume A, the first law is,

$$q = h_{1,out} - h_{1,in} \quad \text{..... (4.65)}$$

and for control volume B,

$$\dot{m}_1 h_{1,in} + \dot{m}_2 h_{2,in} = \dot{m}_1 h_{1,out} + \dot{m}_2 h_{2,out} \quad \text{..... (4.66)}$$

A steam generator, as shown in **Figure 4.8(b)**, and a condenser, as shown in **Figure 4.8(c)** are types of heat exchangers that perform a specific task. A steam generator uses an energy source such as the burning pulverized fuel or a nuclear reactor to create a high-temperature fluid. There is heat transfer from this fluid to water which passes through tube banks to form a superheated vapor. A condenser circulates cold water to condense a high-quality mixture to the liquid state. For the steam generator and condenser, the first law reduces to

$$q = h_{out} - h_{in} \quad \text{..... (4.67)}$$

Diffusers and nozzles are used to control the velocity of the fluid. Diffuser decreases the velocity while increasing the pressure, and the opposite is true for a nozzle. These devices are simply area changing devices, as shown in **Figure 4.8(c)**. Since the kinetic energy is typically important and there is typically single inlet and outlet, **Equation (4.64)** reduces to

$$q = (h_{out} - h_{in}) + \frac{1}{2}(\bar{V}_{out}^2 - \bar{V}_{in}^2) \quad \text{..... (4.68)}$$

where q might be negligible.

A throttling valve is simply a flow construction, as shown in **Figure 4.8(d)**. This reduces the pressure but performs no work. Generally, the heat transfer is small. If a control volume sufficiently far from the actual construction is chosen, then the kinetic energy is small. **Equation (4.64)** then yields

$$h_{in} = h_{out} \quad \text{..... (4.69)}$$

This is a very important result which is applied to any valve.

One of the steady state flow devices is a simple pipe. Assuming that potential energy changes are important and there is no work or heat transfer, **Equation (4.62)** reduces to

$$\left(h + \frac{1}{2}\bar{V}^2 + gz\right)_{in} = \left(h + \frac{1}{2}\bar{V}^2 + gz\right)_{out} \quad \text{..... (4.70)}$$

OR,

$$h_{in} - h_{out} + \frac{1}{2}(\bar{V}_{in}^2 - \bar{V}_{out}^2) + g(z_{in} - z_{out}) = 0 \quad \text{..... (4.71)}$$

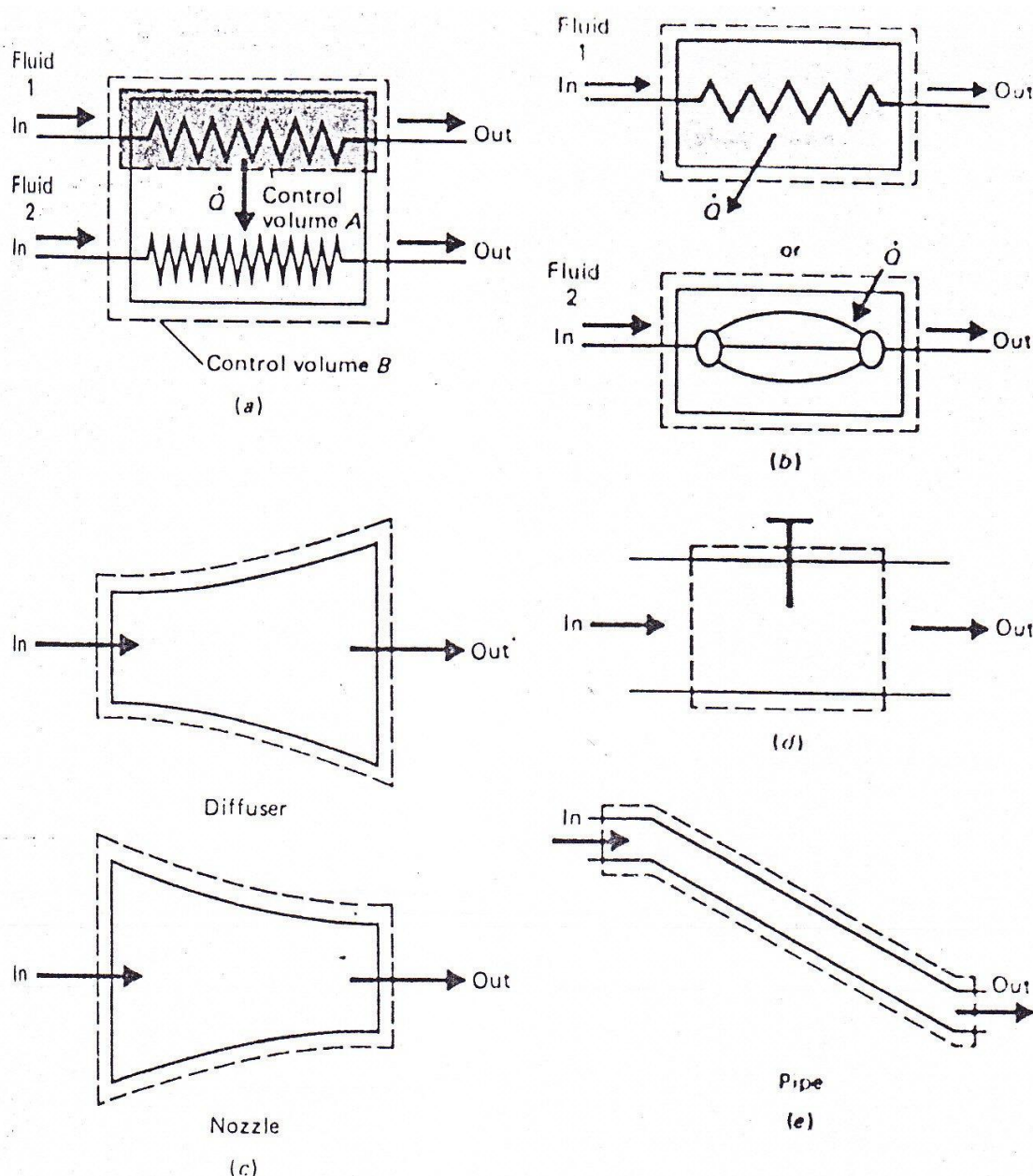


Figure 4.8: Steady State Flow Applications. (a) Heat exchanger; (b) Steam generator; (c) Nozzle and Diffuser; (d) Throttling valve; (e) Pipe.

4.4.3 Unsteady State Work Applications

Short operation turbines and the start-up of systems require unsteady analysis. Another type of unsteady work problem involves the increase of fluid within a tank required to

move a piston, as shown in **Figure 4.9**. Conservation of mass and energy for this type of problem (for negligible kinetic and potential energy in the flow streams compared with enthalpy) will be

Mass:

$$m_2 - m_1 = m_{in} \quad \dots\dots\dots (4.72)$$

Energy:

$$u_2 m_2 - u_1 m_1 = h_{in} m_{in} + {}_1Q_2 - {}_1W_2 \quad \dots\dots\dots (4.73)$$

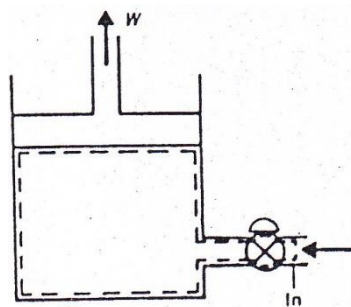


Figure 4.9: Unsteady State Work Application

4.4.4 Unsteady State Flow Applications

Charging and discharging of tanks without work are important applications. The purpose is to add or with draw fluid from a tank. The discharge process is diagrammed in **Figure 4.10**. The kinetic and potential energy changes are usually small, so conservation of mass and conservation of energy for the diagrammed problem will be

Mass:

$$m_2 - m_1 = -m_{out} \quad \dots\dots\dots (4.74)$$

Energy:

$$u_2 m_2 - u_1 m_1 = {}_1Q_2 - h_{out} m_{out} \quad \dots\dots\dots (4.75)$$

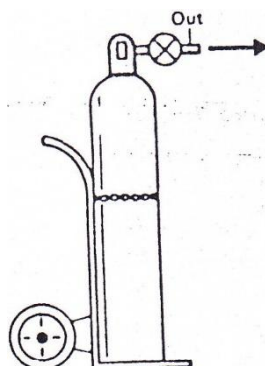


Figure 4.10: Unsteady State Flow Application

4.5 Other Statements of The First Law

The first law of thermodynamics has many different forms, and all are equivalent.

(a) *The internal energy of a closed system remains unchanged if the system is isolated from its surroundings.*

The control mass form of conservation of energy is

$$dE = \delta Q - \delta W \quad \text{..... (4.76)}$$

This expresses the change in energy for a process. This form reduces to the alternative statement that the energy is constant for an isolated system. An isolated system is one with no energy transfers with the surroundings. Thus, if $\delta Q = \delta W = 0$, then above equation reduces to $dE = 0$, as stated.

(b) *Another form states that an extensive property E exists whose increment is equal to the work received by the system while surrounded by an adiabatic wall.*

For an adiabatic wall, $\delta Q = 0$, and the work received is negative by our sign convention, so

$$dE = -\delta W \quad \text{..... (4.77)}$$

Therefore, the work is equal to the change in total energy for an adiabatic process.

(c) *A perpetual motion machine of the first kind is impossible.*

The perpetual motion machine was originally conceived as a purely mechanical contrivance which, when once set in motion, would continue to run ever. Such a machine would be merely a curiosity of no practical value, and we know that the presence of friction makes it impossible. What would be of immense value is a machine producing a continuous supply of work without absorbing energy from the surroundings; such a machine is called a perpetual motion machine of the first kind (PMM- I).

If a device has no energy transfers, then $\delta Q = 0$ and $\delta W = 0$. Thus it is impossible to have dE equal anything but zero.

CHAPTER 5

SECOND LAW OF THERMODYNAMICS

5.1 Necessity of the Formulation of the Second Law

The First Law of Thermodynamics is a theorem of the conservation of energy. It makes no distinction between various forms of energy and declares that all forms of energy are equivalent, that is if one kind of energy disappears, an equal amount of some other kind of energy must appear. As far as the First Law is concerned, all processes are possible provided that energy is neither created nor destroyed.

The Second Law of Thermodynamics states that all forms of energy are not equivalent in their ability to do work. It declares that certain processes are impossible to perform even though these processes do not violate the First Law.

The First Law states that in every cyclic process either work is converted into heat or heat is converted into work. The Second Law, on the other hand, makes the distinction between these two quantities by stating that heat from a single source whose temperature is uniform cannot be completely converted into work in any cyclic process.

Two more examples are presented to demonstrate the inability of the first law of thermodynamics and the conservation of mass statement to completely describe systems. Both example consider isolated systems, so the heat transfers and work across the system boundary are zero. **Figure 5.1(a)** presents an isolated chamber with steam on one side of a partition and a vacuum on the other side. This is the initial state of the system and is indicated as 1. The partition is suddenly removed, and the system reaches equilibrium in the final state, where the steam fills the entire chamber. This final state is denoted as state 2. The conservation of mass and the first law are presented in the figure, indicating that the transformation is a constant-mass and constant-internal energy process. Clearly these conservation statements are satisfied, and the process is physically reasonable. In **Figure 5.1(b)**, the steam completely fills the chamber in its initial state, state 1. The system transforms itself to two subsystems with the steam residing in one-half of the chamber and a perfect vacuum in the other half. A partition is inserted, and this final state is state 2. This case is also described as constant-mass and constant-internal energy process, as indicated in **Figure 5.1(a)**. Again the principles of conservation of mass and energy are satisfied; yet, from our experience, the process in **Figure 5.1(b)** is impossible. Regardless of the time allowed for this process, it will never occur in an isolated system. The inability of the first law of thermodynamics and the conservation of mass to explain this impossibility indicates the need for another fundamental law.

Figure 5.2(a) shows an isolated system containing two blocks of material labeled A and B. In **Figure 5.2(a)** block A is at a high temperature T_H , and block B is at a low temperature T_L . They are initially brought together and isolated from all surroundings, as indicated by state 1. There is a subsequent heat transfer between block A and block B until they reach a common intermediate equilibrium temperature T . This final state is denoted as state 2. The statements of conservation of mass and energy are presented in **Figure 5.2**, indicating that this is a constant-mass and constant-internal energy process. The temperature of the hotter block is expected to decrease and the temperature of the colder block is expected to increase to the equilibrium temperature. The process satisfies

the conservation statements as well as our physical expectations. In **Figure 5.2(b)**, the two blocks are initially brought together at the same temperature and the final state results when block A is at a high temperature T_H and block B is at the low temperature T_L . Again the conservation statements given in **Figure 5.2** are satisfied, yet our experience leads us to expect that this is an impossible process.

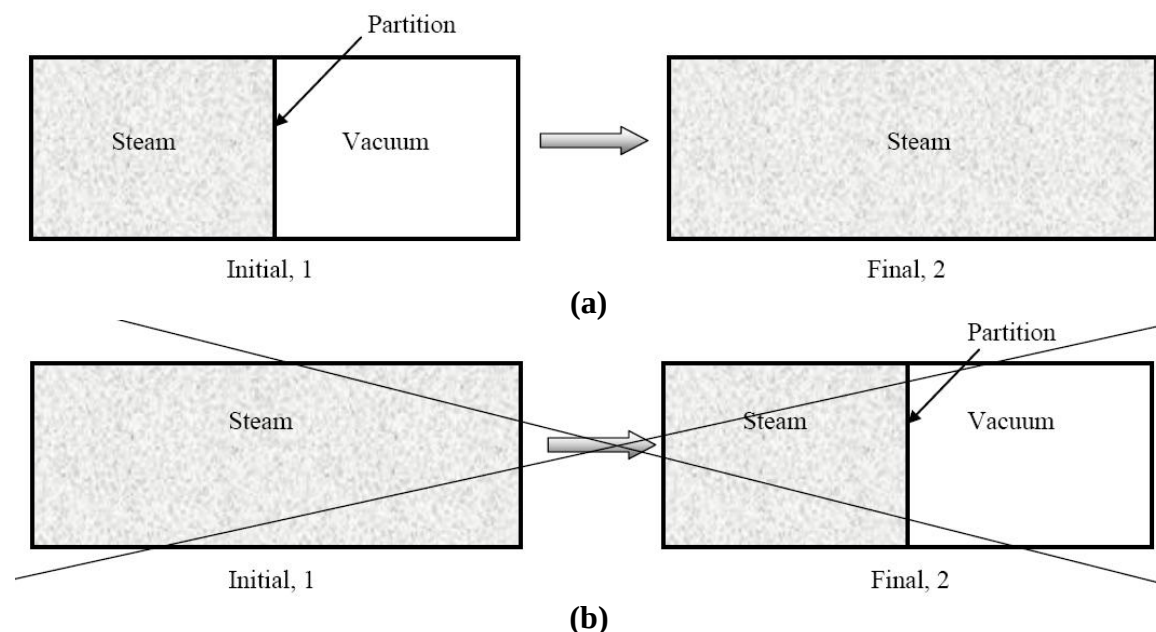


Figure 5.1: Expansion into a vacuum for an isolated system

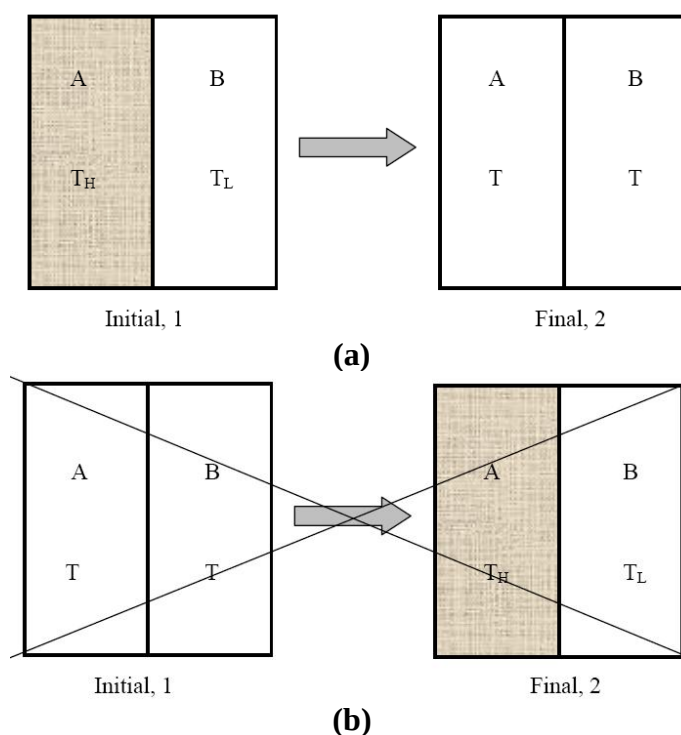


Figure 5.2: Heat transfer between two blocks within an isolated system

These are only few examples that demonstrate that there is a fundamental principle, the second law of thermodynamics, which is not described by the principles of conservation of mass and energy. These examples present the two key features of the second law of

thermodynamics. First, the second law prescribes the *direction* for the processes; second, some thermodynamic characteristics of the system, a system *property*, exists and is always changing in a specified direction.

The example in **Figure 5.1** indicates that the system in (**Figure 5.1a**) tends to proceed to a more random state and that we expect the reverse process (**Figure 5.1b**), which proceeds to a more organized state (i.e., a state with more individual control masses, each with different intensive properties), to be impossible. Thus there is a specific direction to the process. The direction appears to be related to the disorder, randomness, or uncertainty of the microscopic scale of the system. There is an inability of the system to self-organize. The example in **Figure 5.2** also demonstrates that the organized, or structured, state of a separate high-temperature material and low-temperature material in state 1 tends toward a more disorganized, or less structured, state of a single homogeneous temperature in state 2. Again the reverse process is not possible. Thus, the second law needs to describe the direction for processes, and the direction is described in terms of a system property that characterizes the system's randomness, disorganization, or uncertainty.

Observe that the randomness, disorganization, or uncertainty in the two systems of **Figure 5.1** and **5.2** was changed in different ways. In **Figure 5.1**, the state of the steam was changed by rearranging the positions of the molecules of the steam. Because more positions are available for a given molecule after the expansion than before it, our uncertainty as to the position of a given molecule is greater following the expansion. In **Figure 5.2**, the state of the system was changed by allowing heat transfer between subsystems, with no spatial arrangement of the system. By statistical mechanics, the microscopic disorder of a material depends on the temperature. The heat transfer process between the subsystems increases uncertainty about the molecular disorder in the two subsystems compared with the disorder before the process.

Thus, the disorder of the system can be brought to change in at least two ways: geometric rearrangement and heat transfer.

5.2 Entropy and the Second Law for an Isolated System

The above examples focus on isolated systems, and this is the starting point for the second law presentation; yet, energy transfers are fundamental to thermodynamics and require special attention. The example in **Figure 5.2** indicates that **Figure 5.2a** is the possible process and the isolated system tends to a less structured, or more random, state. If the system is redefined to be either block A or block B, then a heat transfer across the system boundary exists between the blocks. This is diagrammed in **Figure 5.3**. For a control mass A, the process starts from a high-temperature initial state and ends at a lower-temperature final state as a result of heat transfer from this control mass B. Physically, block A proceeds from a high temperature, or less organized, state (from molecular point of view) to a lower temperature, or more organized, state (on the molecular scale). This process, which tends to organize or reduce the random molecular nature of block A, is a result of heat loss from this control mass, or a negative energy transfer. As heat transfer is associated with a disorganized energy transfer, so a heat loss from the control mass is consistent with a decrease in the randomness of block A. However, control mass B receives the heat transfer from block A, and this is interpreted as positive disorganized energy transfer. Thus control mass B increases from an initial

low-temperature organized state to a more random high-temperature state as a result of heat transfer. The possible process is one where the complete isolated system tends to a more random state, but the subsystems can individually experience an increase or a decrease in disorder.

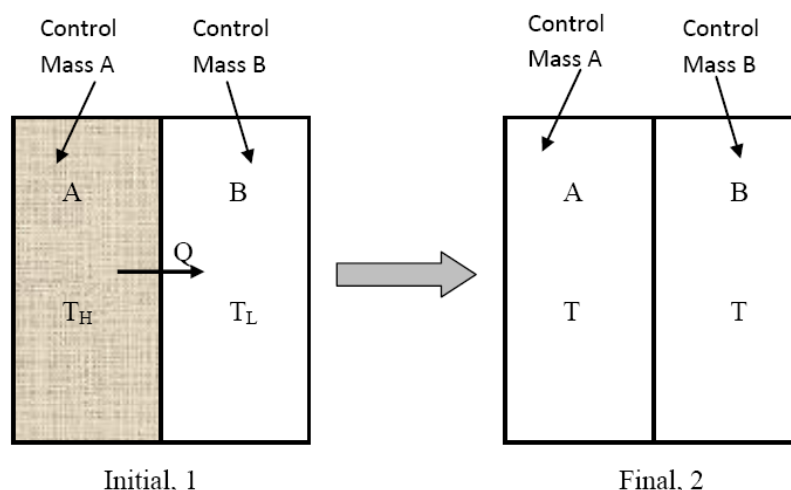


Figure 5.3: Heat transfer between control mass A and control mass B

Hence, heat transfer or disorganized energy transfer to the system property characterizes the randomness, uncertainty, or disorder of the state. This connection between heat transfer and an uncertainty property which shows the directionality of a process is the second law of thermodynamics. The property that describes the randomness or uncertainty is called *entropy*.

The previous examples indicate that an isolated system, after removal of internal constraints, undergoes processes that are not quantified by the first law of thermodynamics. These are combined into the statement of the second law of thermodynamics:

The entropy S , an extensive property, must always increase or remain constant for an isolated system.

This is expressed mathematically as

$$dS_{isolated} \geq 0 \quad \dots\dots\dots (5.1)$$

or,

$$(S_{final} - S_{initial})_{isolated} \geq 0 \quad \dots\dots\dots (5.2)$$

Entropy, like other thermodynamic properties, is defined only at equilibrium states or for quasi- equilibrium processes. **Equations (5.1) and (5.2)** show that the entropy of the final state is never less than that of the initial state for any process, which an isolated system undergoes. In the examples previously explained the final state of the isolated system results after the removal of internal constraint. The final state has the largest value of entropy for the complete isolated system. Thus, an isolated system composed of subsystems tend towards a state which increases the total entropy of the composite or isolated system. This is the maximum possible entropy subject to any constraints that are imposed.

Expressions for change in the property S as a function of other system properties must be developed before we can apply the second law. However, some characteristics of entropy can be examined first. Entropy is an extensive property of the system, so any system composed of subsystems yields

$$S = m_A s_A + m_B s_B + \cdots = \sum m_i s_i \quad \text{..... (5.3)}$$

where s is the specific entropy or entropy per unit mass and the subscript denotes the specific subsystem. The second law, expressed as an integral over all the elements of an isolated system, is

$$dS_{\text{isolated}} = d\left(\int_m s dm\right)_{\text{isolated}} = d\left(\int_m \rho s dV\right)_{\text{isolated}} \geq 0 \quad \text{..... (5.4)}$$

This property is specified by the state of the system, or it is used to specify the state of the system. As prescribed by the state postulate for a compressible substance, the entropy is completely specified by two independent properties of the system; or it can be used as an independent property, combined with another property, to specify all other properties. Entropy is a measure of the molecular disorder of the substance. Larger values of entropy imply the greater disorder or uncertainty, and lower values imply more microscopically organized states. The unit of entropy S is kilojoules per Kelvin.

A key difference between the second law and first law is the inequality sign. The first law specifies a direct relationship between energy and energy transfer, while the second law indicates only the direction for the change in molecular disorder or uncertainty. An alternative, yet equivalent, presentation which is quite useful is to define the term *entropy production* or *entropy generation* S_{gen} and to eliminate the inequality sign:

$$(dS - \delta S_{\text{gen}})_{\text{isolated}} \geq 0 \quad \text{..... (5.5)}$$

or

$$(S_{\text{final}} - S_{\text{initial}} - \delta S_{\text{gen}})_{\text{isolated}} \geq 0 \quad \text{..... (5.6)}$$

Here δS_{gen} is the entropy generated during a change in system state and is always positive or zero. **Equation (5.5)** is exactly the same as **Equation (5.1)**, where δS_{gen} represents the 'amount of inequality' as prescribed by the second law. The negative sign preceding the entropy generation is necessary to reduce the change in entropy to zero.

The second law is also expressed on a rate basis by considering the time interval dt . This law is then

$$\left.\frac{dS}{dt}\right|_{\text{isolated}} = \frac{d}{dt}\left(\int_V \rho s dV\right)_{\text{isolated}} \geq 0 \quad \text{..... (5.7)}$$

or

$$\left.\frac{dS}{dt}\right|_{\text{isolated}} - \dot{S}_{\text{gen}} = 0 \quad \text{..... (5.8)}$$

where $\dot{S}_{\text{gen}}$ indicates the rate of entropy generation.

5.3 Reversible and Irreversible Processes

The concepts of reversible and irreversible processes play a fundamental role in thermodynamics, and the reduction of irreversibilities is a goal of many analyses. To better understand these concepts, consider the isolated system diagrammed in **Figure 5.4**. The isolated system is composed of number of subsystems, four in this example, labeled A, B, C, and D. The subsystems might communicate thermally or mechanically, or they might even transfer mass across their boundaries (except for the outside boundary of the isolated system). Therefore, the total system is isolated when all four subsystems are considered. But if the system is taken as one subsystem, then the other three are viewed as surroundings to the chosen subsystem. If subsystem A has energy transfers in the form of heat transfer and work with the other subsystems, then the subsystem A is a control mass and subsystem B, C, and D are the surroundings to control mass A. Reversible and irreversible processes are strictly applicable for only isolated systems and must be interpreted for subsystems (such as subsystem A) with appropriate terminology. This leads to a set of subcategories of reversible and irreversible processes.

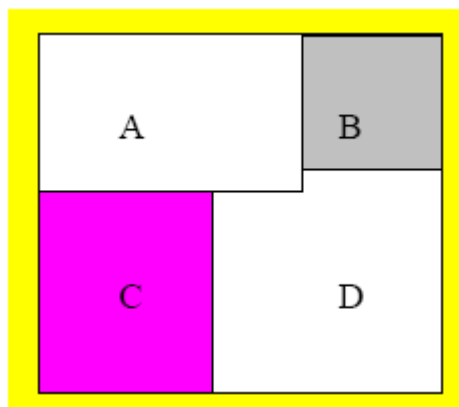


Figure 5.4: An isolated system composed of four subsystems

The second law puts a direction on all isolated system processes: $S_{final} > S_{initial}$. Thus there is a forward and backward direction to all processes. When $S_{final} > S_{initial}$, the forward direction is defined and the backward direction is impossible, as stated by the second law. Such a process is termed as *irreversible*. When $S_{final} = S_{initial}$, the forward and backward processes are equally possible, and the process is termed *reversible*. The inequality sign is the key distinguishing feature of reversible and irreversible processes. Alternatively, a reversible process produces no entropy generation, so S_{gen} is zero; and an irreversible process has a net entropy generation, so S_{gen} is finite and positive.

The previous discussion pertains to an isolated system which could be composed of subsystems. Reversible and irreversible processes are defined in terms of the composite entropy of all the subsystems as required by the second law. These processes can also be referred to as *totally reversible* or *totally irreversible*.

If a particular subsystem is considered as the system, say, subsystem A in **Figure 5.4**, then the concepts of internally reversible or irreversible and externally reversible or irreversible are needed. An *internally reversible process* considers the system (not surroundings), and the process can be reversed at any point, and the system then proceeds in reverse through the same states. This requires that all heat transfers with the

surroundings be reversible, but the surroundings do not necessarily undergo reversible process. An *internally irreversible* process has irreversibilities within the system boundaries. An *externally reversible process* (or *externally irreversible process*) pertains to the surroundings of the system. In the previous example, the surroundings are subsystems B, C, and D, so an externally reversible process (or externally irreversible process) specifies the particular type of process for these subsystems.

Physically, a totally reversible process can be reversed at any point, and the system and surroundings (making up the isolated system) proceed backward along the same process. The totally reversible or simply reversible process must include both the system and surroundings. This differs significantly from an internally reversible process, which views only the system as reversible and excludes the surroundings from consideration. Clearly the totally reversible process is more difficult to achieve.

The reversible process is an idealization, and all actual processes are irreversible and approach reversible processes only in special cases. **Figure 5.5** shows some examples of irreversible and corresponding reversible processes. The first example considers the expansion of a substance against a piston and inelastic spring. As the substance expands, energy is dissipated by the frictional work during the interaction of the piston and wall. The disorder within the piston and wall has increased, so the entropy of is increased and the process is irreversible. Friction is a very common contributor to irreversible processes. The inelastic spring also dissipates energy and increases the disorder. If the ideal case of frictionless motion and an elastic spring are considered, then the process can be reversed at any point in the process and restored to its original state. Thus this process is reversible. The unstrained expansion of steam into a vacuum increases the disorder and is clearly irreversible, whereas a controlled, slow expansion does work against the surroundings, which can be reversed to achieve a reversible process. The heat transfer across a finite temperature difference is irreversible since the disorder of the entire system increases.

In each of the above cases, the viewpoint can also be a subsystem denoted by subsystem A in each case. It is possible for subsystem A to be internally reversible while the reversibilities occur exterior to the system boundary. Specifically in the steam subsystem A undergoing an unrestrained gas expansion could be proceeding through an internally reversible processes, but the process for the complete isolated system is irreversible. This case indicates that this is an externally irreversible process. Similar subsets of reversible and irreversible processes are possible for other examples.

The main features of irreversible processes are that there are dissipative effects which are not directly useful and that the processes can produce through nonequilibrium states. These both tend to increase the molecular disorder, or entropy, of the systems. Both contributions make processes irreversible. However, the reversible process progresses through equilibrium states, and there are no dissipative effects. Therefore, reversible processes are idealizations of real situations.

A clear understanding of the relationship of reversible and irreversible processes with the quasi-equilibrium process is important. The previous definition of a quasi-equilibrium process pertained to only the system and indicated nothing about the surroundings. A *totally reversible process must specify reversibility for both the system and the surroundings*. Therefore, a reversible process is always a quasi-equilibrium process since both the system and the surroundings undergo reversible processes and thus pass through

a series of equilibrium states. Also all internally reversible processes are quasi-equilibrium processes. Conversely, a quasi-equilibrium process is an internally reversible process since these both pertain to the system only. A quasi-equilibrium process is not necessarily a reversible process (totally reversible), since irreversibilities can occur outside the system boundaries.

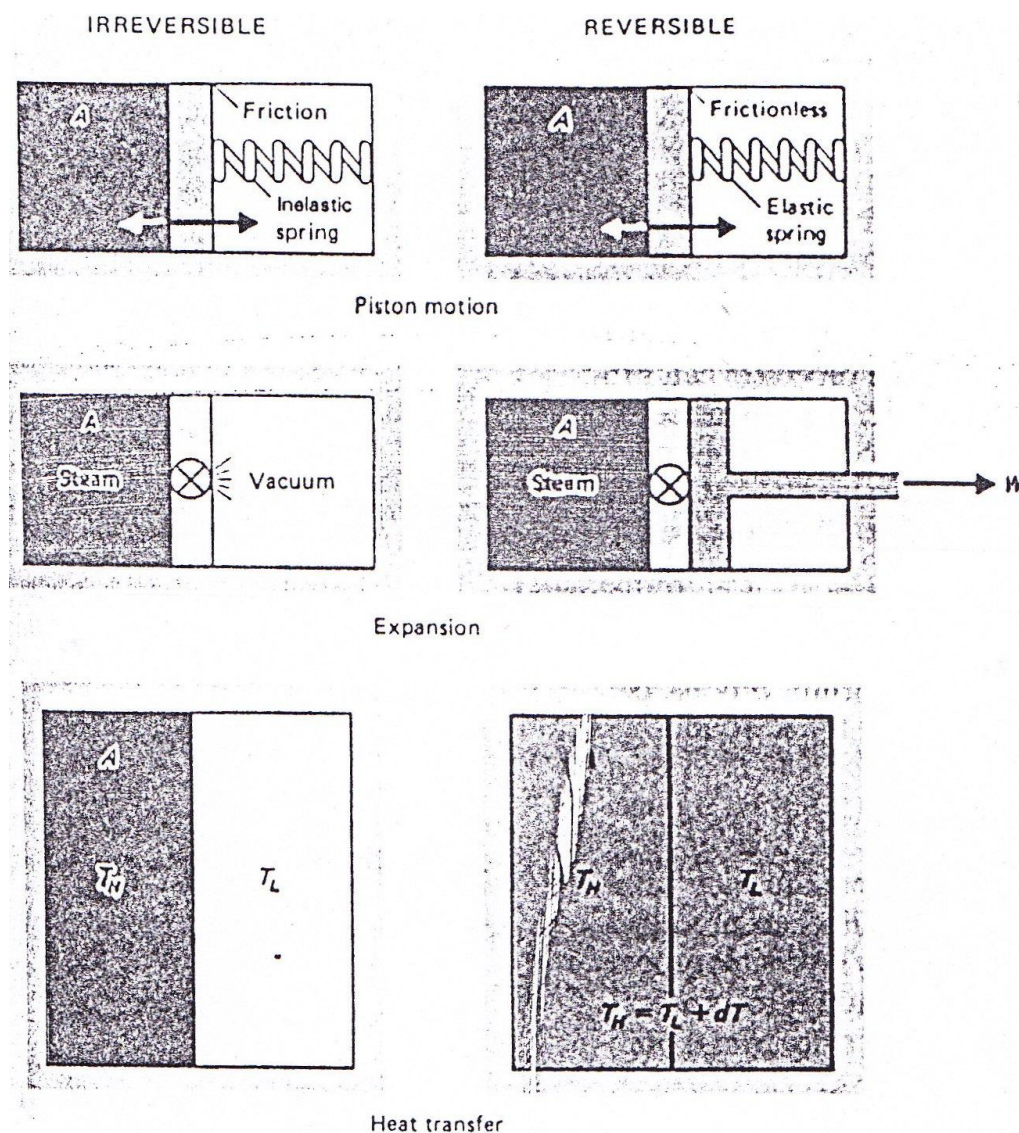


Figure 5.5 Reversible and Irreversible Processes

Hence, the causes of irreversibilities may be

- Friction
- Heat transfer across a finite temperature difference
- Free expansion
- Mixing
- Inelastic deformation

5.4 Entropy: Property and Relations

Entropy is a property that is specified for every equilibrium state of a substance. Entropy represents the disorder, or uncertainty, of the microscopic scale, yet macroscopically it is

used as all other properties. The state postulate for a pure simple compressible substance states that two independent intensive properties specify a state. Therefore, entropy is an additional property that can be used to specify a state.

As indicated by the state postulate, the energy of a pure simple compressible substance is specified by any two independent properties. Thus

$$E = E(S, V) = U(S, V) + KE + PE \quad \text{..... (5.9)}$$

or, if we exclude the potential and kinetic energies,

$$E = U = U(S, V) \quad \text{..... (5.10)}$$

The differential of this property, an exact differential, is

$$dU = \left(\frac{\partial U}{\partial S}\right)_V dS + \left(\frac{\partial U}{\partial V}\right)_S dV \quad \text{..... (5.11)}$$

The two partial derivatives in this expression are intensive parameters since the ratio of two extensive parameters is an intensive parameter.

Temperature is defined in classical thermodynamics as the first partial derivative in **Equation (5.11)**

$$T \equiv \left(\frac{\partial U}{\partial S}\right)_V \quad \text{..... (5.12)}$$

The other partial derivative in **Equation (5.11)** is used to define pressure. The formal definition is

$$P \equiv -\left(\frac{\partial U}{\partial V}\right)_S \quad \text{..... (5.13)}$$

The definitions for temperature and pressure given in **Equations (5.12)** and **(5.13)** respectively, are substituted in **Equation (5.11)** to yield

$$dU = TdS - PdV \quad \text{..... (5.14)}$$

This is an extremely important relation in thermodynamics and is used extensively. It is also known as *Gibbs equation*. It applies generally to a control mass.

5.4.1 Entropy Relations for Ideal Gases

The state postulate yields $U = U(S, V)$, so the change in internal energy combined with the temperature and pressure definitions yields

$$dU = TdS - PdV$$

This is the Gibbs equation which relates equilibrium thermodynamic properties. Thus this expression depends not on the path between any two states but on the substance only. A companion expression is obtained from enthalpy as

$$dH = d(U + PV) = dU + PdV + VdP \quad \text{..... (5.15)}$$

Substituting **Equation (5.14)** in **Equation (5.15)** yields

$$dH = TdS + VdP \quad \text{..... (5.16)}$$

These expressions are also represented on an intensive basis as

$$du = Tds - Pdv \quad \text{..... (5.17)}$$

$$dh = Tds + vdP \quad \text{..... (5.18)}$$

The changes in entropy are obtained directly from these equations as

$$dS = \frac{dU}{T} + \frac{P}{T}dV \quad \text{..... (5.19)}$$

$$dS = \frac{dH}{T} - \frac{V}{T}dP \quad \text{..... (5.20)}$$

or

$$ds = \frac{du}{T} + \frac{P}{T}dv \quad \text{..... (5.21)}$$

$$ds = \frac{dh}{T} - \frac{v}{T}dP \quad \text{..... (5.22)}$$

As we know, $du = c_v dT$ and $dh = c_p dT$. Substituting these relations in above **Equations (5.21)** and **(5.22)** yields

$$ds = c_v \frac{dT}{T} + R \ln \frac{dv}{v} \quad \text{..... (5.23)}$$

$$ds = c_p \frac{dT}{T} - R \ln \frac{dP}{P} \quad \text{..... (5.24)}$$

Finite changes in entropy are evaluated by integrating these last equations to yield

$$s_2 - s_1 = \int_{T_1}^{T_2} c_v \frac{dT}{T} + R \ln \frac{v_2}{v_1} \quad \text{..... (5.25)}$$

$$s_2 - s_1 = \int_{T_1}^{T_2} c_p \frac{dT}{T} - R \ln \frac{P_2}{P_1} \quad \text{..... (5.26)}$$

Either of the previous expressions for the change in entropy is used, depending on what information is known. Unlike the internal energy and enthalpy for an ideal gas which requires only temperature for their evaluation, entropy requires two properties [**Equations (5.23)** and **(5.24)**]. Completion of the evaluation of the change in entropy requires the functional form of the specific heat variation with temperature.

The specific heats are often accurately approximated by a single average value. Thus, for an ideal gas with constant specific heats,

$$s_2 - s_1 = c_v \ln \frac{T_2}{T_1} + R \ln \frac{v_2}{v_1} \quad \text{..... (5.27)}$$

$$s_2 - s_1 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1} \quad \dots\dots\dots (5.28)$$

5.4.2 Incompressible Fluid or Solid Relations

The equation of state for an incompressible fluid or solid is $v = \text{constant}$. Then,

$$du = c_v dT = c dT \quad \dots\dots\dots (5.29)$$

This expression is substituted in **Equation (5.21)** to give the change in entropy

$$ds = c \frac{dT}{T} + \frac{P}{T} dv \quad \dots\dots\dots (5.30)$$

but $v = \text{constant}$, so

$$ds = c \frac{dT}{T} \quad \dots\dots\dots (5.31)$$

Finite changes in entropy are

$$s_2 - s_1 = \int_{T_1}^{T_2} c \frac{dT}{T} \quad \dots\dots\dots (5.32)$$

If the specific heat is approximated by a constant, then

$$s_2 - s_1 = c \ln \frac{T_2}{T_1} \quad \dots\dots\dots (5.33)$$

5.4.3 Entropy for steam

As with all previous properties, data for entropy can be found in tabular, graphical, equation. The steam tables provide tabular information for the entropy of steam. Within the saturation region, the entropy is calculated with the quality as

$$s = s_l + xs_{lg} \quad \dots\dots\dots (5.34)$$

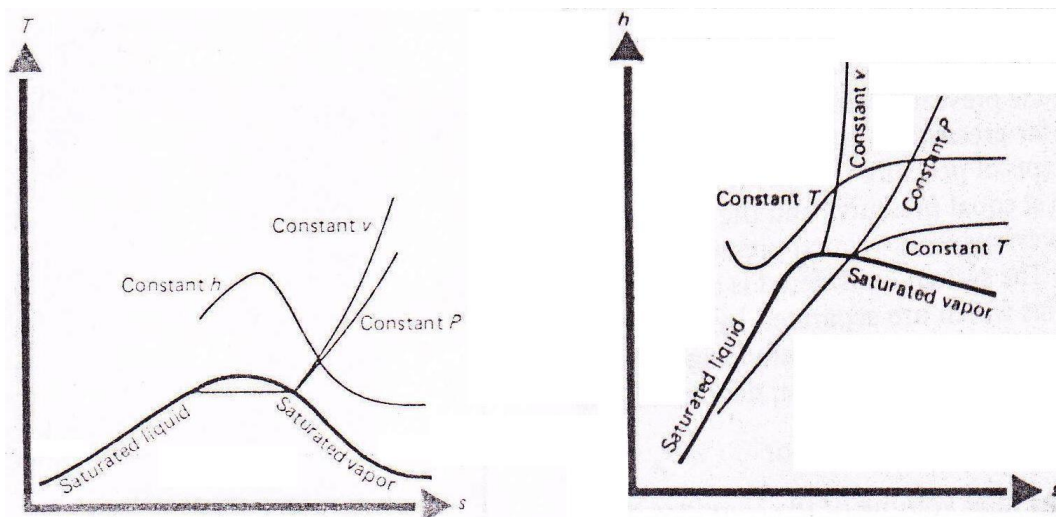


Figure 5.6: The T-s and h-s diagrams

The graphical presentation of property information is used extensively in thermodynamic analysis. The analysis is generally indicated on the $P - v$ diagram. The $P - v$ diagram is very important to simple reversible compressible work calculation since the graphical area under the process representation is the work done. The most thermodynamic process is explained with the help of $P - v$, $T - s$, and $h - s$ diagrams. Typical $T - s$ and $h - s$ diagrams are shown in **Figure 5.6**. The lines of constant volume, pressure and enthalpy are shown on $T - s$ diagram. The $h - s$ diagram is called *Mollier chart*, and the constant-volume, constant-pressure, and constant temperature lines are indicated. For ideal gases with temperature-independent c_p , the $T - s$ and $h - s$ diagrams are similar, since h is directly proportional to T .

5.5 Control Mass and Control Volume Formulation

The second law is presented for an isolated system, and as indicated by reversible and irreversible processes, the isolated system processes are directional. This directionality of the entropy change of an isolated system is a fundamental and important aspect of the second law. But the isolated system is somewhat abstract, and the primary interest of much thermodynamic analysis lies in control mass and control volume systems. Therefore, the second law must be transformed to these viewpoints.

5.5.1 Control Mass Formulation

A control mass exchanges energy with the surroundings by work and heat transfer. Therefore, two types of idealized subsystems of the surroundings are conceived: a *reversible work reservoir* and a *reversible heat transfer reservoir*. Both reservoirs are reversible, indicating that they proceed slowly through equilibrium states and can be restored to the original state at any point in the process. Both are control masses and are termed *reservoir* to indicate a very large system that is unchanging during work or heat transfer.

The reversible heat transfer reservoir participates in heat transfer only (not in work transfer). Therefore, this reservoir is a constant volume control mass. All the heat transfers to the reversible heat transfer reservoir transforms this system through equilibrium states which are specified by a uniform temperature. The heat transfers can be to or from this reservoir. From Gibb's **Equation (5.14)**, for constant volume processes,

$$dS = \frac{dU}{T} \quad \text{..... (5.34)}$$

and the first law for a control mass that does no work is

$$dU = \delta Q \quad \text{..... (5.35)}$$

Thus for the reversible heat transfer reservoir, the entropy change is

$$dS = \frac{\delta Q}{T} \quad \text{..... (5.36)}$$

where δQ is positive into the reversible heat transfer reservoir. The temperature is uniform throughout the reservoir.

The reversible work reservoir participates in only work interactions (no heat transfers), and for a compressible system, the work is a result of boundary expansion. The work,

either positive or negative, is a quasi-equilibrium process. The Gibbs equation for the reversible work reservoir is given as

$$dS = \frac{dU}{T} + \frac{P}{T}dV \quad \dots\dots\dots (5.37)$$

but the first law with only PdV work and no heat transfer is

$$dU = -\delta W = -PdV \quad \dots\dots\dots (5.38)$$

Thus substituting **Equation (5.38)** into **(5.37)** yields

$$dS = \frac{-PdV}{T} + \frac{P}{T}dV = 0 \quad \dots\dots\dots (5.39)$$

so the reversible work reservoir has no entropy change.

In summary, the control mass form of the second law is obtained by considering a general control mass that undergoes both work interactions with reversible work reservoirs and heat transfers with reversible heat transfer reservoirs. The composite system of control mass, reversible work reservoirs, and reversible heat transfer reservoirs forms an isolated system. The work reservoirs have no entropy change, and the heat transfer reservoirs' entropy change is $dS_i = \delta Q_i/T_i$. A general control mass is shown in **Figure 5.7**, where there could be an arbitrary number of work and heat transfer reservoirs.

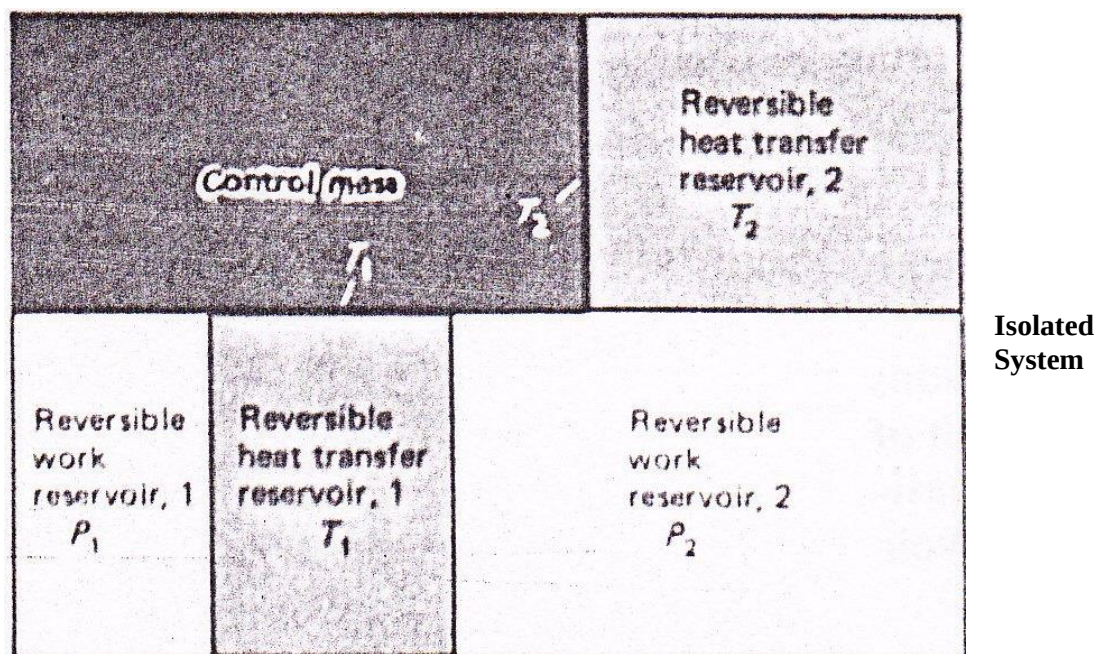


Figure 5.7: Control Mass, reversible heat reservoirs, and reversible work reservoirs within an isolated system

The second law for the isolated system [control mass (CM) plus reservoir] is

$$dS_{isolated} \geq 0 \quad \dots\dots\dots (5.40)$$

but

$$dS_{isolated} = dS_{CM} + \sum dS_{reversible\ work\ transfer\ reservoirs} + \sum dS_{reversible\ heat\ transfer\ reservoirs} \quad \dots\dots\dots (5.41)$$

The entropy change for each reversible heat transfer reservoir is $dS_i = \delta Q_i/T_i$, where δQ_i is positive into the reservoir and T_i is the reservoir temperature and so is the temperature of the boundary between the reversible heat transfer reservoir and the control mass. This, in terms of the control mass, the heat transfer to the control mass from each reservoir is

$$\delta Q_{CM,i} = -\delta Q_{reversible\ heat\ transfer\ reservoir,i} \quad \dots\dots\dots (5.42)$$

Thus, substituting **Equation (5.42)** into **(5.41)** yields

$$dS_{isolated} = dS_{CM} - \sum_i \left(\frac{\delta Q_i}{T_i} \right)_{CM} \geq 0 \quad \dots\dots\dots (5.43)$$

or

$$dS_{CM} \geq \sum_i \left(\frac{\delta Q_i}{T_i} \right)_{CM} \quad \dots\dots\dots (5.44)$$

since the change in entropy of all reversible work reservoirs is zero. Note especially that the temperature in the second law is the boundary temperature interfacing the control mass and heat transfer reservoirs. Thus this is the control mass form of the second law:

The change in entropy S , an extensive property of the control mass, is greater than or equal to the sum of the heat transfers divided by the corresponding absolute temperatures of the reservoirs at the boundary.

Note that dS_{CM} can be positive or negative since the heat transfer can be positive or negative. Recall that heat transfer is a microscopically disorganized energy transfer, so positive heat transfer increases microscopic disorder and increases entropy while negative heat transfer decreases microscopic disorder and decreases entropy. Also there is no contribution to entropy change by work interactions. Since work is an organized energy transfer mechanism and entropy describes microscopic disorder, this is appropriate.

Equation (5.44) is the control mass form for the second law. The inequality applies to irreversible processes. Since the development is in terms of reversible surroundings, the irreversibilities are within the control mass; so this type of process is termed *internally irreversible*. The equality applies to an internally reversible process. The inequality is eliminated if the concept of entropy generation is used. Thus

$$(dS - \delta S_{gen})_{isolated} \geq 0 \quad \dots\dots\dots (5.45)$$

and with **Equation (5.43)**,

$$dS_{CM} - \sum_i \left(\frac{\delta Q_i}{T_i} \right)_{CM} - \delta S_{gen} \geq 0 \quad \dots\dots\dots (5.46)$$

or

$$\delta S_{gen} = dS_{CM} - \sum_i \left(\frac{\delta Q_i}{T_i} \right)_{CM} \geq 0 \quad \dots\dots\dots (5.47)$$

This is the entropy generation by the process. **Equation (5.44)** and **(5.47)** are the forms of the second law that are the starting point for subsequent analysis.

Alternative forms can be obtained with simplifying assumptions. If the control mass has a uniform distribution of mass, then

$$dS_{CM} \geq \sum_i \left(\frac{\delta q_i}{T_i} \right)_{CM} \quad \dots\dots\dots (5.48)$$

In terms of entropy generation, this is

$$\delta S_{gen} = dS_{CM} - \sum_i \left(\frac{\delta q_i}{T_i} \right)_{CM} \geq 0 \quad \dots\dots\dots (5.49)$$

The heat transfers are relative to the control mass, and T_i corresponds to the temperature at boundary i of the control mass where the heat transfer occurs. The sum includes all possible heat transfers with the surroundings. All real processes are irreversible to some extent; so the inequality sign is applicable, and the entropy generation is greater than zero. Only for the idealization of a reversible process is the equals sign appropriate, and $\delta S_{gen} = 0$. The control mass forms of the second law are expressed on a rate basis as

$$\frac{dS_{CM}}{dt} \geq \sum_i \left(\frac{\dot{Q}_i}{T_i} \right)_{CM} \quad \dots\dots\dots (5.50)$$

or

$$\dot{S}_{gen} = \frac{dS_{CM}}{dt} - \sum_i \left(\frac{\dot{Q}_i}{T_i} \right)_{CM} \geq 0 \quad \dots\dots\dots (5.51)$$

The second law connects the system property, entropy to the heat transfer. If the *process is reversible*, the entropy generation is zero, and

$$dS_{CM,rev} \geq \sum_i \left(\frac{\delta Q_i}{T_i} \right)_{CM,rev} \quad \dots\dots\dots (5.52)$$

For a single heat transfer

$$dS_{CM,rev} \geq \left(\frac{\delta Q_i}{T_i} \right)_{CM,rev} \quad \dots\dots\dots (5.53)$$

or, since the reversible heat transfer occurs across an infinitesimal temperature difference,

$$\delta Q_{rev} = T dS \quad \dots\dots\dots (5.54)$$

This shows the importance of $T - s$ diagram to thermodynamic analysis. The heat transfer for a specified reversible process between states 1 and 2 is

$${}_1Q_{2,rev} = \int_1^2 T ds \quad \dots\dots\dots (5.54)$$

or

$${}_1q_{2,rev} = \int_1^2 T ds \quad \dots\dots\dots (5.55)$$

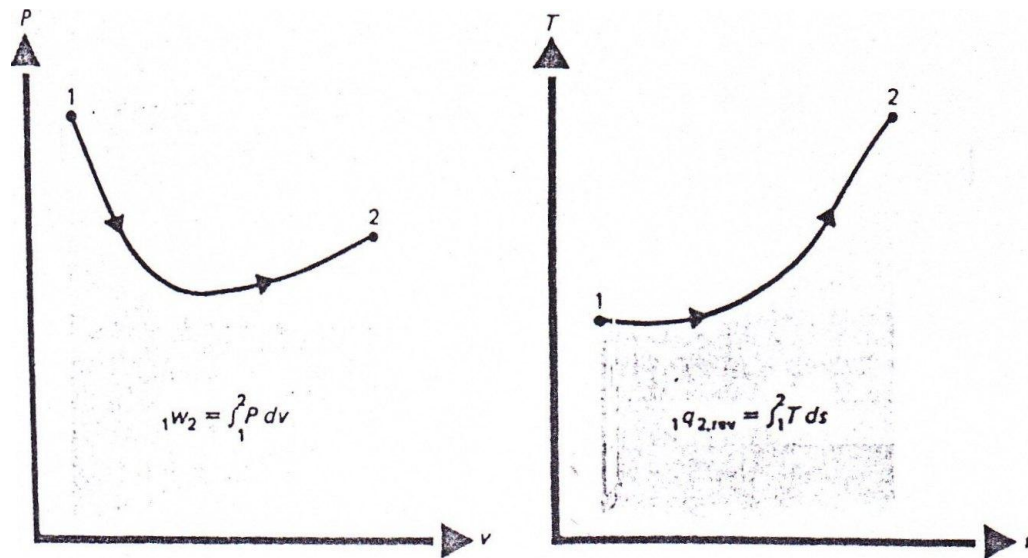


Figure 5.8: Heat transfer and work transfer represented on areas on $T - s$ and $P - v$ diagrams, respectively

5.5.2 Control Volume Formulation

Control volume analysis permits consideration of systems in which mass crosses the boundary. This mass transfer carries the property entropy into and out of the volume. Therefore, the change in entropy within the control volume is modified by the mass transport. The control mass form of the second law states that the change in entropy within the boundary is related to the quantity $\delta Q/T$ at the boundary. Thus, this is the control volume statement of the second law:

The change in entropy within the control volume minus the net entropy transported into the control volume is greater than or equal to the sum of the heat transfers divided by the corresponding boundary absolute temperatures.

Let S denote the rate at which entropy is transported with the fluid at the inlets and outlets, where the overdot indicates the rate of flow of entropy. The entropy within the control volume is denoted by S_{CV} , so the second law is

$$\frac{dS_{CV}}{dt} - \dot{S}_{gen} \geq \sum_i \left(\frac{\dot{Q}_i}{T_i} \right)_{CV} \quad \dots\dots\dots (5.56)$$

Or, with the net rate of flow of entropy equal to the inlet minus the outlet contributions,

$$\frac{dS_{cv}}{dt} - \dot{S}_{in} + \dot{S}_{out} \geq \sum_i \left(\frac{\dot{Q}_i}{T_i} \right)_{cv} \quad \text{..... (5.57)}$$

The terms on the left-hand side of the inequality require detailed consideration. The mass transport contributions into and out of the control volume, denoted by $\dot{S}_{in}$ and $\dot{S}_{out}$, respectively, follow the development presented in the control volume formulation of conservation of mass and energy. The fluid element carries entropy across each boundary element, so the transported entropy is the mass flow rate multiplied by the entropy per unit mass, or

$$\dot{S}_{in} = (s\dot{m})_{in} \quad \text{..... (5.58)}$$

and

$$\dot{S}_{out} = (s\dot{m})_{out} \quad \text{..... (5.59)}$$

Thus the net transport is

$$\dot{S}_{net} = (s\dot{m})_{in} - (s\dot{m})_{out} \quad \text{..... (5.60)}$$

The control volume form of the second law is given as

$$\frac{dS_{cv}}{dt} - (s\dot{m})_{in} + (s\dot{m})_{out} \geq \sum_i \left(\frac{\dot{Q}_i}{T_i} \right)_{cv} \quad \text{..... (5.61)}$$

For multiple inlets and outlets, this is

$$\frac{dS_{cv}}{dt} - \sum_{in} (s\dot{m})_{in} + \sum_{out} (s\dot{m})_{out} - \sum_i \left(\frac{\dot{Q}_i}{T_i} \right)_{cv} \geq 0 \quad \text{..... (5.62)}$$

Equation (5.62) is the general control volume form of the second law. The first term is the entropy storage term within the control volume. The next two terms are the entropy transported by mass transport into and out of the control volume. The right-hand side represents the entropy transport via heat transfer. The equivalent expression corresponding to Equation (5.92) from using the entropy generation concept is

$$\dot{S}_{gen} = \frac{dS_{cv}}{dt} - (s\dot{m})_{in} + (s\dot{m})_{out} - \sum_i \left(\frac{\dot{Q}_i}{T_i} \right)_{cv} \geq 0 \quad \text{..... (5.63)}$$

For the case of no mass transport into or out of the control volume, these expressions reduce to

$$\frac{dS_{cv}}{dt} \geq \sum_i \left(\frac{\dot{Q}_i}{T_i} \right)_{cv} \quad \text{..... (5.64)}$$

or

$$\dot{S}_{gen} = \frac{dS_{cv}}{dt} - \sum_i \left(\frac{\dot{Q}_i}{T_i} \right)_{cv} \geq 0 \quad \dots\dots\dots (5.65)$$

These are the same expressions given in **Equations (5.50)** and **(5.51)**. If the system is isolated so that there are no interactions with the surroundings, the heat transfers are zero, and

$$\frac{dS_{cv}}{dt} \geq 0 \quad \dots\dots\dots (5.66)$$

or

$$\dot{S}_{gen} \geq 0 \quad \dots\dots\dots (5.67)$$

These expressions are equivalent to **Equations (5.7)** and **(5.8)**.

5.6 Isentropic Process for an Ideal Gas

An isentropic process is a constant-entropy process. If a control mass undergoes a process which is both reversible and adiabatic, then the second law [**Equation (5.44)**] specifies the entropy change to be zero. A steady state reversible flow through an adiabatic control volume also has no entropy change from inlet to outlet [**Equation (5.63)**]. Both examples are isentropic process. Although an isentropic process might be an idealization of an actual process, it serves as a limiting case, for particular applications.

The comparison between the isentropic processes and the adiabatic but irreversible processes are shown in **Figure 5.9**. The processes denoted by solid lines are the reversible processes, and the dashed lines indicate the irreversible processes. The irreversible processes proceed from initial state of the process to larger values of entropy. The process for state 1 to state 2 is an expansion process while the process from state 3 to state 4 is a compression process.

The entropy change for an ideal gas is presented in **Equations (5.25)** and **(5.26)**. For an isentropic process $s_2 - s_1 = 0$, so these expressions become

$$0 = \int_{T_1}^{T_2} c_v \frac{dT}{T} + R \ln \frac{v_2}{v_1} \quad \dots\dots\dots (5.68)$$

$$0 = \int_{T_1}^{T_2} c_p \frac{dT}{T} - R \ln \frac{P_2}{P_1} \quad \dots\dots\dots (5.69)$$

The final form used depends on the approximation made for the temperature dependence of the specific heats.

Assuming that the specific heats are accurately approximated by constant values eliminates the integrals in the above equations. The *constant-specific-heat* equations are

$$0 = c_v \ln \frac{T_2}{T_1} + R \ln \frac{v_2}{v_1} \quad \dots\dots\dots (5.70)$$

$$0 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1} \quad \dots\dots\dots (5.71)$$

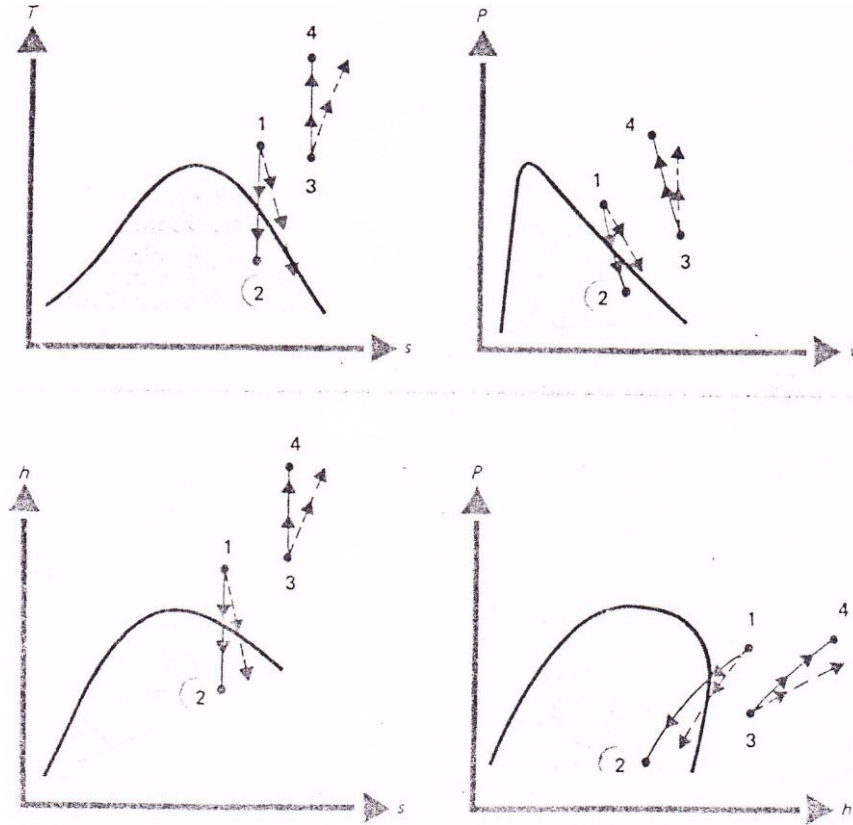


Figure 5.9: Process comparison

Dividing by the specific heats and taking the exponential of each expression yield

$$\left(\frac{T_2}{T_1}\right)_s = \left(\frac{v_2}{v_1}\right)_s^{-\frac{R}{c_v}} = \left(\frac{v_1}{v_2}\right)_s^{\frac{R}{c_v}} \quad \dots\dots\dots (5.72)$$

$$\left(\frac{T_2}{T_1}\right)_s = \left(\frac{P_2}{P_1}\right)_s^{\frac{R}{c_p}} \quad \dots\dots\dots (5.73)$$

where the subscript indicates that the process occurs at constant entropy. The power on each expression is rewritten in terms of $\gamma = c_p/c_v$ by noting that $R = c_p - c_v$, so $R/c_v = \gamma - 1$ and $R/c_p = (\gamma - 1)/\gamma$. Thus, Above Equations reduce to

$$\left(\frac{T_2}{T_1}\right)_s = \left(\frac{v_1}{v_2}\right)_s^{\gamma-1} \quad \dots\dots\dots (5.74)$$

$$\left(\frac{T_2}{T_1}\right)_s = \left(\frac{P_2}{P_1}\right)_s^{\frac{\gamma-1}{\gamma}} \quad \dots\dots\dots (5.75)$$

Isentropic Process for an Incompressible Fluid or Solid

The entropy change for an incompressible fluid or solid is given in **Equation (5.31)** as

$$ds = c \frac{dT}{T} \quad \dots\dots\dots (5.76)$$

For an isentropic process, $ds = 0$, so $dT = 0$. Thus an isentropic process is an isothermal process for an incompressible fluid or solid. Also the internal energy is given as

$$du = cdT \quad \dots\dots\dots (5.77)$$

So $du = 0$ for an isentropic process. The change in enthalpy is

$$dh = du + Pdv + vdP \quad \dots\dots\dots (5.78)$$

But $du = 0$ for this process and $dv = 0$ for an incompressible fluid or solid, so

$$dh = vdP \quad \dots\dots\dots (5.79)$$

This last expression is integrated to yield

$$h_2 - h_1 = v(P_2 - P_1) \quad \dots\dots\dots (5.80)$$

5.7 Cyclic Processes and Carnot Cycle

Processes that return to their initial state are called *cyclic processes*. The individual processes that make up the elements of the cyclic process vary, and they depend on the particular application. An ideal steam power cycle is composed of constant pressure heat transfer processes (to the working fluid in the steam generator and from the working fluid in the condenser) and adiabatic work processes (work addition by the pump and work output by turbine). The idealized spark-ignition engine is composed of adiabatic and constant volume processes. The fuel and air are compressed adiabatically, and the subsequent combustion is idealized as a constant volume heating. The hot gases expand adiabatically, doing work. Then the exhaust gases are cooled at constant volume. In these idealized examples, the processes are usually assumed to be reversible. These examples indicate that a cyclic process is composed of many different individual processes, and the combination depends on application.

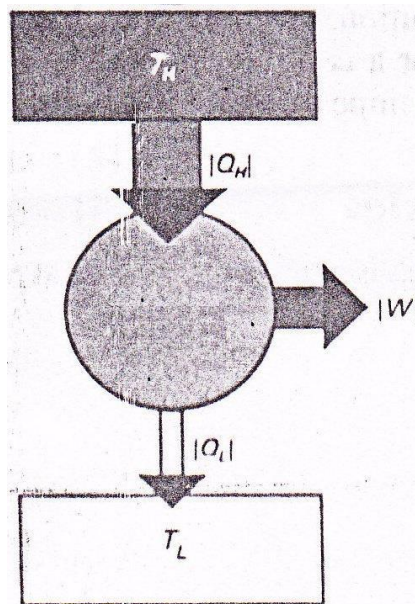


Figure 5.10: Engine Representation

These example cycles have a common feature – they operate between two limiting temperatures. The high temperature results from the combustion process in the steam generator or within the cylinder. The low temperature results from the cooling process. The characteristics of these two temperature cycles are shown from a general viewpoint as a high temperature heat transfer reservoir or source at T_H and a low temperature heat transfer reservoir or sink at T_L . The cycle operating between these two temperatures is arbitrary. The general power cycle or engine is shown schematically in **Figure 5.10**.

The first law for an arbitrary cycle states

$$\oint \delta W = \oint \delta Q \quad \text{..... (5.81)}$$

which is valid for an arbitrary collection of processes and for both reversible and irreversible cycles. For the two-heat transfer reservoir cycle shown in **Figure 5.10**,

$$W = Q_H - Q_L \quad \text{..... (5.82)}$$

The second law for a cycle [**Equation (5.44)**] states

$$\oint dS_{CM} = 0 \geq \oint \sum_i \left(\frac{\dot{Q}_i}{T_i} \right)_{CM} \quad \text{..... (5.83)}$$

where the zero results from the consideration of a cycle. **Equation (5.81)** and **Equation (5.83)** are general equations for cycles. These expressions lead to a very important statement about cycles operating between two heat transfer reservoirs. For reversible heat transfer with the two heat transfer reservoirs, the second law is

$$0 \geq \frac{Q_H}{T_H} - \frac{Q_L}{T_L} \quad \text{..... (5.84)}$$

or

$$\frac{Q_L}{Q_H} \geq \frac{T_L}{T_H} \quad \text{..... (5.85)}$$

This last expression could also be obtained from the entropy generation expression in **Equation (5.47)**.

The *efficiency* of a cycle η is defined as

$$\eta = \frac{\text{desired output}}{\text{required input}} \quad \text{..... (5.86)}$$

A power cycle or engine as diagrammed in **Figure 5.10** has a work output W and an input from the high-temperature reservoir Q_H . Thus, this efficiency is

$$\eta = \frac{W}{Q_H} \quad \text{..... (5.87)}$$

Equation (5.82) yields

$$\eta = 1 - \frac{Q_L}{Q_H} \quad \text{..... (5.88)}$$

The ratio of heat transfer is eliminated with **Equation (5.85)** to yield

$$\eta \leq 1 - \frac{T_L}{T_H} \quad \dots\dots\dots (5.89)$$

where the equality applies to a reversible cycle and the inequality applies to an irreversible cycle. Thus

$$\eta_{irrev} < \eta_{rev} = 1 - \frac{T_L}{T_H} \quad \dots\dots\dots (5.90)$$

or the maximum efficiency of a heat engine operating between two heat transfer reservoirs occurs for a reversible cycle. This statement applies regardless of the specific process of the cycle, but they must be reversible.

A very important reversible cycle that uses two heat transfer reservoirs is shown in **Figure 5.11a** and **5.11b**. The two specific examples undergo the same processes but operate with different components. The cycle includes isothermal energy exchange with the two heat transfer reservoirs, and the other two processes take place adiabatically. All processes are reversible. The heat transfer occurs across infinitesimal temperature differences, so the temperature within the system is the same as the source or sink. The other two processes are reversible and adiabatic, so they are isentropic. The state diagram in **Figure 5.11c** shows the following four processes:

- | | |
|--------|-----------------------------------|
| 1 to 2 | Isothermal heat transfer at T_H |
| 2 to 3 | Isentropic expansion process |
| 3 to 4 | Isothermal heat transfer at T_L |
| 4 to 1 | Isentropic compression process |

This cycle is called a *Carnot cycle*.

A control mass undergoing a Carnot cycle is shown in **Figure 5.11a**. There is a heat transfer to the control mass from state 1 to state 2 at T_H , and some expansion work is obtained. The process from state 2 to state 3 is an adiabatic expansion to temperature T_L . Then there is a heat transfer from the control mass from state 3 to state 4 which occurs reversibly at T_L . The final process takes the control mass adiabatically from state 4 back to state 1. An alternate situation is a power plant which includes four control volume elements with the working fluid undergoing the processes as it moves through the elements (**Figure 5.11b**). There is an isothermal heat transfer to the fluid at the high temperature in the boiler from state 1 to state 2. The fluid is expanded adiabatically through the turbine from state 2 to 3 to obtain a work output. The condenser is used for the isothermal heat transfer from state 3 to 4 at T_L . The adiabatic compression of the fluid to state 4 is achieved by the pump. Both types are represented by the general power cycle diagram shown in **Figure 5.10**.

The above analysis considers a power cycle or engine used to produce a net output of work. A refrigerator is a cyclic device that seeks a heat transfer from a low temperature region and requires work input. This device is shown schematically in **Figure 5.12**. The performance for a refrigerator is termed as the *coefficient of performance* and is given as

$$COP_R = \frac{Q_L}{W} \quad \dots\dots\dots (5.91)$$

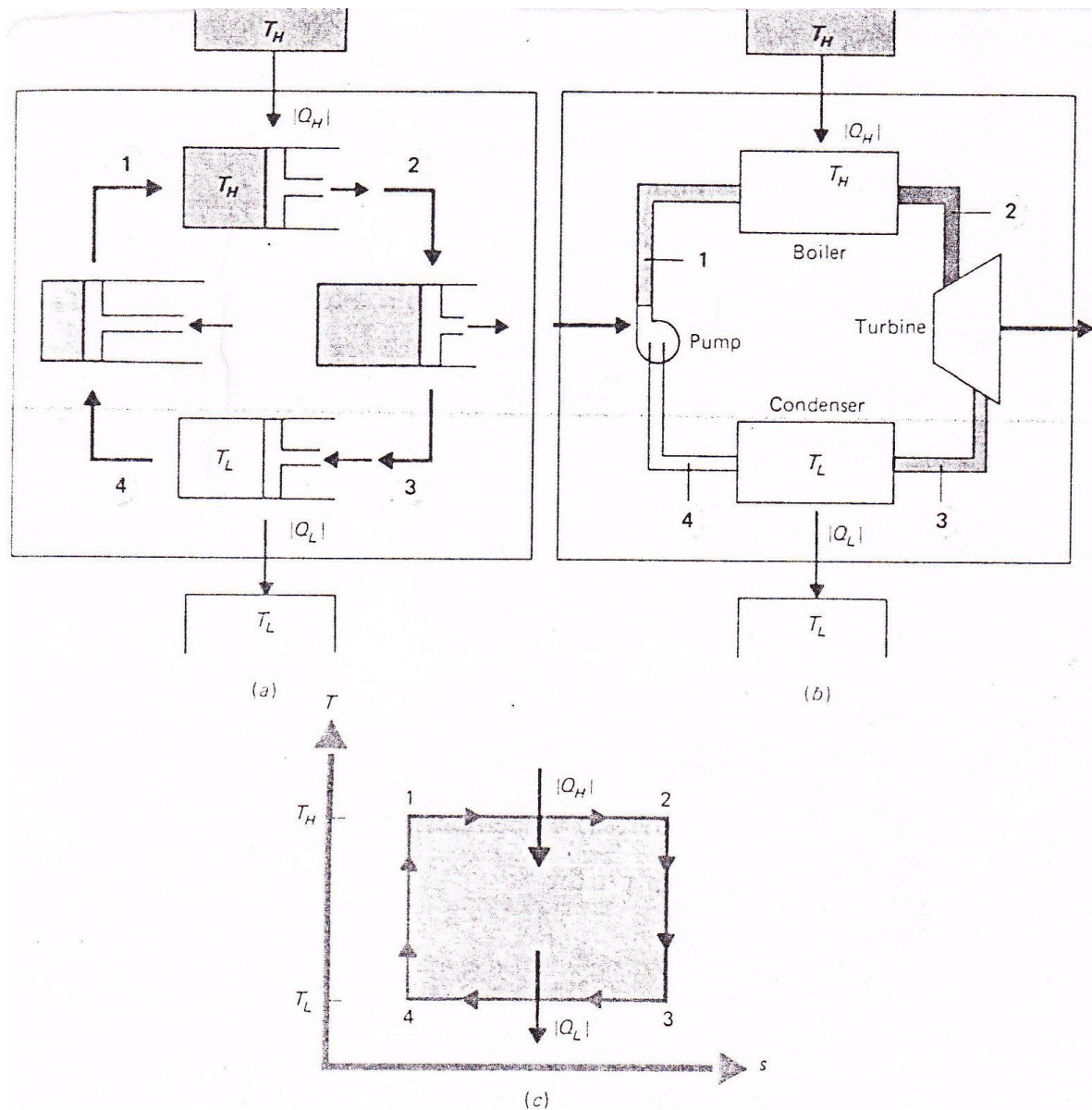


Figure 5.11: Carnot Cycle

The first law states that

$$-W = Q_L - Q_H \quad \dots\dots\dots (5.92)$$

or

$$COP_R = \frac{Q_L}{Q_H - Q_L} \quad \dots\dots\dots (5.93)$$

The second law yields

$$0 \geq \frac{Q_L}{T_L} - \frac{Q_H}{T_H} \quad \dots\dots\dots (5.94)$$

or

$$\frac{Q_H}{Q_L} \geq \frac{T_H}{T_L} \quad \dots\dots\dots (5.95)$$

Thus

$$COP_R \leq \frac{1}{\frac{T_H}{T_L} - 1} \quad \dots\dots\dots (5.96)$$

The equality applies to the reversible refrigerator and the inequality to the irreversible refrigerator. Thus

$$COP_{R,irrev} < COP_{R,rev} = \frac{1}{\frac{T_H}{T_L} - 1} \quad \dots\dots\dots (5.97)$$

The COP theoretically ranges from 0 to large values > 1 .

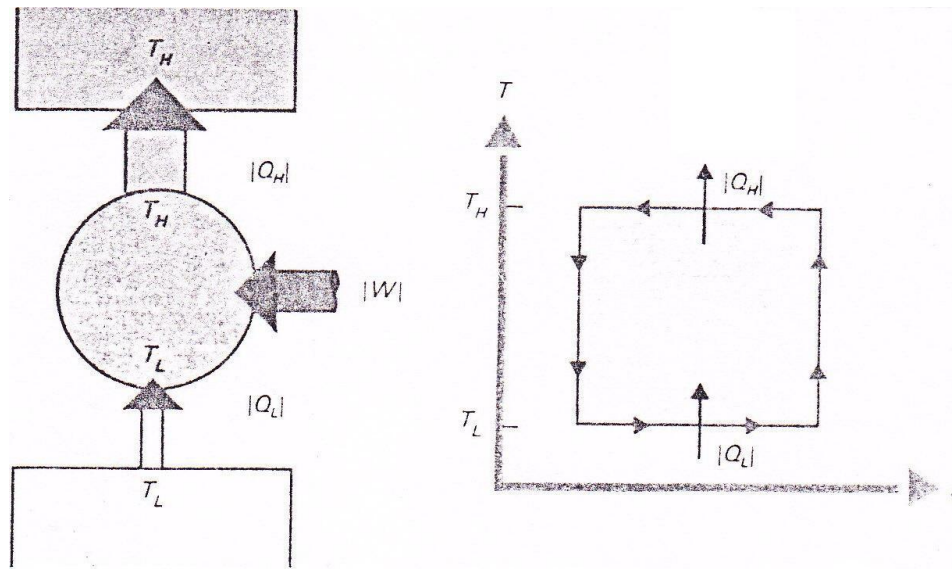


Figure 5.12: Refrigerator representation

A heat pump (HP) operates as indicated in **Figure 5.12**, but it seeks a heat transfer to a high-temperature region so the desired output is Q_H . The above analysis yields

$$COP_{HP} = \frac{Q_H}{W} = \frac{Q_H}{Q_H - Q_L} = \frac{1}{1 - \frac{Q_L}{Q_H}} \leq \frac{1}{1 - \frac{T_L}{T_H}} \quad \dots\dots\dots (5.98)$$

where the equality sign applies to the reversible cycle and the inequality applies to the irreversible cycle. Therefore

$$COP_{HP,irrev} < COP_{HP,rev} = \frac{1}{1 - \frac{T_L}{T_H}} \quad \dots\dots\dots (5.99)$$

Comparing **Equations (5.93)** and **(5.98)** gives

$$COP_{HP} - COP_R = 1 \quad \dots\dots\dots (5.100)$$

5.8 Other Statements of Second Law of Thermodynamics

Some of the classical statements of the second law of thermodynamics are given below.

The first statement is popularly known as Kelvin-Planck statement of the second law of thermodynamics. The statement is:

It is impossible to construct a heat engine to work in a cyclic process whose sole effect is to convert all the heat supplied to it into an equivalent amount of work.

The other statement of the second law of thermodynamics known as Clausius statement is given below:

It is impossible to construct a device to work in a cyclic process whose sole effect is the transfer of heat from a body at a lower temperature to a body at a higher temperature.

Another form of the statement is "It is impossible for heat energy to flow from a body at a lower temperature to a body at a higher temperature without the aid of an external work".

Both the statements are shown diagrammatically in **Figure 5.13**. The two statements of the Second Law lead to same conclusions even though they appear different. It is impossible to have a system which satisfies one of the statements and violates the other.

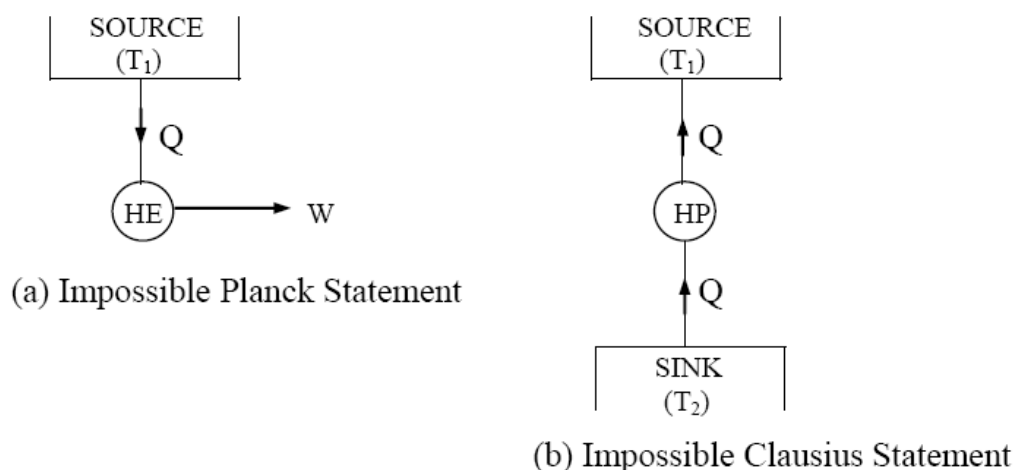


Figure 5.13: Models of Second Law Statements

Consider a system as shown in **Figure 5.14(a)**. It is obvious that, both the heat engine and heat pump obey the first law of thermodynamics but the pump obeys Clausius statement and engine violates Planck statement. If we consider this system as a whole, we find that Q_2 units of heat are continuously transferred from the heat sink at T_2 to heat source at T_1 ($T_1 > T_2$) without any external input to the system, which is against Clausius statement. Thus a violation of Kelvin-Planck statement leads to the violation of Clausius statement.

Considering the system shown in **Figure 5.14(b)** the heat pump violates the Clausius statement while the engine satisfies the Kelvin-Planck statement. But taken together the net effect is that an amount of heat ($Q_1 - Q_2$) is taken from the source at T_1 and is being completely and continuously converted into work (as no net exchange takes place with the sink) and this is against Kelvin-Planck statement. Thus a violation of Clausius statement leads to the violation of Kelvin-Planck statement.

Therefore it is obvious that the two statements are complementary to each other. The truth of the first implies the truth of the second and *vice-versa*.

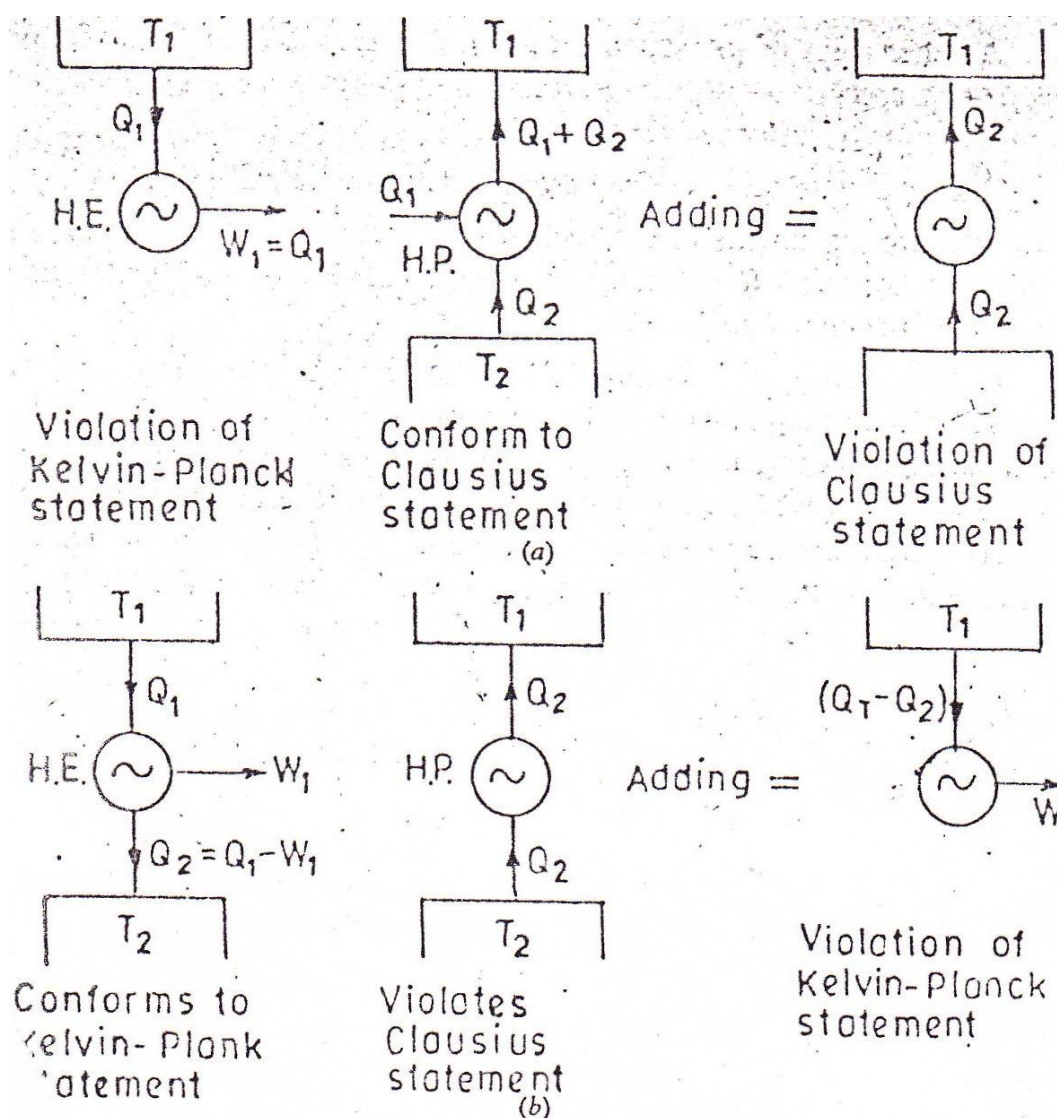


Figure 5.15: Equivalence of Kelvin-Planck and Clausius statements

CHAPTER 6

THERMODYNAMIC CYCLES

6.1 Introduction

Heat engine is a device that converts heat transfer into mechanical work. If a series of individual processes that starts at some initial state and return to the initial state, combination of such processes is known as a cyclic process.

Based on working fluid heat engine cycles can be categorized into the following:

- (a) Heat engines that use working fluid which do not change phase during the whole cycle (Gas powered engines).
- (b) Heat engines that uses working fluid which changes phase during cycle (Vapor powered engines).

Based on energy sources used in heat engines, they can be categorized as:

- (a) External energy source (burning of fossil fuels in a boiler).
- (b) Internal combustion energy (burning of fossil fuel within engine itself).

The Carnot cycle is an ideal heat engine that is free from all the defects of practical engines. Such ideal cycle was developed by Sadi Carnot, a French engineer. The Carnot cycle consists of two isothermal heat transfer (one at higher temperature another at lower temperature) and one isentropic expansion and one isentropic compression processes.

There are some practical difficulties in constructing heat engines that operates on Carnot cycle. For instance, in order to design heat exchanger that allow heat transfer at constant temperature of working fluid in a reversible process will require extremely large surface area of heat exchanger. If the processes have to be reversible the temperature difference between working fluid and surroundings (source and sink) should be infinitesimally small, which means the time required for heat transfer will be extremely long. On the other hand the process in compression and expansion to be adiabatic and reversible i.e., isentropic, which is not possible in practice.

Although the Carnot cycle is not practical for implementation, it is useful for determining the theoretical (ideal) behavior of heat engine.

For any engine, following three items are essential:

- | | |
|---------------------------|---|
| Source: | The source have constant temperature usually temperature at higher level (T_H) from which heat can be added to heat engine. It should have infinite thermal capacity. |
| Sink: | The sink should have constant temperature at lower level (T_L) to which heat can be rejected from the heat engine. It also should have infinite thermal capacity. |
| Working Substance: | In order transfer heat from source to sink and produce useful work a working substance is required. |

As Carnot cycle is not possible to use as heat engine in practice, others cycles have been developed. Most of these cycles operate with nonisothermal heat transfer. However they have higher thermodynamic efficiencies in practice compared to the engines with near-isothermal heat transfer.

The other reasons for developing heat engines that do not operate on the Carnot cycle are as follows:

- Characteristics of the energy source available.
- Characteristics of the working fluid chosen.
- Limitation in fabricating hardware.
- Requirement for steady and varying work output.

In order to analyze cycles, we should know the states of the working fluid at the beginning and end of each process of a cycle. One standard measure of performance for heat engines is the efficiency.

Efficiency

It is a ratio of the net work output (desired output) per cycle to the heat transfer (required input) into the engine per cycle. It is denoted by η and generally expressed in percentage.

$$\eta = \frac{\text{desired output}}{\text{required input}} = \frac{w_{net}}{q_{in}} \quad \dots\dots\dots (6.1)$$

Mean Effective Pressure

It is the constant pressure which, if exerted through the work-producing expansion stroke of an engine between the largest specific volume v_f and the smallest volume v_i gives the same net specific work for the entire cycle as produced by the actual engine.

$$w_{net} = P_{MEP}(v_f - v_i) = P_{MEP}v_i\left(\frac{v_f}{v_i} - 1\right) = P_{MEP}v_i(r_v - 1) \quad \dots\dots\dots (6.2)$$

where $r_v = v_f/v_i$ is called volumetric expansion ratio for the engine. Then expression for mean effective pressure becomes

$$P_{MEP} = \frac{w_{net}}{v_i(r_v - 1)} = \frac{q_{in}\eta}{v_i(r_v - 1)} \quad \dots\dots\dots (6.3)$$

6.2 External Combustion Brayton Cycle

Originally the Brayton cycle also known as the Joule cycle was developed in order to use in a piston engine with fuel injection. However, nowadays it is widely used in open cycle as well as closed-cycle turbine engines.

The working fluid of closed Brayton cycle is air. This cycle consists of two constant pressure (isobaric) heat transfer processes and two isentropic processes (one is isentropic expansion and another is isentropic compression) as shown in **Figure 6.1 (a), (b) and (c)**.

Note that the heat transfer is no longer isothermal process as in Carnot cycle. Thus the efficiency of closed-cycle air-standard Brayton engine will be less than ideal Carnot cycle efficiency.

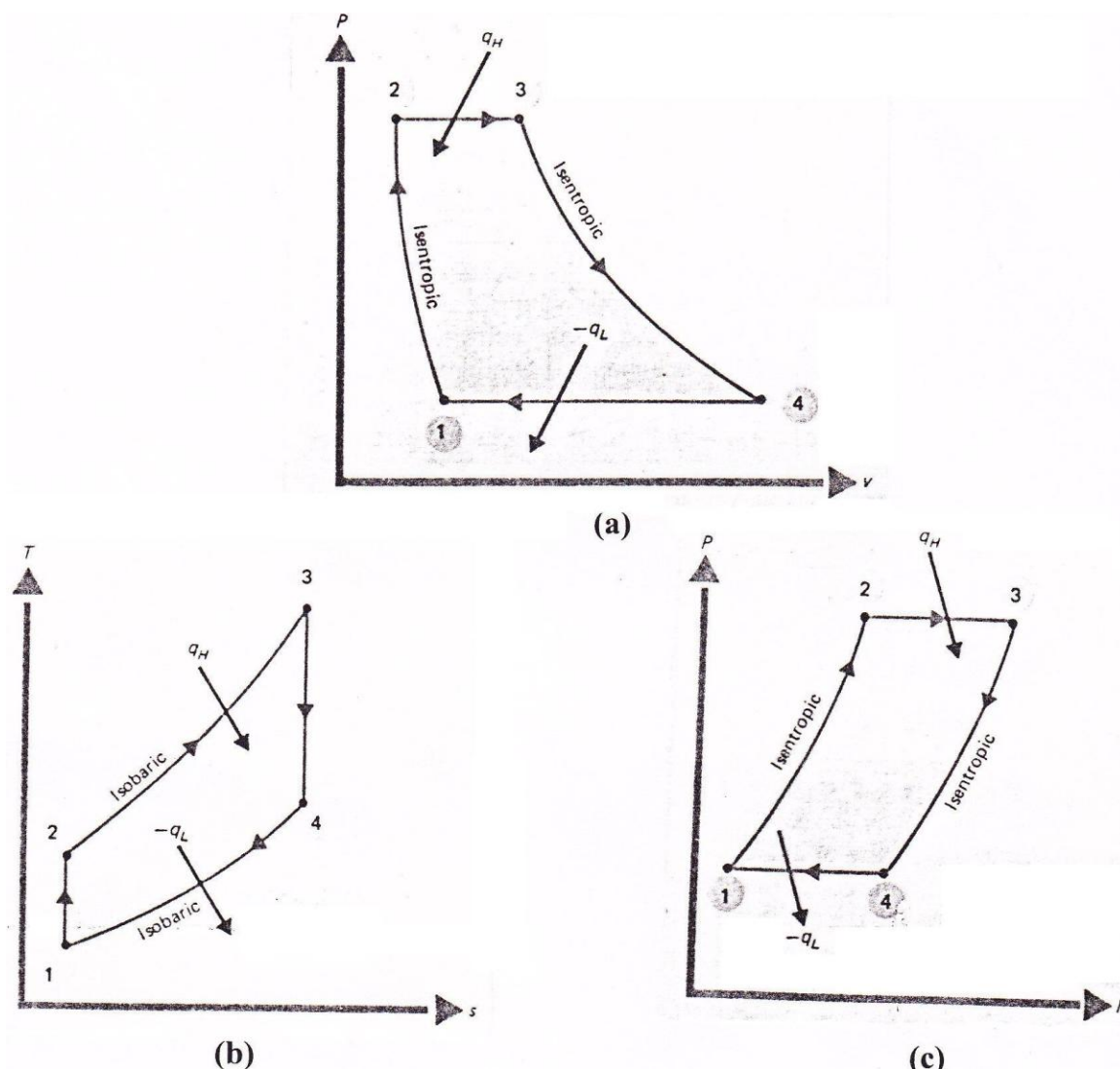


Figure 6.1: $P-v$, $T-s$ and $P-h$ diagrams for Brayton cycle

The hardware of closed-cycle Brayton engine is shown in **Figure 6.2**. The working fluid air enters the compressor in state 1, where it is compressed isentropically until state 2 is reached and enters high temperature heat exchanger. In this heat exchanger heat will be added to the fluid at constant pressure until state 3 is reached. Now high temperature air enters the turbine, where an isentropic expansion occurs, producing mechanical work. The working fluid (air) leaves the turbine at state 4 and enters low temperature heat exchanger, where heat will be rejected from the fluid until state 1 is reached. After completing a cycle previous processes will be repeated in same order.

Compression in compressor and expansion in turbine are assumed to be isentropic processes in the ideal closed cycle air standard Brayton cycle. It is easier to construct compressor and turbine which operates nearly adiabatic. However, it is difficult to approach reversibility.

Analysis of Closed Brayton Cycle

Work done on the compressor is given by

$$w_C = c_p(T_2 - T_1) \quad \dots\dots\dots (6.4)$$

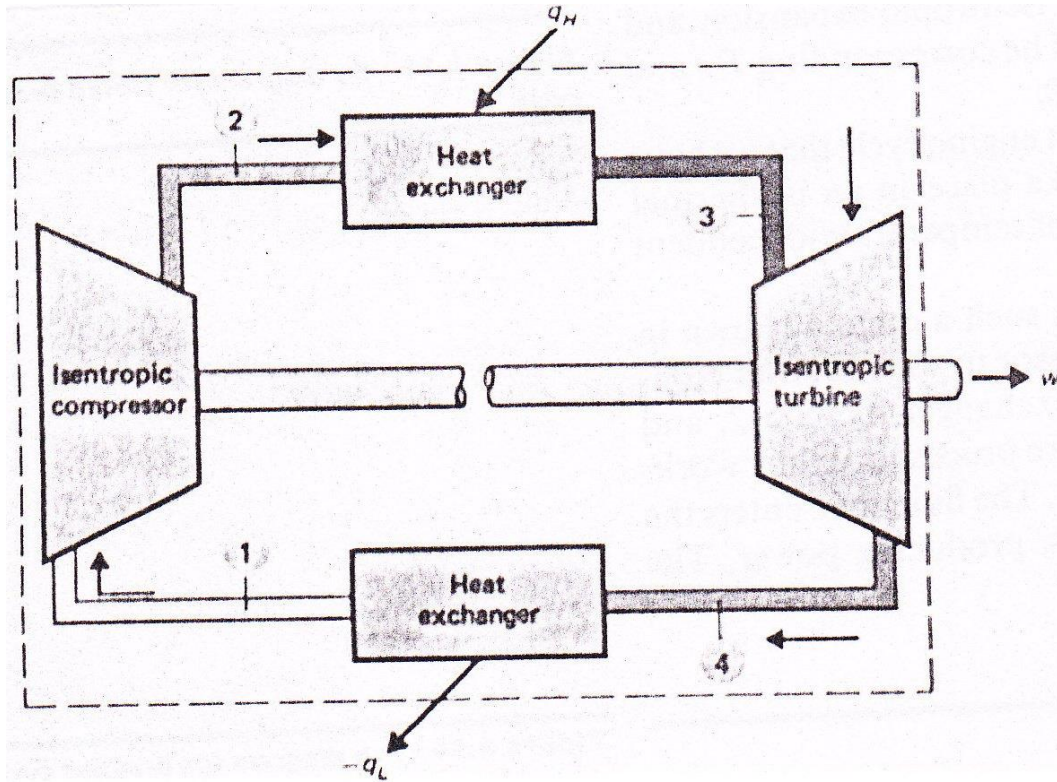


Figure 6.2: Diagram of simple Brayton cycle hardware

Work done by the turbine is given by

$$w_T = c_p(T_3 - T_4) \quad \dots\dots\dots (6.5)$$

Thus, net work available is

$$w_{net} = w_T - w_C = c_p[(T_3 - T_4) - (T_2 - T_1)] \quad \dots\dots\dots (6.6)$$

Heat supplied to the fluid during the cycle is

$$q_H = q_{23} = h_3 - h_2 = c_p(T_3 - T_2) \quad \dots\dots\dots (6.7)$$

The thermodynamic efficiency of the cycle is given by

$$\eta = \frac{w_{net}}{q_H} = \frac{c_p[(T_3 - T_4) - (T_2 - T_1)]}{c_p(T_3 - T_2)} = 1 - \frac{T_4 - T_1}{T_3 - T_2} \quad \dots\dots\dots (6.8)$$

For the isentropic processes 1-2 and 3-4, pressure temperature relationships are

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} \quad \dots\dots\dots (6.9)$$

and

$$\frac{T_3}{T_4} = \left(\frac{P_3}{P_4}\right)^{\frac{\gamma-1}{\gamma}} \quad \dots\dots\dots (6.10)$$

Since $P_2 = P_3$ and $P_4 = P_1$

$$\frac{T_2}{T_1} = \frac{T_3}{T_4} \quad \text{or} \quad \frac{T_4}{T_3} = \frac{T_1}{T_2} = \frac{T_4 - T_1}{T_3 - T_2} \quad \dots\dots\dots (6.11)$$

Substituting the ratios of **Equation (6.11)** in **Equation (6.8)**,

$$\eta = 1 - \frac{T_1}{T_2} = 1 - \frac{T_4}{T_3} \quad \dots\dots\dots (6.12)$$

Let $r_p = \frac{P_2}{P_1} = \text{pressure ratio}$, then

$$\frac{T_2}{T_1} = \frac{T_3}{T_4} = (r_p)^{\frac{\gamma-1}{\gamma}} \quad \dots\dots\dots (6.13)$$

Hence, efficiency equation reduces to

$$\eta = 1 - \left(\frac{1}{r_p}\right)^{\frac{\gamma-1}{\gamma}} \quad \dots\dots\dots (6.14)$$

The efficiency of the closed cycle Brayton engine is found to depend on the isentropic pressure ratio only. If we increase the inlet pressure to the turbine, the inlet temperature to the turbine is also increased. The inlet turbine temperature is usually limited by material properties of the turbine blades, which places a practical upper limit on the cycle efficiency.

6.3 Internal Combustion Cycle

In many power producing devices, such as the automotive gasoline engine, the diesel engine, and the gas turbine, the working fluid is a gas. Broadly speaking, these devices take in either mixture of fuel and air or fuel and air separately, compress this charge to a high pressure and cause the fuel to burn. A part of the energy liberated as heat as a result of combustion is converted into useful work by causing the high pressure and temperature products of combustion to expand in the engine or turbine while the remainder is carried away with the exhaust gas leaving the device.

In general, the mass of the fuel per unit mass of air i.e. *fuel-air ratio*, is very small (0.08 to 0.025) and therefore the properties of the fuel air mixture before combustion approximates closely to those of air. The properties of the products of combustion also do not differ much from those of pure air. For this reason, it is convenient to analyze the performance of these devices by devising the idealized cycles known as *air standard cycles*.

The air-standard cycles are based on the following hypothetical assumptions:

1. The working substance consists of a fixed mass of air. This *closed system* undergoes a cycle of processes so that the system is restored to its initial state at the end of each cycle.
2. The combustion process is replaced by an equivalent heat addition process from an external source. Thus there is *no change* in the *chemical composition* of the working fluid.
3. The exhaust process is replaced by an *equivalent heat rejection process*.

4. Compression and expansion processes in the cycle are *reversible adiabatic* processes.
5. The *specific heats* C_p and C_v of air do not vary with temperature.

Because of the above assumptions the air cycle approximation represents the limit towards which an actual engine may aim. The main use of the ideal cycle is that it is amenable to simple mathematical analysis based on perfect gas laws and can give rough answer to complex engine problems. The analysis can also indicate the relative effect of the principal variables such as compression ratio, inlet pressure, inlet temperature, etc.

6.3.1 Air-standard Otto Cycle

In previous heat engine cycles, we have thought of the cycle comprised of different processes being carried out in a separate component. The turbine has been regarded as power producing component. The efficiency of turbine decreases rapidly as the size decreases. Therefore, it is only feasible to use turbine power plant, if the power required is at least several hundred kilowatts.

For small power, it is usually preferable to use a cycle comprised of series of non-steady flow processes. A given mass of working fluid can be taken through desired processes in a cylinder fitted with reciprocating piston. Such reciprocating engines are preferable for comparatively low power output. For high flow rate i.e., high output such engine will be excessively heavy and bulky.

The air-standard Otto cycle is an idealized cycle used in spark-ignition gasoline engines. Basically, Otto cycle consists of two isentropic processes and two isochoric (constant volume) processes as shown in **Figure 6.3**. The mixture of air and fuel at state 1 enters the cylinder and the piston begins the compression stroke at constant entropy until the piston reaches the top dead position at state 2. The spark plug ignites and the combustion of fuel will add energy at constant volume resulting increase in pressure till state 3 is reached. Now the piston begins the power stroke.

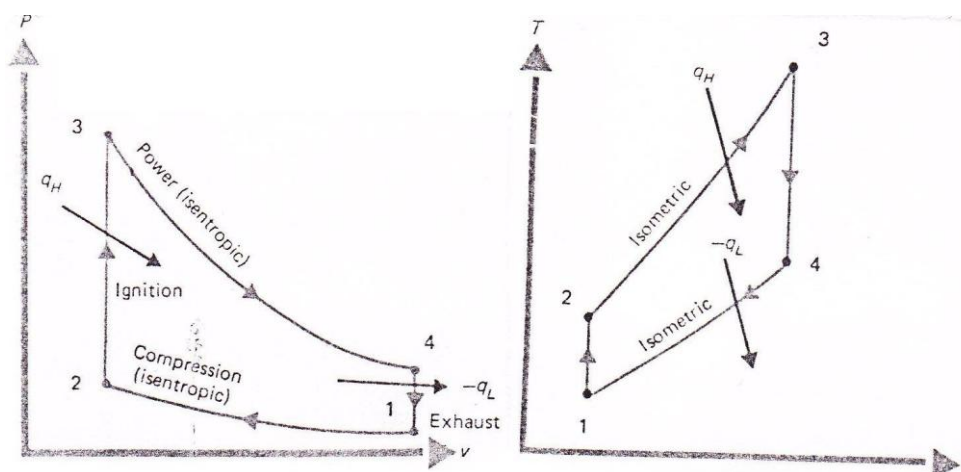


Figure 6.3: $P - v$ and $T - s$ diagrams for Otto cycle

The expansion of working fluid is also an isentropic process, which continues till state 4 (bottom dead position is reached). Then the exhaust valve opens, reducing the pressure instantaneously till state 1 is reached at constant volume. It is a heat rejection at constant volume. Then again a mixture of fuel air will enter the cylinder for next cycle and the cycle is repeated.

The efficiency of the ideal air-standard Otto cycle is easily found by

$$\eta = 1 - \frac{q_L}{q_H} = 1 - \frac{c_v(T_4 - T_1)}{c_v(T_3 - T_2)} = 1 - \frac{T_4 - T_1}{T_3 - T_2} \quad \dots\dots\dots (6.15)$$

For isentropic processes,

$$\frac{T_4}{T_3} = \left(\frac{v_3}{v_4}\right)^{\gamma-1} = \left(\frac{v_2}{v_1}\right)^{\gamma-1} = \frac{T_1}{T_2} \quad \dots\dots\dots (6.16)$$

and substituting the resulting ratio of temperatures into the efficiency relation gives

$$\eta = 1 - \frac{T_4}{T_3} = 1 - \frac{T_1}{T_2} = 1 - \left(\frac{v_2}{v_1}\right)^{\gamma-1} = 1 - \left(\frac{1}{r}\right)^{\gamma-1} \quad \dots\dots\dots (6.17)$$

where $r = \frac{v_1}{v_2}$ is the compression ratio.

The efficiency of the ideal air-standard Otto cycle is thus dependent on only the compression ratio of the engine.

Mean Effective Pressure for Otto Cycle

As a matter of fact, pressure in the cylinder keeps on changing with the position of the piston. For all sorts of calculations, we need the mean effective pressure. Mean effective pressure is defined as the constant net pressure acting on the piston that will produce same amount of work as that produced by the actual varying pressure produces during the cycle. It is the ratio of work done to the stroke volume or piston displacement volume. Mathematically,

$$MEP = \frac{\text{work done}}{\text{stroke volume}} = \frac{w_{net}}{v_1 - v_2} = \frac{\eta q_H}{v_1 - v_2} \quad \dots\dots\dots (6.18)$$

Improving the Performance of Otto Cycle Engines

Improving the efficiency of an air-standard Otto cycle engine depends chiefly on increasing the compression ratio according to **Equation (6.17)**.

Changes in the compression ratio can be examined in terms of the piston diameter D_P , the length of the piston stroke L , and the clearance volume V_C remaining in the cylinder when the piston is at top dead center as shown in **Figure 6.4**. The compression ratio is given in terms of these variables by

$$r = \frac{V_1}{V_2} = \frac{\frac{\pi}{4} D_P^2 L + V_C}{V_C} = 1 + \frac{\pi D_P^2 L}{4 V_C} \quad \dots\dots\dots (6.19)$$

which shows that increasing the piston diameter or piston stroke, or decreasing the clearance volume, will increase the compression ratio.

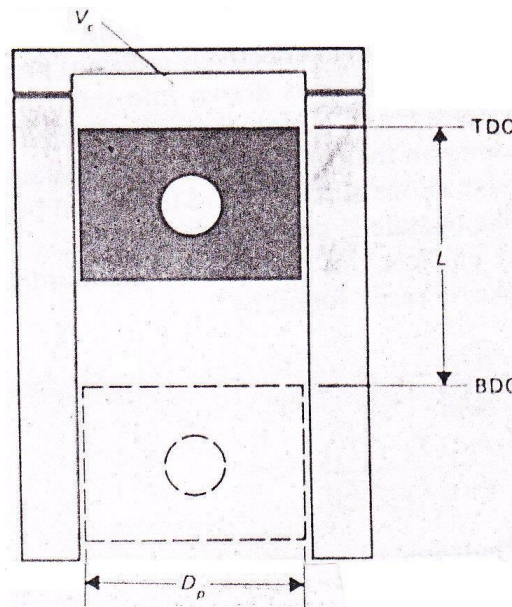


Figure 6.4: Compression ratio quantities

6.3.2 Air-standard Diesel Cycle

The purpose of the Diesel cycle is to get higher compression ratio using air at the start of the cycle. The $P - v$ and $T - s$ diagrams are shown in Figure 6.5.

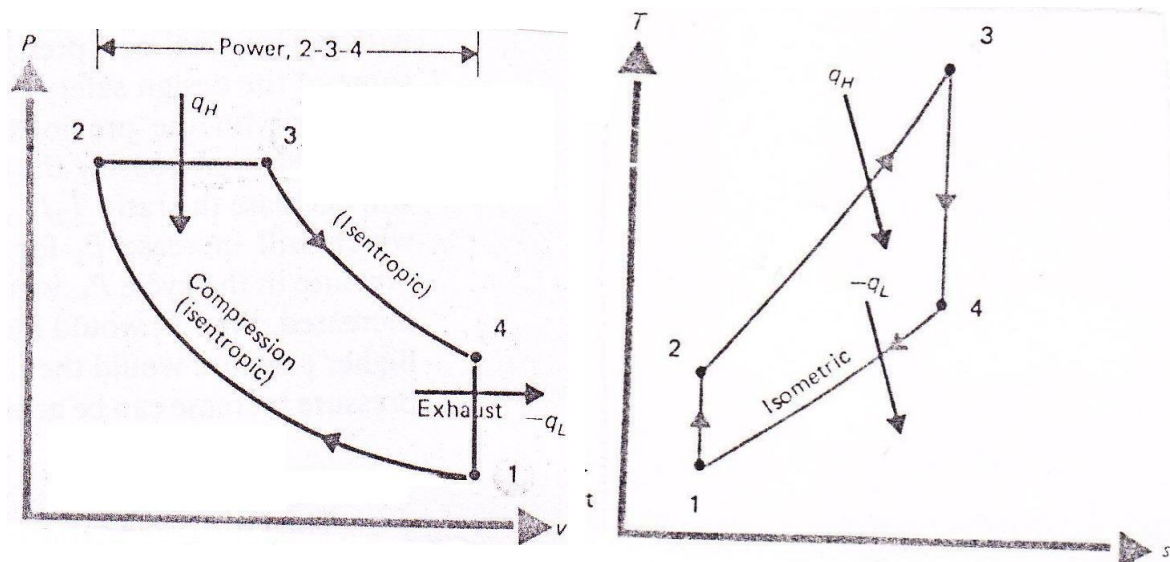


Figure 6.5: $P - v$ and $T - s$ diagrams for Diesel Cycle

Basically, the diesel cycle consists of two isentropic processes, one isobaric and one isochoric process. Pure air at state 1 enters the cylinder and the piston begins the compression stroke at constant entropy until the piston reaches the top dead position at state 2. As the compression ratio is quite high, the temperature of air at state 2 will be very high. At state 2, fuel is injected into the air. Because of the high temperature of the air, fuel will ignite spontaneously without any external source of ignition. The combustion of fuel will add the energy to expand the working fluid in order to move the piston at constant pressure, until state 3 is reached. Then working fluid will expand

isentropically, until state 4 is reached. At the end of this process, the exhaust valve opens and the pressure drops to the initial condition at state 1. The cycle will then repeat.

The diesel cycle efficiency is

$$\eta = 1 - \frac{q_L}{q_H} = 1 - \frac{c_v(T_4 - T_1)}{c_p(T_3 - T_2)} \quad \dots\dots\dots (6.20)$$

For the ideal air-standard diesel cycle assuming c_p and c_v are not temperature-dependent, and substituting $\gamma = c_p/c_v$, and rearranging them

$$\eta = 1 - \frac{1}{\gamma} \frac{T_1}{T_2} \left[\frac{T_4/T_1 - 1}{T_3/T_2 - 1} \right] \quad \dots\dots\dots (6.21)$$

If the compression ratio v_1/v_2 is denoted by r and the expansion ratio v_4/v_3 is denoted by r_e , then

$$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2} \right)^{\gamma-1} = (r)^{\gamma-1} \quad \text{and} \quad \frac{T_3}{T_4} = \left(\frac{v_4}{v_3} \right)^{\gamma-1} = (r_e)^{\gamma-1} \quad \dots\dots\dots (6.22)$$

For constant pressure process 2-3

$$\frac{T_3}{T_2} = \frac{v_3}{v_2} = \frac{v_4}{v_2} \frac{v_3}{v_4} = \frac{v_1}{v_2} \frac{v_3}{v_4} = \frac{r}{r_e} \quad \dots\dots\dots (6.23)$$

Then

$$\frac{T_4}{T_1} = \frac{T_4}{T_3} \frac{T_3}{T_2} \frac{T_2}{T_1} = \left(\frac{1}{r_e} \right)^{\gamma-1} \left(\frac{r}{r_e} \right) (r)^{\gamma-1} \quad \dots\dots\dots (6.24)$$

Now, substituting above ratios in **Equation (6.21)**, then the diesel ideal air-standard efficiency can be rewritten as

$$\eta = 1 - \frac{1}{\gamma} \left(\frac{1}{r} \right)^{\gamma-1} \frac{(r/r_e)^{\gamma} - 1}{(r/r_e) - 1} = 1 - \frac{1}{\gamma} \left(\frac{1}{r} \right)^{\gamma-1} \left[\frac{\alpha^{\gamma} - 1}{\alpha - 1} \right] \quad \dots\dots\dots (6.25)$$

where $\alpha = r/r_e = v_3/v_2$ is called the cutoff ratio.

The compression ratio of the diesel cycle is always greater than the expansion ratio. So for a given compression ratio r_1 , the diesel engine always has lower efficiency than an Otto engine operating at the same compression ratio. The closer r_e is made r , the closer diesel efficiency will approach that of Otto engine.

6.4 Rankine Cycle

The simple Rankine cycle is the common cycle used in all steam power plants. The Rankine cycle was devised to make use of the characteristics of water as working fluid and to handle the phase change between liquid and vapor. Of course, many other materials can be chosen instead of water as working fluids. The choice depends on many factors, including the need to accommodate the temperatures of heat transfer to and from the vapor and liquid states while maintaining low vapor pressures in the system.

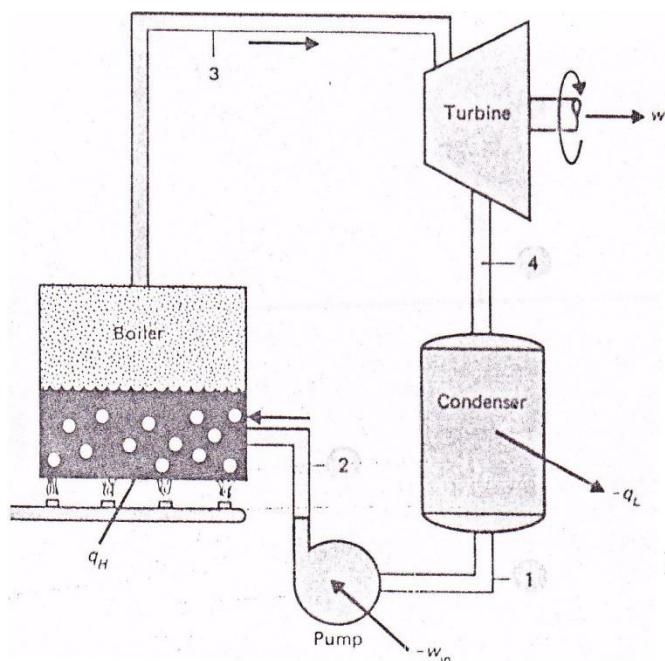
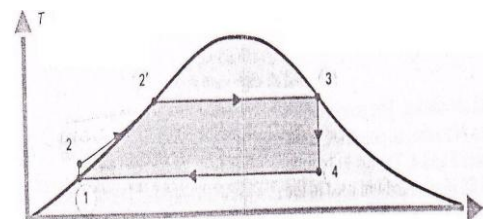
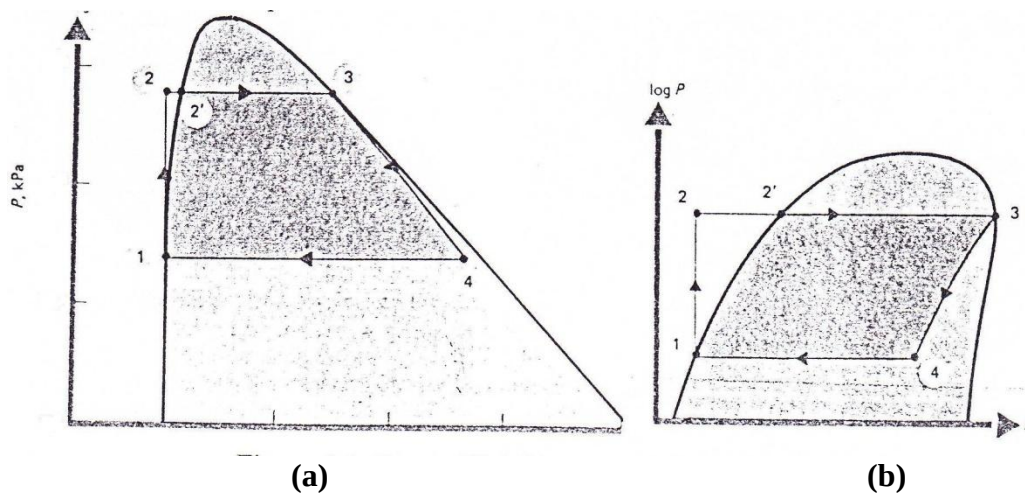


Figure 6.6: Idealized simple Rankine cycle

Figure 6.7: $T - s$ diagram

In an idealized simple Rankine cycle as shown in **Figure 6.6**, heat transfer takes place to the working fluid in a constant-pressure process in the boiler. The $T - s$ diagram for the cycle is shown in **Figure 6.7**. Liquid enters the pump and is increased in pressure to state 2. It enters the boiler from the pump at a relatively low temperature (state 2) and is heated to saturation along line 2-2'. The liquid experiences a slight increase in volume owing to thermal expansion. At point 2', the saturation temperature is reached. Since the boiler is operated at effectively constant pressure, further heat transfer takes place at constant temperature. And the energy added to the working fluid goes into the heat of vaporization, producing vapor until a quality of 100 percent is reached at point 3. Then the vapor is expanded through a turbine to produce work and leaves at state 4. The expansion is limited in the practical cycle by the saturation pressure available at the temperature of the cooling medium used in the condenser. If excessive condensation is allowed to occur, the condensed liquid droplets will rapidly erode the turbine blades.

Figure 6.8: $P - v$ and $P - h$ diagrams for ideal Rankine cycle

At the end of the expansion process (state 4), low-pressure but fairly high-quality vapor leaves the turbine. The vapor is then condensed to a liquid as the vapor comes into contact with surfaces in the condenser that are cooled, usually by a cold water stream. Because the condenser operates near the temperature of the cooling water, the condensation process occurs at temperatures well below the normal (atmospheric) boiling point of most working fluids. And the condenser pressure on the working fluid side is, therefore, often subatmospheric. The liquid leaves the condenser at state 1.

Following condensation, the liquid enters a pump. The working fluid is returned to the high pressure necessary for energy addition at the higher boiler temperature, and the cycle is repeated.

Note that the $P - v$ diagram used to illustrate the Rankine cycle is a log-log plot. This is necessary to illustrate the cycle because the specific volume ranges over four decades over the pressure range of interest in steam cycles. The $P - v$ and $P - h$ diagrams for the cycle are shown in **Figure 6.8(a)** and **(b)** respectively.

Analysis of Rankine Cycle

For the expansion process 3-4, work produced by the turbine is

$$w_T = h_3 - h_4 \quad \dots\dots\dots (6.26)$$

For the compression process 1-2, work absorbed by the pump is

$$w_P = h_2 - h_1 \quad \dots\dots\dots (6.27)$$

For the process 2-3, the heat supplied to the fluid in the boiler is

$$q_H = h_3 - h_2 \quad \dots\dots\dots (6.28)$$

The efficiency of the Rankine cycle is then given by

$$\eta = \frac{w_{net}}{q_H} = \frac{w_T - w_P}{q_H} = \frac{(h_3 - h_4) - (h_2 - h_1)}{h_3 - h_2} \quad \dots\dots\dots (6.29)$$

Usually, the pump work is quite small compared to the turbine work and is sometimes neglected. The $h_1 \approx h_2$, and the cycle efficiency approximately becomes

$$\eta \cong \frac{h_3 - h_4}{h_3 - h_2} \quad \dots\dots\dots (6.30)$$

The idealized Rankine cycles approach the Carnot efficiency rather closely. This is not true for actual engines that operate on the Rankine cycle. For real engines, many irreversibilities are present, chiefly in turbine and pump.

6.5 Vapor Compression refrigeration Cycle

The most common method of providing air conditioning and chilling as well as heat pumping is probably by the vapor compression cycle. The cycle hardware is shown in **Figure 6.9**, and the $P - h$ and $T - s$ diagrams are shown in **Figure 6.10 (a)** and **(b)**. In this cycle, the working fluid is initially a saturated or slightly superheated vapor at a relatively low pressure (state 1), and it is compressed to a high pressure (state 2), where it

is condensed to a saturated or slightly subcooled liquid (state 3). Condensation may occur either by heat loss to the surroundings, if the saturation temperature of the condensing vapor is sufficiently greater than that of the surroundings, or by the use of a cooling water to provide a lower-temperature heat sink. The condensed liquid is then expanded adiabatically through a throttling valve to a low pressure (state 4), where it becomes a mixture of saturated liquid and vapor of low quality. Its saturation temperature at this pressure is low, and a heat transfer from the surroundings can be used to evaporate the liquid fraction, thus providing the cooling effect and leaving the working fluid at initial state 1. The resulting saturated or slightly superheated vapor is fed to the compressor, completing the cycle.

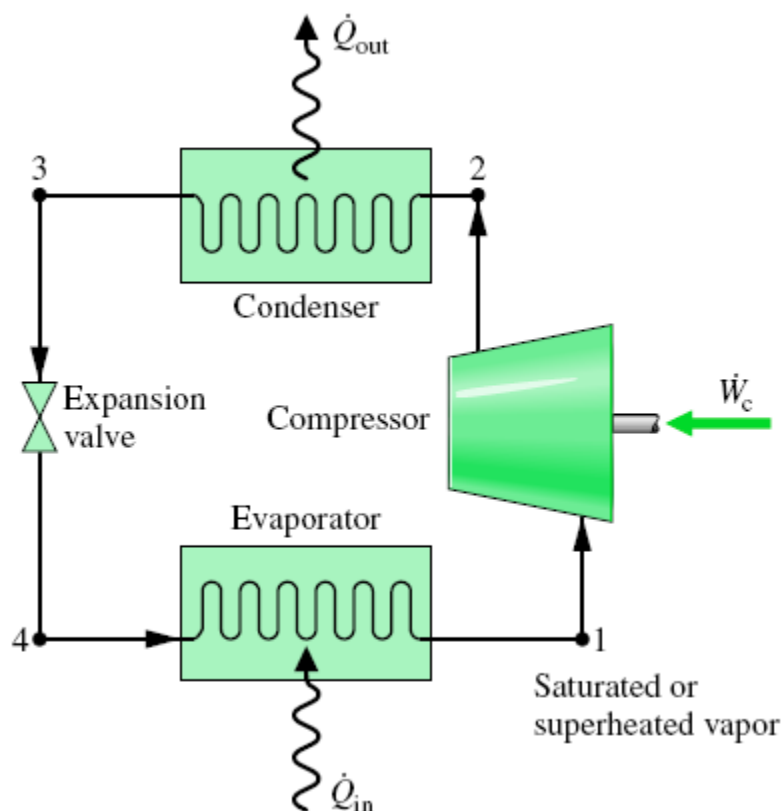


Figure 6.9: Vapor-compression cycle

The working fluid for this cycle is chosen so that the heat transfer in the condenser can take place at pressures obtainable from economical compressors, and the evaporation temperature in the evaporator can be easily be compatible with the inlet pressure to the compressor while allowing efficient heat transfer from the surroundings. For residential and commercial air conditioners, various chlorofluorinated hydrocarbons have the necessary properties; where lower temperatures are required, such as in commercial freezing equipment, ammonia is often used.

The vapor compression heat pump cycle is exactly as shown in **Figure 6.9** and **6.10**; only the objective of the cycle is changed. Now the evaporator is placed outdoors where there is a heat transfer to the working fluid at environmental temperatures (i.e., heat transfer from the environment). The evaporator may be placed in conjunction with any energy source that is at a lower than desired from the cycle. In domestic heating systems, the heat transfer can be obtained from the outdoor air (an *air-source* heat pump) or from a

convenient body of water such as lake, river, or well (a *water-source* heat pump). The condenser is placed where the higher-temperature heat transfer being rejected from the cycle is to be used. This can be in the residence for heating or in a heat exchanger in an industrial process.

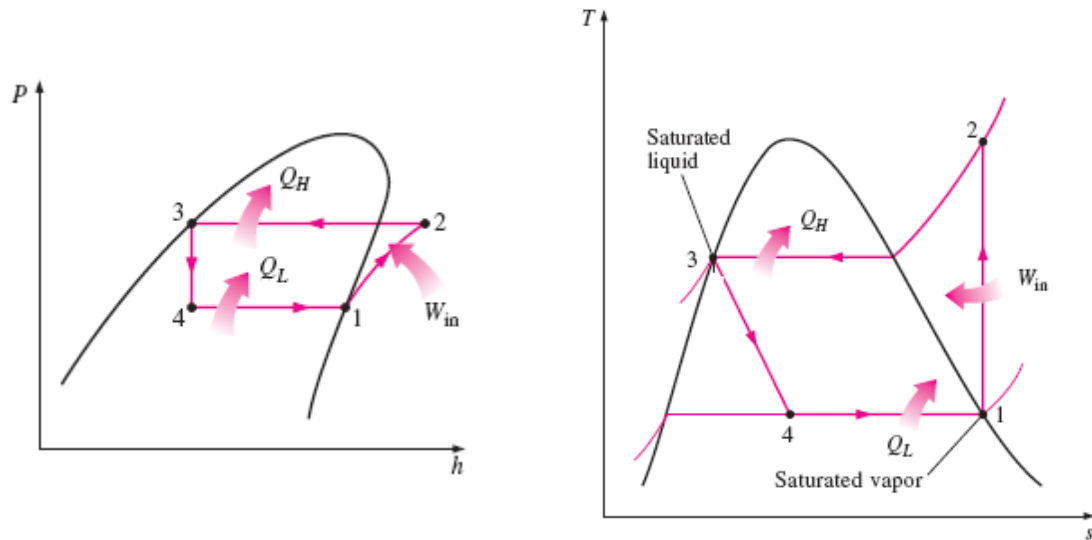


Figure 6.10: $P - h$ and $T - s$ diagrams for vapor-compression cycle

The measure of performance of a refrigeration system is given in terms of the coefficient of performance, which was defined as

$$COP_R = \frac{q_L}{w_c} = \frac{h_1 - h_4}{h_2 - h_1} \quad \dots\dots\dots (6.31)$$

Similarly, for the heat pump

$$COP_{HP} = \frac{q_H}{w_c} = \frac{h_2 - h_3}{h_2 - h_1} \quad \dots\dots\dots (6.32)$$

CHAPTER 7

INTRODUCTION TO HEAT TRANSFER

7.1 Basic Concepts and Modes of Heat Transfer

During the study of thermodynamics it is found necessary to distinguish between the thermal energy storage and that in transit from one system to another. The former is referred to as *internal energy* while the latter as *heat*. Thermal energy can originate from any kind of energy according to the first law of thermodynamics which governs all energy transformations quantitatively, without any restriction on the direction. Transfer of heat is due to the unique property of matter, *temperature*, and is governed by the second law of thermodynamics which dictates that free flow of heat is possible only from a body of higher temperature to that at a lower temperature.

All heat transfer processes, therefore, involve the transfer of energy, and obey the first as well as the second law of thermodynamics. It may, initially, appear that the principles of heat transfer are obtainable from the basic laws of thermodynamics. It is however not true. The science of thermodynamics deals with equilibrium. Since the heat flow will occur due to temperature non-equilibrium, its quantitative analysis could be derived on the basis of some other branch of science designated as *Heat Transfer*. The subject of heat transfer is, in reality, firmly grounded in the sciences of the *thermodynamics* and *fluid mechanics*.

As an example of the different kinds of problems which are treated by thermodynamics and heat transfer, consider the cooling of a hot steel bar which is placed in a pail of water. Thermodynamics may be used to predict the final equilibrium temperature of the steel bar-water combination. Thermodynamics will not tell us how long it takes to reach this equilibrium condition or what the temperature of the bar will be after a certain length of time before the equilibrium condition is attained. Heat transfer may be used to predict the temperature of both the bar and the water as a function of time.

The design of boilers, refrigerators or other types of heat exchangers say, for the storage, transfer and use of cryogenic substances and for a number of similar applications involve consideration of heat transfer rates. These are only a few examples of the engineering heat transfer in day-to-day practice.

There are three different modes of heat transfer: *conduction*, *convection*, and *radiation*. They have in common that a temperature difference must exist and that heat is always transferred in the direction of decreasing temperature. On the other hand, they differ entirely in the physical mechanisms and laws by which they are governed.

7.1.1 Conduction

Heat conduction is due to the property of matter which allows the passage for heat energy even its parts are not in motion relative to one another. The heat transfer rate per unit area is proportional to the normal temperature gradient:

$$\frac{\dot{Q}}{A} \propto \frac{\partial T}{\partial x} \quad \dots\dots\dots (7.1)$$

When the proportionality constant is inserted,

$$\dot{Q} = -KA \frac{\partial T}{\partial x} \quad \dots\dots\dots (7.2)$$

where $\dot{Q}$ is the heat transfer rate and $\frac{\partial T}{\partial x}$ is the temperature gradient in the direction of heat flow. The positive constant K is called the *thermal conductivity* of the material, and the minus sign is inserted so that the second law of thermodynamics will be satisfied; i.e., heat must flow downhill on the temperature scale. **Equation (7.1)** is called Fourier's law of heat conduction after the French mathematical physicist Joseph Fourier, who made very significant contributions to the analytical treatment of conduction heat transfer. It is important to note that **Equation (7.1)** is the defining equation for the thermal conductivity and that K has the units of *watts per meter per Celsius degree* in a typical system of units in which the heat flow is expressed in watts.

Heat is conducted in solids in two ways: transport of energy by free electrons and lattice vibration. In good conductors, a large number of free electrons move about in the lattice structure of the material which transport heat from high temperature region to the low temperature region. The portion of energy transported by free electrons is larger than that by lattice vibration. An increase in temperature causes increase in both lattice vibration and speed of electrons, but increased vibration of lattice disturbs the movement of free electrons causing reduction in transport of energy by free electrons which means the overall conduction is reduced. In insulators and alloys, the transport of energy is mainly due to lattice vibration and an increase in temperature increases conduction.

Heat conduction in liquid and gases are based on the movements of atoms and molecules. In case of liquids, random translatory motion is small and it appears that translation of energy occurs by longitudinal vibrations, similar to the propagation of sound.

7.1.2 Convection

Thermal convection is a process of energy transport affected by the circulation or mixing of a fluid medium (gas, liquid or powdery substance). Convection is possible only in fluid medium and is directly linked with the transport of medium of itself. Macroscopic particles of a fluid moving in space cause the heat exchange, and thus convection constitutes the *macroform* of the heat transfer. The effectiveness of heat transfer by convection depends largely upon the mixing motion of the fluid. With respect to origin, two types of convection are distinguished: *forced*, and *natural* or *free* convection.

In natural or free convection, the circulation of the fluid medium is caused by buoyancy effects, i.e., by the difference in the densities of the cold and heated particles. Consider the heat flow from a hot plate to atmosphere. The stagnant layer of air in the immediate vicinity of the plate gets thermal energy by conduction. The energy thus transferred serves to increase the temperature and internal energy of the air particles. Because of temperature rise these particles become less dense (and therefore lighter) than the surrounding air. The lighter air particles move upwards to a region of low temperature where they mix with and transfer a part of their energy to the cold particles. Simultaneously the cold air particles descend downwards to fill the space vacated by the

hot air particles. The circulation pattern, upward movement of the warm air and the downward movement of the cool air, is called the *convection currents*.

In *forced convection*, the flow of fluid is caused by a pump, fan, or by atmospheric winds. These mechanical devices provide a definite circuit for the circulating currents and that speed up the heat transfer rate. Example of forced convection are: flow of water in condenser tubes, fluid passing through the tubes of a heat exchanger; cooling of internal combustion engine; air conditioning installation and nuclear reactors.

Regardless of the particular nature, the appropriate rate equation for the convective heat transfer between a surface and an adjacent fluid is prescribed by *Newton's Law of cooling*:

$$\dot{Q} = hA(T_s - T_\infty) \quad \dots\dots\dots (7.3)$$

where, $\dot{Q}$ is the convective heat flow rate, A is the area exposed to heat transfer, T_s and T_∞ are the surface and fluid temperatures respectively. The heat transfer coefficient h depends upon the thermodynamic and transport properties (e.g. density, viscosity, specific heat and thermal conductivity of the fluid), the geometry of the surface, the nature of fluid flow, and the prevailing thermal conditions.

Convection mechanisms involving phase change leads to the important fields of boiling (evaporation) and condensation.

7.1.3 Radiation

Thermal radiation is the transmission of heat in the form of radiant energy or wave motion from one body to another across an intervening space. Unlike heat transfer by conduction and convection, transport of thermal radiation does not necessarily affect the material medium between the heat source and the receiver. An intervening medium is not even necessary and the radiation can be affected by through vacuum or a space devoid of any matter. Radiation exchange, in fact, occurs most effectively in vacuum. A material present between the heat source and the receiver would either reduce or eliminate entirely the propagation of radiation energy.

The mechanism of the heat flow by radiation consists of three distinct phases:

- (i) *Conversion of thermal energy of the hot source into electromagnetic waves*: All bodies above absolute zero temperature are capable of emitting radiant energy. Energy released by a radiating surface is not continuous but in the form of successive and separate (discrete) packets or quanta of energy called *photons*. The photons are propagated through the space as rays; the movement of swarm of photons is described as the electromagnetic waves.
- (ii) *Passage of wave motion through intervening space*: The photons, as carriers of energy, travel with unchanged frequency in straight paths with speed equal to that of light.
- (iii) *Transformation of wave into heat*: When the photons approach the cold the cold receiving surface there occurs reconversion of wave motion into thermal energy which is partly absorbed, reflected or transmitted through the receiving surface.

The maximum rate of radiation that can be emitted from a surface at an absolute temperature T_s (in K or R) is given by the *Stefan–Boltzmann law* as

$$\dot{Q}_{max} = \sigma AT_s^4 \quad \dots\dots\dots (7.4)$$

where $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$ is the *Stefan–Boltzmann constant*. The idealized surface that emits radiation at this maximum rate is called a *blackbody*, and the radiation emitted by a blackbody is called blackbody radiation. The radiation emitted by all real surfaces is less than the radiation emitted by a blackbody at the same temperature, and is expressed as

$$\dot{Q}_{emit} = \varepsilon \sigma AT_s^4 \quad \dots\dots\dots (7.5)$$

where ε is the *emissivity* of the surface. The property emissivity, whose value is in the range $0 < \varepsilon < 1$, is a measure of how closely a surface approximates a blackbody for which $\varepsilon = 1$.

7.2 Linear One dimensional Steady State Heat Conduction

One dimensional steady state heat conduction includes heat flow through a wall or along an insulated rod. For one dimensional, steady state heat conduction through a solid with no heat generation, **Equation (7.2)** reduces to

$$\dot{Q} = -KA \frac{dT}{dx} \quad \dots\dots\dots (7.6)$$

Integrating with boundary conditions ($T = T_1$ at $x = 0$ and $T = T_2$ at $x = L$), we get

$$\begin{aligned} \dot{Q} \int_0^L dx &= -KA \int_{T_1}^{T_2} dT \\ \therefore \dot{Q} &= \frac{KA}{L} (T_1 - T_2) \quad \dots\dots\dots (7.7) \end{aligned}$$

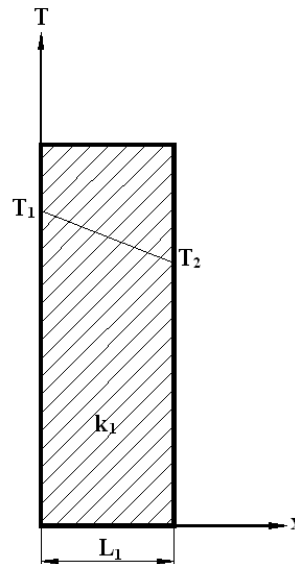


Figure 7.1: One dimensional Heat Conduction

7.3 Radial Heat Conduction Through Tubes

Let us assume the cylinder to be of infinite length and the flow to be only in radial direction. If there is no heat generation, then for steady state heat flow equation becomes

$$\dot{Q} = -KA \frac{dT}{dr} \quad \dots\dots\dots (7.8)$$

$$\text{or, } \dot{Q} = -K(2\pi rL) \frac{dT}{dr}$$

$$\text{or, } \dot{Q} \frac{dr}{r} = -2\pi KL dT$$

Integrating with boundary conditions ($T = T_1$ at $r = r_1$ and $T = T_2$ at $r = r_2$), we get

$$\dot{Q} \int_{r_1}^{r_2} \frac{dr}{r} = -2\pi KL \int_{T_1}^{T_2} dT$$

$$\text{or, } \dot{Q} \int_{r_1}^{r_2} \frac{dr}{r} = -2\pi KL \int_{T_1}^{T_2} dT$$

$$\text{or, } \dot{Q} \ln\left(\frac{r_2}{r_1}\right) = -2\pi KL(T_2 - T_1) = 2\pi KL(T_1 - T_2)$$

Hence radial heat flow through a tube under steady condition is given by

$$\therefore \dot{Q} = \frac{2\pi KL}{\ln\left(\frac{r_2}{r_1}\right)} (T_1 - T_2) \quad \dots\dots\dots (7.9)$$

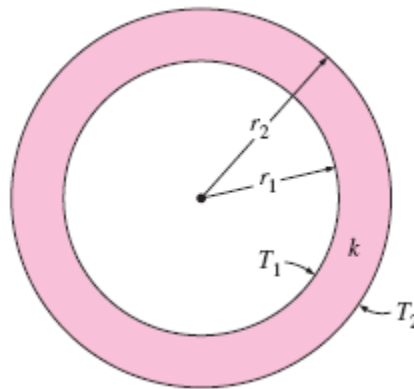


Figure 7.2: Radial heat flow through a tube

7.4 Heat Flow Through Composite Structures

There are a few systems of considerable practical utility which are made up of two or more layers of different materials. For example, cold storage walls have a layer of bricks, a layer of thick insulation and plaster on both sides. Similarly steam pipes have the cylindrical steel wall of the pipe, a layer of insulating material and then a layer of protecting plaster. The treatment of conductive heat flow in such structures may be considered as an extension of the single wall structures discussed earlier.

In such composite structures each layer has a different value of thermal resistance; the overall cumulative effect of these resistances has to be considered. Usually the resistances of the free surfaces also have to be accounted for as additional resistances in series, in the

calculations. This is due to the fact the actual surface temperatures, which are different than the ambient values, are not known.

It is generally assumed that heat flow is steady and one-dimensional, there are no heat sources in the structure and that the resistance due to interface contact is negligible.

Interface contact depends upon total force exerted on the contacting surfaces, physical nature of the bond, the quality of the fluid trapped in the contacting surface etc. The mechanism of heat transfer across the interface contact is therefore complicated. However the values of resistances at interface contact obtained experimentally are so small as to justify the above assumption.

7.4.1 Composite Plane Wall

Figure 7.3 shows a system consisting of three layers of different materials placed in series so as to make a composite plane structure. The temperature gradients in the three materials are also shown, and the heat flow may be written as

$$\dot{Q} = \frac{K_1 A}{L_1} (T_1 - T_2) = \frac{K_2 A}{L_2} (T_2 - T_3) = \frac{K_3 A}{L_3} (T_3 - T_4)$$

Note that the heat flow must be same through all sections.

From the above equations we write

$$T_1 - T_2 = \frac{\dot{Q} L_1}{K_1 A} \quad T_2 - T_3 = \frac{\dot{Q} L_2}{K_2 A} \quad \text{and} \quad T_3 - T_4 = \frac{\dot{Q} L_3}{K_3 A}$$

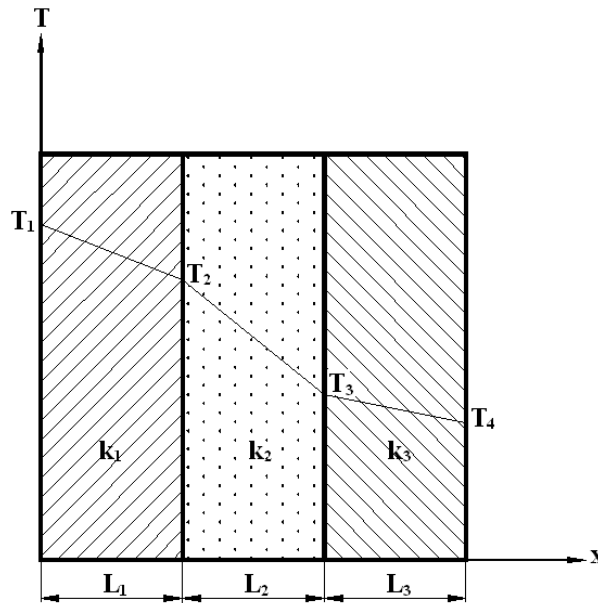


Figure 7.3: One-dimensional heat transfer through a composite wall

Adding both sides of all the above equations we get,

$$\frac{\dot{Q}}{A} \left[\frac{L_1}{K_1} + \frac{L_2}{K_2} + \frac{L_3}{K_3} \right] = T_1 - T_4$$

$$\dot{Q} = \frac{A(T_1 - T_4)}{\frac{L_1}{K_1} + \frac{L_2}{K_2} + \frac{L_3}{K_3}} \quad \dots\dots\dots (7.10)$$

7.4.2 Composite Cylinders

Let us assume, for a general case, a composite cylindrical tube made of three layers having thermal conductivity K_1 , K_2 and K_3 respectively. A practical application of such a system is usually found in industrial installations where one fluid at a certain temperature flows inside the tube while the outer surface is exposed to some other fluid at different temperature. Let these temperatures be T_1 and T_4 in an insulated pipe of length L shown in **Figure 7.4**.

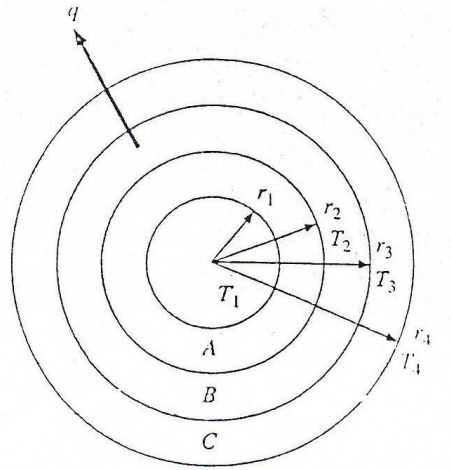


Figure 8.4: Composite cylindrical wall

If the cylinder is too long then the heat flow may be assumed to be radial only. In steady state the rate of flow of heat is same for through each section and is given by:

$$\therefore \dot{Q} = \frac{2\pi K_1 L}{\ln\left(\frac{r_2}{r_1}\right)} (T_1 - T_2) = \frac{2\pi K_2 L}{\ln\left(\frac{r_3}{r_2}\right)} (T_2 - T_3) = \frac{2\pi K_3 L}{\ln\left(\frac{r_4}{r_3}\right)} (T_3 - T_4)$$

From the above equations we can write down

$$T_1 - T_2 = \frac{\dot{Q}}{\frac{K_1 \cdot 2\pi L}{\ln\left(\frac{r_2}{r_1}\right)}} \quad T_2 - T_3 = \frac{\dot{Q}}{\frac{K_2 \cdot 2\pi L}{\ln\left(\frac{r_3}{r_2}\right)}} \quad \text{and} \quad T_3 - T_4 = \frac{\dot{Q}}{\frac{K_3 \cdot 2\pi L}{\ln\left(\frac{r_4}{r_3}\right)}}$$

Adding the both sides

$$\frac{\dot{Q}}{2\pi L} \left[\frac{\ln\left(\frac{r_2}{r_1}\right)}{K_1} + \frac{\ln\left(\frac{r_3}{r_2}\right)}{K_2} + \frac{\ln\left(\frac{r_4}{r_3}\right)}{K_3} \right] = T_1 - T_4$$

$$\therefore \dot{Q} = \frac{2\pi L(T_1 - T_4)}{\frac{\ln\left(\frac{r_2}{r_1}\right)}{K_1} + \frac{\ln\left(\frac{r_3}{r_2}\right)}{K_2} + \frac{\ln\left(\frac{r_4}{r_3}\right)}{K_3}} \quad \dots\dots\dots (7.11)$$

7.5 Electric Analogy for Thermal Resistance

A review of above equations lead that heat transfer rate may be considered as a flow, and the combination of thermal conductivity, thickness of material, and area as a resistance to flow, and the Fourier equation may be written

$$\text{Heat flow} = \frac{\text{thermal potential difference}}{\text{thermal resistance}} \quad \dots\dots\dots (7.12)$$

a relation quite like Ohm's law in electric-circuit theory. In **Equation (7.7)** the thermal resistance is L/KA , and in **Equation (7.10)** it is the sum of the three terms in the denominator. We should expect this situation in **Equation (7.10)** because the three walls side by side act as three thermal resistances in series. The equivalent electrical circuit is shown in **Figure 7.3**.

The electrical circuit analogy may be used to solve more complex problems involving both series and parallel thermal resistances. A typical problem and its analogous electric circuit are shown in **Figure 7.5**. The one-dimensional heat flow equation for this type of problem may be written

$$\dot{Q} = \frac{\Delta T_{\text{overall}}}{\sum R_{th}} \quad \dots\dots\dots (7.12)$$

where the R_{th} are the thermal resistances of various materials. The units for the thermal resistance are $^{\circ}\text{C}/\text{W}$.

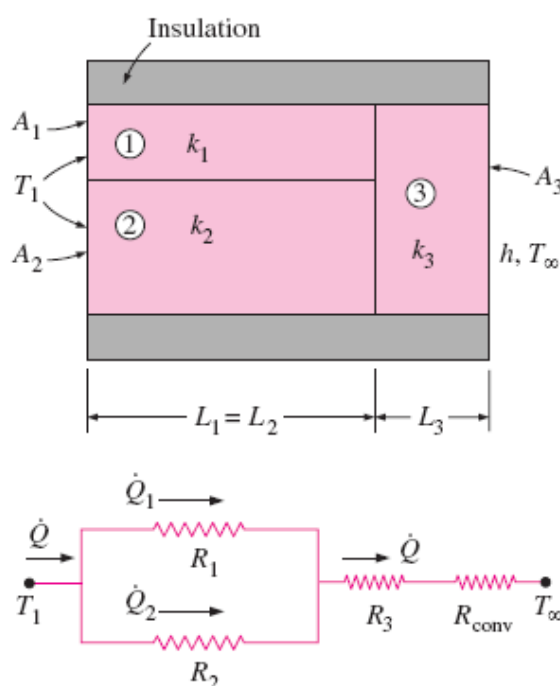


Figure 8.5: Series and parallel one-dimensional heat transfer through a composite wall and electrical analog

7.6 The Overall Heat Transfer Coefficient

When a moving fluid comes in contact with a stationary surface, a thin boundary layer develops adjacent to the wall and in this layer there is no relative velocity with respect to

surface. In a heat exchange process, this layer is called stagnant film and heat flow in the layer is covered both by conduction and convection processes. Since thermal conductivity of fluid is low, the heat flow from the moving fluid to the wall is mainly due to convection. The rate of convective heat transfer between a solid boundary and adjacent fluid is given by **Equation (7.3)**

$$\dot{Q} = hA(T_s - T_\infty) \quad \dots\dots\dots (7.3)$$

The resistance of the stagnant boundary layer is included in the convection coefficient. The factor $1/hA$ represents the thermal resistance of the film heat transfer process.

The heat transfer through a wall separating two moving fluids involves: (a) flow of heat from the fluid of high temperature to the wall, (b) heat conduction through the wall and (c) transport of heat from the wall to the cold fluid (**Figure 7.6**)

Under steady state conditions, the heat transfer is expressed as

$$\dot{Q} = h_1A(T_{\infty 1} - T_1) = \frac{KA}{L}(T_1 - T_2) = h_2A(T_2 - T_{\infty 2})$$

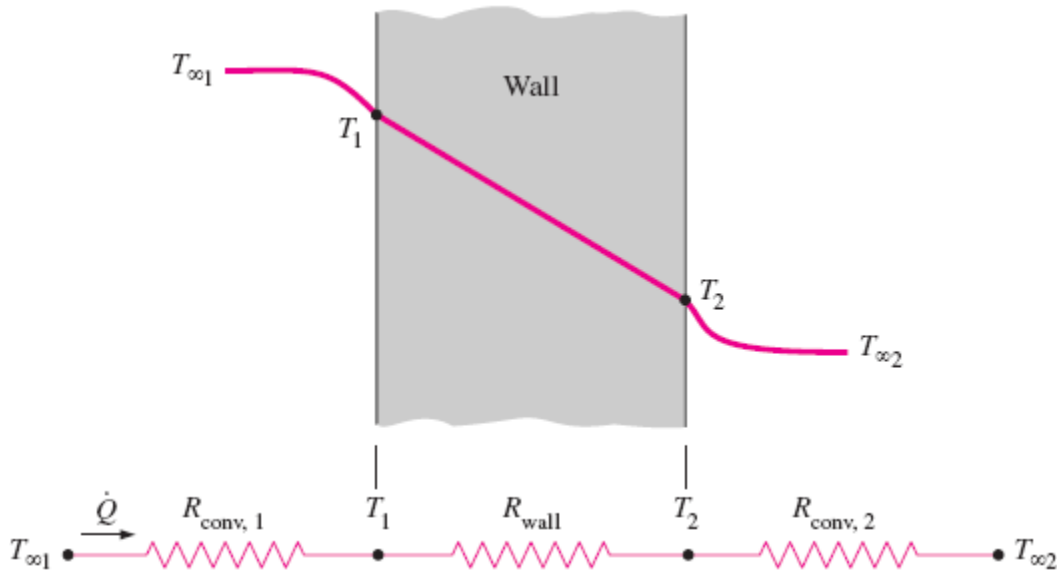


Figure 7.6: Overall heat transfer through a plane wall

The heat transfer process may be represented by the resistance network in **Figure 8.6**, and the overall heat transfer is calculated as the ration of the overall temperature difference to the sum of the thermal resistances:

$$\dot{Q} = \frac{A(T_{\infty 1} - T_{\infty 2})}{\frac{1}{h_1} + \frac{L}{K} + \frac{1}{h_2}} \quad \dots\dots\dots (7.13)$$

The overall heat transfer by combined conduction and convection is frequently expresses in terms of an overall heat transfer coefficient U , defined by the relation

$$\dot{Q} = UA\Delta T \quad \dots\dots\dots (7.14)$$

where A is some suitable area for the heat flow. In accordance with **Equation (7.13)**, the overall heat transfer coefficient would be

$$U = \frac{1}{\frac{1}{h_1} + \frac{L}{K} + \frac{1}{h_2}} \quad \dots\dots\dots (7.15)$$

For a hollow cylinder exposed to a convection environment on its inner and outer surfaces, the electric-resistance analogy would appear as shown in **Figure 7.7** where, again, T_A and T_B are the two fluid temperatures. Note that the area for convection is not the same for both fluids in this case, these areas depending on the inside tube diameter and wall thickness. In this case overall heat transfer coefficient would be expressed by

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{\frac{1}{h_1 A_1} + \frac{\ln(r_2/r_1)}{2\pi KL} + \frac{1}{h_2 A_2}} \quad \dots\dots\dots (7.16)$$

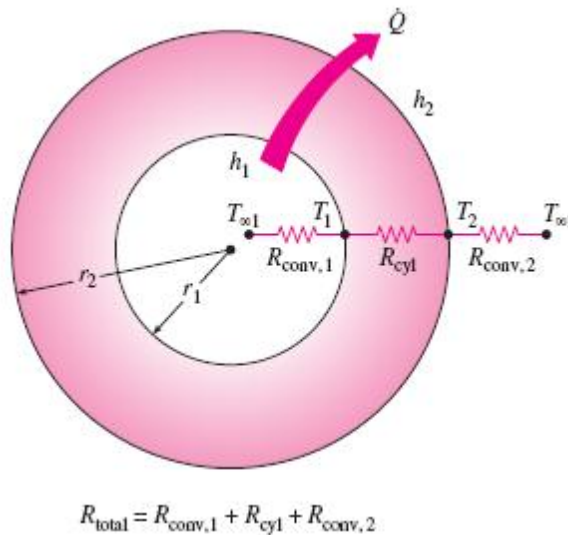


Figure 7.7: Overall heat transfer through a hollow cylinder

in accordance with the thermal network shown in **Figure 7.7**. The terms A_1 and A_2 represent the inside and outside surface areas of the inner tube. The overall heat transfer coefficient may be based on either the inside or the outside area of the tube. Accordingly,

$$U_1 = \frac{1}{\frac{1}{h_1} + \frac{A_1 \ln(r_2/r_1)}{2\pi KL} + \frac{A_1}{A_2} \frac{1}{h_2}} \quad \dots\dots\dots (7.17)$$

$$U_2 = \frac{1}{\frac{A_1}{A_2} \frac{1}{h_1} + \frac{A_2 \ln(r_2/r_1)}{2\pi KL} + \frac{1}{h_2}} \quad \dots\dots\dots (7.18)$$

7.7 The Nature of Heat Convection

One of the major problems encountered in engineering practice is the calculation of heat exchange between a *solid surface* and *fluid* in contact with it. If the fluid is at *rest*, the problem reduces to that of *conduction* only. Such a situation, however, is difficult to visualize. Even if we consider water being heated in a vessel, it will be found that it is *not*

at rest because of density gradients being developed in the whole body of water. The water in the bottom is hotter and therefore lighter than that at the top, thus it rises up replacing the colder water which flows down. Therefore even in this case the water is in motion due to heating. The heat exchange by *convection*, strictly speaking, refers to that occurring between a *solid surface* and the *fluid flowing* along it.

Convection is not fully a heat transfer mechanism (which will depend on the mere existence of temperature differential) because it depends for its operation on the *motion of fluid* acting as a carrier of energy. The fluid motion along the solid surface may be motivated by any of the two processes given below:

- (a) It may be caused by *density difference* produced by *temperature gradient*. Heat exchange in such a situation is known as *free or natural convection*.
- (b) It may be caused by some external agency such as a *pump* or *blower*. The motion of *fluid* in such cases will be faster than that in *free motion* and therefore will result in increased rate of heat transfer. This type of heat transfer is known as *forced convection*.

It is obvious that the transfer of energy in a convection process occurs mechanically and depends upon the motion of fluid particles. Thus in addition to the law of heat conduction and geometry of the system, the heat exchange by convection will also depend upon the laws of *fluid dynamics*.

The Newton's heat convection equation $\dot{Q} = hA\Delta T$ is not as simple as it appears. This equation is actually a definition of the *convective heat transfer coefficient* or *unit thermal convection conductance*, h , rather than representation of the heat convection law. The determination of the *local value* h_x or *average value* h , of this coefficient forms the basis of the computation of heat transfer by convection.

7.8 Introduction to Radiation Heat Transfer

Radiation is an electromagnetic phenomenon of varying wavelengths closely allied to the transmission of light and radio. It proceeds in straight lines at the speed of light, (3×10^8 m/s). This speed is the product of the wavelength and frequency of the radiation

$$C = \lambda \nu \quad \dots\dots\dots (7.19)$$

where C is the speed of light, λ is wavelength and ν is the frequency.

Unlike conduction, radiation requires no transfer medium between the emitting and receiving surfaces. In fact, any material medium between such surfaces could impede radiation transfer of energy. The classic example of energy transmission by radiation is the sun, which transmits abundant energy to the earth by this means. Radiation is independent of mass, except for nuclear reactions.

The amount of energy (heat) transferred by conduction largely depends upon temperature difference rather than temperature level. But in radiation it is the temperature of the emitting surface that controls the quality of the energy transmitted.

Radiation is very much a surface phenomenon and will leave the transmitting surface through a wide wavelength band. A surface will emit or absorb radiant energy without a

temperature difference, but in order for the energy transfer to occur there must be a temperature difference between the exchange surfaces.

Figure 7.8 shows the general shape of the spectromagnetic curve of radiant power against wavelength for a *black body*. A black body which is defined as being *black* is a perfect radiator energy.

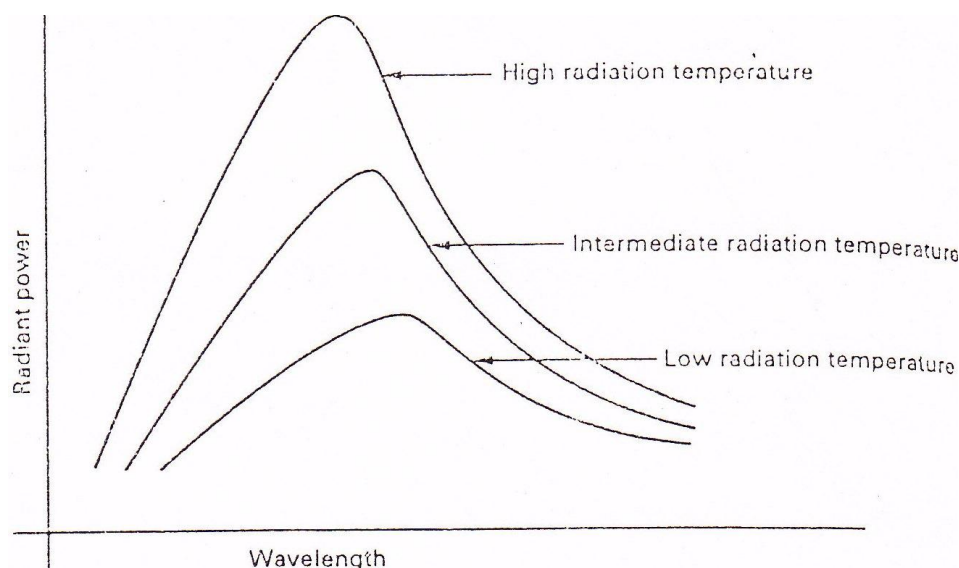


Figure 7.8: Spectromagnetic curves

In **Figure 7.8** the height of the curve defines the quantity, and the general shape shows the quality of the emitted radiant energy. At the lower radiation temperature there is a shift towards emissions of lower wavelength.

A radiating surface at a high temperature, perhaps over 800°C , will emit some wavelengths which are within the visible light spectrum, approximately 10^{-6} to 10^{-7} meters. At lower temperatures, less than 800°C , the radiation will be in *infrared* range, approximately 10^{-2} to 10^{-6} meters.

If body emits energy across the black body wavelength spectrum but only a fraction of the power of the power of the black body, then is called a *grey body*. The term *black* and *grey* do not necessarily refer to the color of the body; they merely describe its effectiveness as a radiator. A black body is a perfect radiator; a grey body is not perfect radiator.

Radiation Surface Parameters

Surfaces are capable of emitting, absorbing, reflecting or transmitting radiant energy. It can be shown that the emissivity and absorptivity of any given surface are of equal value. For any given, the radiation received must be reflected, absorbed or transmitted through the material.

For a black surface, therefore, the net sum of the maximum reflected, absorbed and transmitted energy must be equal to unity;

$$\rho + \alpha + \tau = 1 \quad \dots\dots\dots (7.20)$$

where ρ = reflectivity
 α = absorptivity energy
 τ = transmissivity

For materials that are opaque, the transmissivity is zero, so $\tau = 0$ and **Equation (7.20)** becomes

$$\rho + \alpha = 1 \quad \text{..... (7.21)}$$

which becomes

$$\rho + \varepsilon = 1 \quad \text{..... (7.22)}$$

where ε = emissivity.

Because emissivity and absorptivity can be shown to be equal, for a black surface, the reflectivity is also considered to be zero. Thus from **Equation (7.22)** the emissivity of a perfect black surface, ε_B is unity.

For a grey surface the emissivity, ε_S , which is a ratio of the grey surface emission to the black surface emission, must therefore be of the form

$$\varepsilon_S = \frac{E_S}{E_B} < 1 \quad \text{..... (7.23)}$$

For a grey surface with emissivity ε_S the Stefan-Boltzmann equation becomes

$$\dot{Q} = \varepsilon_S \sigma A T_s^4 \quad \text{..... (7.24)}$$

Emissivity, and hence absorptivity, vary somewhat with surface temperature and are also a function of surface finish.

The radiant heat-exchange between two grey bodies at temperature T_1 and T_2 depends on how the two bodies view each other and their emissivities, and it is given by

$$\dot{Q}_{1-2} = F_{1-2} \sigma A_1 (T_1^4 - T_2^4) \quad \text{..... (7.25)}$$